

BASE24-pos

APACS 70 DEVICE HANDLER

USER'S GUIDE

Release 6.0



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PREFACE

The BASE24-pos APACS 70 Device Handler is a BASE24 product that provides an interface between a BASE24 acquirer system and POS terminals conforming to the APACS 70 Standard. The term APACS 70 Device Handler or APACS 70 POS Device Handler is used to describe the BASE24 process that interfaces with a terminal.

Intended Audience

This document is primarily intended for staff responsible for the installation, configuration, operation and maintenance of the BASE24-pos APACS 70 Device Handler module. The BASE24-pos APACS 70 Device Handler module supports POS terminals conforming to the APACS 70 Standard, and is compatible with BASE24-pos Release 6.0.

Software

This manual documents standard processing for the Release 6.0 BASE24-pos APACS 70 Device Handler.

Custom software modifications (CSMs) may result in processing that differs from the material presented in this manual. The customer is responsible for identifying and noting these changes.

Location of Information

A list of section headings and related page references are given in the Contents page. The headings are function oriented.

Section 1, “Introduction,” gives a brief overview of the BASE24-pos architecture and the requirements of the APACS 70 Standards.

Section 2, “Processing Overview,” gives a brief and general description of the functionality of the BASE24-pos APACS 70 Device Handler module.

Section 3, “Configuration and Files Maintenance,” describes the XPNET and BASE24 File configuration requirements for the BASE24-pos APACS 70 Device Handler module.

Section 4, “Startup and Control,” details initialisation processing and the operator commands available to control the BASE24-pos APACS 70 Device Handler module.

Section 5, “Message Processing,” provides a description of the message interface between the device and BASE24.

Section 6, “Cardholder Transactions,” details the types of transaction at APACS terminals that are supported by the BASE24-pos APACS 70 Device Handler module.

Section 7, “Reconciliation and Balancing,” details cutover processing performed by the BASE24-pos APACS 70 Device Handler module and BASE24-pos settlement.

Appendix A, “Device Handler Internal Tables,” describes the translations used during internal Device Handler processing.

Appendix B, “Device Handler External Tables,” describes the standard translations used, but customers may modify these tables.

Appendix C, “Log Messages,” describes the log messages supported with the causes and appropriate remedies, where relevant.

Appendix D, “Transaction Key Security,” outlines the security aspects of Standard 70.

Related Documentation

BASE24 Documentation

- NET24-XPNET Network Control Operations Guide (XN-NT000-05)
- NET24-XPNET Network Control Command Reference Manual (XN-NT000-06)
- BASE24 Text Command Reference Manual (BA-AE000-05)
- BASE24 Event Message Reference Manual (BA-AE000-07)
- BASE24 Logical Network Configuration File Manual (BA-AE000-10)
- BASE24 Tokens Manual (BA-AE000-12)
- BASE24-pos Files Maintenance Manual (PS-AE000-02)
- BASE24-pos Settlement and Reporting Manual (PS-AE000-06)
- BASE24-pos Transaction Processing Manual (PS-AE000-07)
- BASE24-pos Address Verification Manual (PS-AE110-01)
- BASE24-pos Multiple Currency Support Manual (PS-AE300-10)
- BASE24-pos EMV Support Manual (PS-CC986-01)

Third Party Documentation

- APACS Standard 70, October 2004
Distributed by APACS Standards Unit, London, UK

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1. INTRODUCTION

The BASE24-pos APACS 70 Device Handler module provides the message interface, specific device controls and terminal reconciliation functions between BASE24-pos and terminals conforming to the APACS 70 Standards. The BASE24-pos release 6.0 APACS 70 Device Handler module supports the functions described in Book 2 of the APACS 70 Standard, up to and including the October 2004 version.

The Device Handler module, which is part of the Device Handler/Router/Authorisation process, supports messages, including EMV and multiple currency transactions, received from APACS 70 terminals. It provides the interface between the terminal message and the internal BASE24-pos Standard Internal Message (PSTM). The Device Handler module also maintains transaction totals for reconciliation between the terminal and BASE24.

1.1 BASE24 SYSTEM ARCHITECTURE

The APACS 70 POS Device Handler module operates in conjunction with the BASE24-pos product.

1.1.1 Integration with BASE24-pos

The APACS 70 POS Device Handler module can be integrated with BASE24-pos. BASE24-pos is a complete point-of-sale authorisation and switching system that uses the technology of the HP NonStop computer to provide reliable, fault-tolerant service 24 hours a day, seven days a week.

BASE24-pos offers routing of transactions based on card type and retailer floor and ceiling limits. If the primary destination for authorisation is not available, BASE24-pos allows the transaction to be sent to up to two additional authorisation centres before a decision is made about approving, denying, or referring the transaction. Destinations to which transactions can be routed include host computers and interchange networks.

In addition, BASE24-pos offers a comprehensive set of authorisation options on the HP NonStop computer. These include local and national negative identification, positive identification, and positive identification with account balances.

BASE24-pos also includes online control of all network communications elements and POS terminals, as well as a comprehensive set of daily and periodic reports for service providers and individual retailers.

1.1.2 Transaction Flow

The XPNET process is the first recipient of a message sent by a POS terminal to BASE24. XPNET is the transport layer, managing communications between the BASE24 application and external entities, such as POS terminals. XPNET then sends the message to the bound Device Handler/Router/Authorization process.

The APACS 70 POS Device Handler module is bound into a Router/Authorization process. Any messages coming into the bound process from a POS device are first received by the Router module, which then routes the message to the Device Handler module. Messages coming into the bound process from any other destination are also received first by the Router module, but are routed instead to the Authorization module to be logged to the POS Transaction Log File (PTLF).

1.1.3 System Overview

When the Device Handler module receives a transaction message from a terminal it performs validation and handles any exception conditions that may arise.

The Device Handler module uses information contained in the terminal request message and on the POS Terminal Data File (PTDF) to create the POS Standard Internal Message (PSTM) for internal use by BASE24 and conversely to format response messages into the native mode for use by the terminal. If immediate confirmation of transaction messages is supported, the terminal sends a confirmation message to BASE24 as soon as a response message is received.

Figure 1-1 shows the Device Handler module in relation to other processes in the BASE24-pos system.

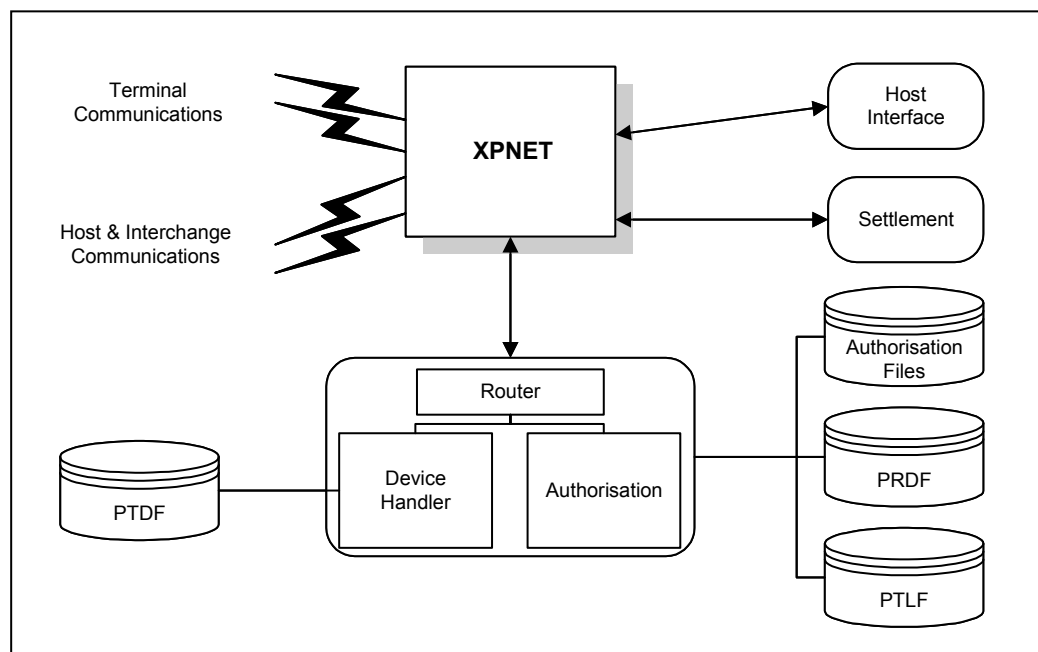


Figure 1-1 Relationship to other BASE24-pos Processes

1.2 APACS STANDARD 70

Standard 70 consists of the following books, under the general title - Card acceptor to acquirer interface standards:

- *Book 1 - Business rules for card processing*
- *Book 2 - Messages, data elements and code values for real-time systems*
- *Book 3 - Messages, data elements and code values for post-event systems*
- *Book 4 - Communications*
- *Book 5 - Security and key management.*
- *Book 6 - Data port interface*
- *Book 7 - Terminal Identities*

Standard 70 is a redrafting of a number of previous standards (Standard 29-4, 30, 40, 50 and the Common Attachment) into a more manageable format.

These standards are defined and maintained by the Association for Payment Clearing Services (APACS), and have become the *de facto* national standard in the U.K. for communications between a POS terminal and a host system.

Multiple terminal vendors have worked with the APACS standards referenced in this manual.

The APACS 70 POS Device Handler module is flexible enough to communicate with any sophisticated third-generation POS terminal having the capability of sending and receiving messages using the formats defined by the APACS standards. Customers can use any POS terminal that conforms to the APACS standards referenced in this manual.

The principle is that a host computer should be able to communicate with any terminal complying with this standard without the need to specifically recognise the manufacturer of that terminal or have to modify its actions based on this knowledge.

1.3 SUPPORTED TERMINALS

The APACS 70 POS Device Handler module supports terminals that conform to the functions defined in Book 2 of Standard 70 for either of the modes of operation described below.

1.3.1 Standard 70 (Authorisation Only)

Standard 70 Book 2 defines the interfaces and basic working of terminals whose function is to initiate card-based payments at the point of sale for authorisation. It specifically does not seek to define how the authorisations are subsequently captured (this is defined in Book 3 of Standard 70).

These types of terminal are described as “APACS 70 (Auth Only)”, when referenced in this manual.

1.3.2 Standard 70 (Financial Presentment)

Standard 70 Book 2 also defines the interfaces and basic working of terminals whose function is to capture card-based transactions at the point of sale. Because these terminals provide capture capabilities, they do not need to support the functions defined in Book 3 of Standard 70.

In order to provide capture capability, these terminals must support a higher level of security than authorisation only terminals. Therefore, these terminals and the APACS 70 POS Device Handler module support the Message Authentication (MAC) and PIN block encryption functions defined in Book 5 of Standard 70.

The financial presentment message formats are based on the authorisation only message formats, and in fact this is one of the specified modes of operation (i.e., a terminal produced to support financial presentments can operate as an authorisation only terminal). Because of this, these types of terminal are described as “APACS 70”, when referenced in this manual.

1.4 SUPPORTED PROTOCOLS

The communications options for APACS 70 terminals are defined in Book 4 of Standard 70.

The BASE24-pos APACS 70 Device Handler module can be configured to use the following communication protocols:

- Asynchronous 3270
- Bisynchronous 3270
- POS-2 DATAPAC 3101/VISA End-to-End
- SNA LU Type 0
- TCP/IP
- VISA
- VISA 1
- VISA 2
- X.21
- X.25.

The protocol selected by the customer depends on the line communications environment established between the BASE24-pos APACS 70 Device Handler module and its terminals. Line communications environments for the BASE24-pos APACS 70 Device Handler module can be dial, leased, X.21 or X.25.

Authorised ACI personnel assist customers in determining the type of protocol best suited for the customer's particular requirements at installation or when a system is configured.

1.5 TRANSACTION KEY SECURITY

The cryptographic security system defined in Book 5 of Standard 70 (known as the Transaction Key management scheme) provides for selective encryption and authentication of messages exchanged in retail Electronic Funds Transfer (EFT) applications.

Secret cryptographic keys must always be protected. However the degree of physical protection that can be provided in a retail terminal environment is likely to offer tamper resistance rather than full tamper proofing.

As a consequence, any key management system that relies on the use of long-term secret keys in the equipment at the retail point of sale is unlikely to satisfy the security requirements of the application.

The Transaction Key management scheme provides keys that are unique to each transaction and which do not require, or result in, any long-term secret keys being held in terminal equipment. Separate cryptographically isolated keys are used for Message Authentication and PIN Block Encryption.

The system produces transaction keys which are updated for each transaction, and which are based on the key for the previous transaction, the current transaction contents and the contents of the previous transaction. Thus transactions are chained providing automatic confirmation of previous transactions.

Further details are provided in Appendix D.

1.6 IMPLEMENTATION CONSIDERATIONS

This section documents environmental issues to be considered when implementing the APACS 70 POS Device Handler module.

1.6.1 Software Dependencies

The APACS 70 POS Device Handler module is compatible with XPNET version 3.1.

The product operates in a release 6.0 BASE24-pos system only.

1.6.2 Disk File Requirements

For each POS terminal supported by the APACS 70 POS Device Handler module, a record must exist in the PTDF. The terminal type in a PTDF record for an APACS terminal must be set to a value of either 70 or 71. It is recommended that a value of 70 is used for an APACS 70 (Auth Only) terminal, and that a value of 71 is used for an APACS 70 terminal.

2. PROCESSING OVERVIEW

The APACS 70 POS Device Handler module is responsible for the online message transfer between BASE24 and POS terminals conforming to the APACS 70 Standard.

2.1 FILE USAGE

This section provides an overview of the files used by the APACS 70 POS Device Handler module.

For further information on configuration requirements for the APACS 70 POS Device Handler module, see section 3.

All BASE24 files used by the APACS 70 POS Device Handler module are listed below. The APACS 70 POS Device Handler module uses both Base files and BASE24-pos files and, therefore, the table indicates whether each file is a Base file or a BASE24-pos file.

Files Used By The APACS 70 POS Device Handler Module	
File Name	File Type
Logical Network Configuration File (LCONF)	Base
Log Message Configuration File (LMCF)	Base
POS Retailer Definition File (PRDF)	POS
POS Terminal Data File (PTDF)	POS

There are also certain XPNET configuration requirements. The Network Environment File (NEF) is used to allow BASE24 to communicate with APACS 70 POS devices.

2.2 PROCESS CONTROL

This section provides an overview of operator interface to the APACS 70 POS Device Handler module.

For further information on startup and control of the APACS 70 POS Device Handler module, see section 4.

2.2.1 Operator Interface

The processing of the APACS 70 POS Device Handler module is controlled using Network Control Point Communications (NCPCOM) commands. For example, a process containing the APACS 70 POS Device Handler module can be started and stopped using NCPCOM.

Other NCPCOM functions supported by the APACS 70 POS Device Handler module include:

- Warmbooting the process
- Activating or deactivating a terminal or range of terminals
- Resetting the key registers in a PTDF record
- Turning on or off a load flag for a terminal or range of terminals

NCPCOM can also be used to modify the attributes of entities (e.g., APACS 70 terminal communications parameters).

For more detailed information on the NCPCOM utility, see the *NET24-XPNET Network Control Operations Guide*.

2.2.2 Event Message Generation

The APACS 70 POS Device Handler module generates event messages that keep operators aware of activity occurring in the system. Event messages can be used to assist operators with their daily tasks or to notify operators of problems occurring in the system. Event messages provide operators with insight as to how the system is running.

The event messages generated by the APACS 70 POS Device Handler module are documented in Appendix C. For a complete explanation of BASE24 event message documentation, refer to the *BASE24 Event Message Reference Manual*.

2.3 MESSAGE PROCESSING

This section provides an overview of the message processing performed by the APACS 70 POS Device Handler module.

For further information on message processing in the APACS 70 POS Device Handler module, see section 5.

2.3.1 APACS 70 Messages

The APACS 70 POS Device Handler module supports authorisation, financial and administrative transactions. Authorisation transactions are transactions performed on behalf of a cardholder, where authorisation processing is performed, but draft capture is not performed. Financial transactions are monetary transactions performed on behalf of a cardholder, and reversals of these transactions. Administrative transactions are monetary transactions performed in support of financial transactions (i.e., reconciliation) typically by supervisory personnel.

The financial transactions and administrative transactions are known collectively as Electronic Funds Transfer (EFT) messages, and are always protected by a message authentication code (MAC).

2.3.1.1 Authorisation Messages

The APACS 70 POS Device Handler module supports the following types of authorisation message:

- Authorisation Request;
- Authorisation Response;
- Reversal of Authorisation Request;
- Confirmation of Authorisation Response.

These message types are defined in Section 8.2 of APACS Standard 70 Book 2.

2.3.1.2 Financial Messages

The APACS 70 POS Device Handler module supports the following types of financial message:

- Data Capture Request
- Data Capture Response
- Reversal of Data Capture Request
- Post-Event Advice of a Data Capture Request

These message types are defined in Section 8.3 of APACS Standard Book 2.

2.3.1.3 Administrative Messages

The APACS 70 POS Device Handler module supports the following types of administrative message:

- Reconciliation Request
- Reconciliation Response
- Post-Event Advice of a Reconciliation Request

These message types are defined in Section 8.4 of APACS Standard Book 2.

2.3.2 Message Flows

All of the transactions supported by the APACS 70 POS Device Handler module must originate with a request from an APACS 70 terminal. BASE24 uses the information in the message to authenticate the message source and verify the MAC (if applicable), and then, by using a number of internal tables, translate the message to the appropriate BASE24 format.

The request message could be a first try, a retry, or a post-event advice. A *first try* is the first attempt of the terminal to send a request to BASE24; a *retry* is a subsequent attempt, after the original communication session with BASE24 has been lost; a *post-event advice* is an advice, where the terminal has authorised the transaction.

A response is sent from the APACS 70 POS Device Handler module for each request received unless the APACS 70 POS Device Handler module is unable to read the POS Terminal Data File (PTDF) record that corresponds to the terminal sending the request. In this case, no response is sent and the terminal times out. All messages that BASE24-pos sends to APACS 70 devices are formatted in accordance with the layouts described in Book 2 of the APACS 70 Standard.

2.3.3 Message Sequencing

The APACS 70 POS Device Handler module supports a method of ensuring that every message being sent from the terminal is being received only once and in the correct order. This method is known as message sequencing. The message number in the APACS 70 message number must be incremented for every transaction successfully processed by a host system.

The APACS 70 POS Device Handler module compares the sequence number in the current message against the sequence number in the previous message received from the terminal. If the numbers match, then this could indicate any of the following:

- The message is a duplicate of a previously transmitted message.
- The message is a retry of a previously transmitted message (i.e., the terminal timed out waiting for a response after successfully connecting to BASE24).
- The message is a post-event advice of a previously transmitted request (i.e., the terminal timed out waiting for a response, and authorised the transaction itself).
- The terminal failed to process the previous response message successfully.

The APACS 70 POS Device Handler module uses other data in the message to determine which of these scenarios has occurred. It then performs the appropriate processing.

2.3.4 Transaction Logging

The POS Transaction Log File (PTLF) contains information about the transactions processed by BASE24-pos.

Unless the APACS 70 POS Device Handler module rejects a message during request processing, transaction details are always written to the PTLF.

When a reconciliation transaction is performed, totals are logged to the PTLF, if logging PTLF totals is configured in the LOG TOTALS field on screen 3 of the PTDF. If logging PTLF totals is not configured, no totals are logged to the PTLF.

2.3.5 Software And Hardware Security

As the function of APACS 70 terminals is to capture card based transactions at the point of sale, the APACS 70 POS Device Handler module includes transaction security processing for message authentication codes (MACs) and PIN Block encryption.

Standard 70 Book 5 mandates the use of the Transaction Key security management scheme, as described in Appendix D. The APACS 70 POS Device Handler module supports this scheme by using either its own software or a Hardware Security Module (HSM).

- For software security processing, the MAC TYPE field on screen 7 of the PTDF must be set to a value of 01. In this case, algorithms coded within the Device Handler module is used to perform the security functions.
- For hardware security processing, the MAC TYPE field on screen 7 of the PTDF must be set to a value of 03. In this case, calls are made to an HSM to perform the security functions.

Notes:

1. The processing required for deriving the PIN Block encryption key is not supported in software.
2. If hardware security processing is required, then a Thales (Racal) HSM must be used, as Atalla HSMs do not support the Transaction Key security management scheme.
3. If hardware security processing is required, then the interface to the Thales (Racal) HSM can be managed either by BASE24-pos itself, or by the BASE24-es Transaction Security Services (TSS). The Transaction Key security management scheme is supported by TSS version 06.2 and later.
4. If the APACS terminal supports Triple DES MACing and Triple DES PIN Block encryption, then TSS must be used to communicate with the Thales (Racal) HSM. The Triple DES extension to the Transaction Key security management scheme is supported by TSS version 06.2 and later.

2.4 CARDHOLDER TRANSACTIONS

This section provides an overview of the cardholder transaction support within the APACS 70 POS Device Handler module.

For further information on cardholder transaction processing performed by the APACS 70 POS Device Handler module, see section 6.

2.4.1 Transaction Types

The complete customer transaction set supported by APACS 70 terminals is listed below. APACS 70 (Auth Only) terminals are limited to authorisation transactions only.

- Authorisation only for purchase of goods and services, cash advances, account funding, refunds, and purchase with cash back
- Authorisation and data capture for purchase of goods and services, cash advances, account funding, refunds, and purchase with cash back
- Reversal of authorisation only for purchase of goods and services, cash advances, account funding, refunds, and purchase of goods and services with cash back
- Reversal of authorisation and data capture for purchase of goods and services, cash advances, account funding, refunds, and purchase of goods and services with cash back
- Offline data capture for purchase of goods and services, cash advances, account funding, refunds, and purchase of goods and services with cash back
- Card verification
- Balance enquiry

Note that Standard 70 no longer supports Preauthorisation transactions. However, some “legacy” APACS 70 terminals may still provide support for these types of transaction, so support has been retained in the APACS 70 POS Device Handler.

2.4.2 Transaction Processing

As transaction details are input, the terminal compiles a message for transmission to BASE24. This message contains the transaction request details that include:

- Terminal identity
- Message sequence number
- Transaction type
- Cardholder's primary account number (PAN)
- Transaction amount

When the APACS 70 POS Device Handler module receives a message from the terminal, it is translated from the into the BASE24-pos Standard Internal Message (PSTM) format. The internal message is then sent to the BASE24-pos Router/Authorisation module.

For more information on the PSTM, see the ***BASE24-pos Transaction Processing Manual***.

When the message has been processed, the authorising entity sends an internal response to the APACS 70 POS Device Handler module. Using information from the internal message and its internal tables, the APACS 70 POS Device Handler module formats an external response message and sends it to the terminal.

On receipt of the response message, the terminal displays a message on its screen, indicating the outcome of the transaction processing (e.g., whether the transaction has been approved).

2.4.3 EMV Support

If you have installed the BASE24-pos Europay, MasterCard, and Visa (EMV) add-on product, the APACS 70 Device Handler module performs the following functions:

- For requests, the module maps EMV data from the request message to tokens and appends these tokens to the PSTM. If EMV terminal capabilities are not provided in the request message, the module checks the EMV terminal capability settings for the terminal on BASE24-pos Terminal Data files (PTD) screen 17 to determine which EMV tokens to include. The module formats the EMV Request data token, the EMV Status token, and the EMV Discretionary Data token as needed.
- For responses, the module maps EMV data from the EMV Response Data token and the EMV Script Data token to the response message sent back to the terminal.

For detailed information on PTD screen 17 and EMV-specific tokens, refer to the *BASE24-pos EMV Support Manual*.

2.4.4 Multiple Currency Transaction Support

If you have installed the BASE24-pos Multiple Currency add-on product, the APACS 70 Device Handler module performs the following multiple currency functions:

- Validates the transaction currency code passed in the request message against the primary terminal currency defined in the CURRENCY CODE field on PTD screen 2.
- Adds the Multiple currency token with an entry for the primary terminal currency.
- If the MC TOTALLING TIPE field on PTD screen 2 is set to a value of 1, amounts in the PSTM are converted into the primary terminal currency before they are stored in the PTDF. Otherwise, amounts in the PSTM are not converted into the primary terminal currency before they are stored in the PTDF.

For detailed information on Multiple Currency support for the APACS 70 Device Handler module, refer to the *BASE24-pos Multiple Currency Support Manual*.

2.4.5 Draft Capture Options

The APACS 70 POS Device Handler module provides an efficient method of draft capture with APACS 70 terminals by performing authorisation and draft capture in a single transaction. If a transaction is not completed correctly, the operator who entered the transaction sends a reversal message.

The APACS 70 POS Device Handler module also supports transactions that require follow-up processing for draft capture. This method is used for terminals operating in APACS 70 (Auth Only) mode. A transaction is authorised online, and then data collection is performed using a mechanism external to BASE24 (e.g., a paper voucher follow-up) or polling the terminal.

2.5 CUTOVER AND RECONCILIATION

This section provides an overview of BASE24-pos settlement and cutover processing, and its application within the APACS 70 POS Device Handler module.

For further information on the cutover and reconciliation transaction processing performed by the APACS 70 POS Device Handler module, see section 7.

2.5.1 BASE24-pos Settlement and Cutover

BASE24-pos provides an efficient method of settlement and cutover. *Settlement*, in terms of BASE24-pos, is actually a reconciliation effort whereby the current day's transaction activity is closed out and reported, and the system is shown to be in balance. The term *cutover* simply refers to the point in time where transactions stop being logged to the current day's business and start being logged to the next day's business. It refers to the closing of the current posting day and the opening of the next.

BASE24-pos settlement differs from the settlement carried out by participating BASE24-pos financial entities (e.g., retailers, retailer sponsors, cardholders, card issuers, interchanges) in that BASE24-pos does not move funds to settle accounts. It does, however, provide the support that allows these financial entities to initiate the movement of funds.

An extensive set of reports provides financial entities with the information they require to settle their accounts. BASE24-pos also provides the POS Transaction Log File (PTLF), which keeps a record of each transaction processed by BASE24-pos during its defined reporting day. In addition, BASE24-pos provides an automated method to cut over terminals, retailers, institutions, interchanges, and the BASE24-pos network to a new posting day. For more information on BASE24-pos settlement, cutover, and reporting, refer to the ***BASE24-pos Settlement and Reporting Manual***.

2.5.2 Terminal Balancing

The APACS 70 POS Device Handler module allows merchants to balance their APACS 70 terminals with BASE24-pos. The APACS 70 POS Device Handler module retains terminal totals and allows the retailer to reconcile with BASE24-pos in order to ensure that the terminal totals balance against BASE24-pos.

The APACS 70 POS Device Handler module supports the APACS 70 reconciliation transaction, which allows merchants to balance terminals with BASE24-pos. Reconciliation messages include the totals accumulated by the terminal, and the APACS 70 POS Device Handler module compares these totals to those maintained in the PTDF (in the batch totals fields). If the totals match, then they are written to the PTLF, and the PTDF totals are cleared. If the totals do not match, then both sets of totals are written to the PTLF, before the PTDF totals are cleared.

In addition, merchants also can receive a report a day later, or contact the service provider to receive terminal balancing information. The BASE24-pos report that contains terminal balancing information is the Retailer Reconciliation Report (B24POS35). This report is used to balance transaction activity at the terminal batch level and can be used by customers to determine whether their terminals balance with BASE24-pos.

3. CONFIGURATION AND FILES MAINTENANCE

This section describes the files used to configure the APACS 70 POS Device Handler module. The descriptions explain the role each file plays in determining the options supported by the APACS 70 POS Device Handler module.

Before a customer can be configured to use the APACS 70 POS Device Handler module, the APACS 70 POS Device Handler module must be properly installed, the module needs to be added to the system, and the BASE24-pos network must be enabled and available for file maintenance.

The NEF and LCONF are BASE24 files vital in determining APACS 70 POS Device Handler module configuration issues.

BASE24-pos files POS Retailer Definition File (PRDF) and POS Terminal Data File (PTDF) also contain information set up by the customer in order to provide the APACS 70 POS Device Handler module with information about retailers and terminals.

The TKN file allows customers to configure which additional transaction data can be logged to the PTLF or subsequently extracted.

3.1 XPNET CONFIGURATION

This section contains examples of the Network Environment File (NEF) definitions that are required to install the Device Handler module. They cover the process, device, station and line definitions for the Device Handler module and communications port (ASYNCR and X.25).

For more details about defining processes, devices, stations and lines, see the *NET24-XPNET Network Control Operations Guide*. For information about each attribute used in the definition, see the *NET24-XPNET Network Control Command Reference Manual*.

3.1.1 Process Definitions

The example below shows two Device Handler/Router/Authorisation processes.

```
RESET PROCESS
SET PROCESS BCPU 6
SET PROCESS LIBRARY $DATA.B24OBJ.AUTHLIB
SET PROCESS PROGRAM $DATA.B24OBJ.APACSDH
SET PROCESS PPD $BDH1
SET PROCESS PRIORITY 150
SET PROCESS CPU 6
SET PROCESS STARTUP DEMAND
SET PROCESS IAUDIT ON
SET PROCESS SERVICE Q1A^APACDHQ1
ADD PROCESS P1A^APACS1, UNDER SYSNAME \SYS1, UNDER NODE
P1A^NODE
```

```
RESET PROCESS
SET PROCESS BCPU 7
SET PROCESS LIBRARY $DATA.B24OBJ.AUTHLIB
SET PROCESS PROGRAM $DATA.B24OBJ.APACSDH
SET PROCESS PPD $BDH2
SET PROCESS PRIORITY 150
SET PROCESS CPU 7
SET PROCESS STARTUP DEMAND
SET PROCESS IAUDIT ON
SET PROCESS SERVICE Q1A^APACDHQ1
ADD PROCESS P1A^APACS2, UNDER SYSNAME \SYS1, UNDER NODE
P1A^NODE
==
```

```
RESET DEVICE
SET DEVICE RECVLEN 200
SET DEVICE TYPE 51
SET DEVICE XMITLEN 200
SET DEVICE ADD DESTSTRING 1,Q1A^APACHQ1, 13, "8"
SET DEVICE ADD PROTOCOLSTRING 1,VISA2,1, "02"
ADD DEVICE ASYNC^DEV, UNDER SYSNAME \SYS1, UNDER NODE
P1A^NODE
```

```
RESET DEVICE
SET DEVICE RECVLEN 128
SET DEVICE XMITLEN 128
SET DEVICE ADD DESTSTRING 1,Q1A^APACDHQ1, 13, "8"
ADD DEVICE X25^DEV, UNDER SYSNAME \SYS1, UNDER NODE
P1A^NODE
==
```

```
RESET LINE
SET LINE ENQ 120
SET LINE ERANALYSIS 30
SET LINE PORT $AT
SET LINE TIMEOUT 150
SET LINE TYPE NOPARITY
SET LINE PROTOCOL VISA
SET LINE MODE ASYNC
SET LINE DATATIMEOUT 0
SET LINE RESPTIMEOUT 0
SET LINE PARITY OFF
ADD LINE L1A^ASYN1, UNDER SYSNAME \SYS1, UNDER NODE
P1A^NODE
```

```
RESET LINE
SET LINE ATIMER 15
SET LINE ERANALYSIS 30
SET LINE ETE TPAD
SET LINE PORT $PSS.#PHONE
SET LINE WFCALL ON
SET LINE PROTOCOL X.25
SET LINE DATATIMEOUT 0
SET LINE RESPTIMEOUT 0
SET LINE PARITY OFF
ADD LINE L1A^PSS1, UNDER SYSNAME \SYS1, UNDER NODE
P1A^NODE
==
```

```
RESET STATION
SET STATION DESTINATION Q1A^APACDHQ1
SET STATION ERRORACT REINEVERY
SET STATION REINSTATE 300
SET STATION RETRY 3
SET STATION DEVICE ASYNC^DEV
SET STATION LINENAME \SYS1.P1A^NODE.L1A^ASYN1
ADD STATION S1AASYN1, UNDER SYSNAME \SYS1, UNDER NODE
P1A^NODE
```

```
RESET STATION
SET STATION DESTINATION Q1A^APACDHQ1
SET STATION RETRY 3
SET STATION DEVICE X25^DEV
SET STATION LINENAME \SYS1.P1A^NODE.L1A^PSS1
ADD STATION S1APSS1, UNDER SYSNAME \SYS1, UNDER NODE
P1A^NODE
==
```

3.1.2 Device Definitions

Below is an example of an ASYNC port device declaration.

```
DEVICE      RACAL^DEV = 51:
            DEVICE TYPE 51,
            TRANSMIT 200,
            RECEIVE 200,
            PROTOCOLSTRING( ID VISA2, OFFSET 1, MATCH
            '02' ),
            DESTSTRING (DIALIN PROCESS Q1A^RCLDHQ1,OFFSET
            13,MATCH '8'),
            NORMAL SELECT;
```

Below is an example of an X.25 port device declaration.

```
DEVICE      X25^DEV = 50:
            TRANSMIT 128,
            RECEIVE 128,
            DESTSTRING (DIALIN PROCESS Q1A^RCLDHQ1,OFFSET
            13,MATCH '8');
```

3.1.3 Station Definitions

Below is an example of an ASYNC station.

```
STATION    S1AASYN1 = 25:
           DEVICE RACAL^DEV,
           DESTINATION PROCESS Q1A^RCLDHQ1,
           LOGICAL NETWORK IS POS1,
           OUTPUT INTERVAL 0,
           MAX CORE QUEUE 2,
           RETRY 3,
           REINSTATE EVERY 300 SECONDS,
           TERMINAL TYPE 0,
           NO LOGICAL ACK;
```

Below is an example of an X.25 station.

```
STATION    S1APSS1 = 5:
           DEVICE X25^DEV,
           DESTINATION PROCESS Q1A^RCLDHQ1,
           LOGICAL NETWORK IS POS1,
           OUTPUT INTERVAL 0,
           MAX CORE QUEUE 2,
           RETRY 3,
           TERMINAL TYPE 0,
           NO LOGICAL ACK;
```

3.1.4 Line Definitions

Below is an example of an ASYNC line.

```
LINE      L1A^ASYN1 = 25:
          DEVICE ID '$AT',
          VISA,
          SPEED 5,
          LINE TYPE 1,
          ASYNC,
          TIMEOUT 1500 MILLISECONDS,
          RETRY 1,
          ENQ DELAY TIMER 1200 MILLISECONDS,
          STATIONS( S1AASYN1 ),
          FULL DUPLEX;
```

Below is an example of an X.25 line.

```
LINE      L1A^PSS1 = 5:
          DEVICE ID '$PSS.#PHONE',
          X25,
          STATIONS ( S1APSS1),
          END TO END PROTOCOL 8,
          WAIT FOR CALL,
          ATIMER 15,
          FULL DUPLEX;
```

3.2 BASE24 FILE CONFIGURATION

This section covers the configuration of BASE24 files that is required to install the Device Handler module. The following files are covered:

- Logical Network Configuration File (LCONF)
- POS Terminal Definition File (PTDF)
- POS Retailer Definition File (PRDF)
- Token File (TKN)

3.2.1 Logical Network Configuration File (LCONF)

The Logical Network Configuration File (LCONF) is the central location of file information and processing information for the BASE24 system. One LCONF exists for each logical network.

The LCONF contains assigns that link the physical names of objects in the logical network to names used by BASE24 code, and parameters that define processing values at the logical network level.

The LCONF assigns and parameters listed in this section are required by BASE24-pos to initialise and process the BASE24-pos APACS 70 Device Handler module.

For more details, refer to the *BASE24 Logical Network Configuration File Manual*.

3.2.1.1 LCONF Assigns

The LCONF process name for each assign is 16 asterisks (*).

POS-PTDF

The fully qualified file name of the POS Terminal Data File (PTDF) used by all Device Handler modules. This assign is mandatory.

3.2.1.2 LCONF Parameters

The LCONF process name for each parameter is specific to the calling Device Handler process. If the specific read fails, a generic read (using a process name of 16 asterisks) is attempted.

FUND-TRAN-TO-CASH-ADV

This parameter indicates how the Device handler processes account funding transactions. The valid values are as follows:

Y = Yes, process account funding transactions as cash advance transactions

N = No, process account funding transactions as normal purchase transactions

Param Default: N

POS-DH-MAX-TDFS

This parameter indicates the maximum number of PTDF keys that a single Device Handler module can store in memory. The Device Handler module uses this value at initialisation to calculate and then allocate extended memory space for the management of PTDF disk addresses. The value should be equal to the total number of devices that are associated with each BASE24-pos Device Handler module. A rule of thumb might be the average maximum number of different terminals that a Device Handler module manages.

Param Default: 20

POS-DH-AUTH-RESP-TIMEOUT

This parameter specifies the number of seconds that the BASE24-pos Device Handler module waits for an authorisation response to a transaction request message before sending an unable to process response message to the device. This parameter can be set to any numeric value greater than zero.

Param Default: 30

POS-DH-LMT-EXCEED-DISP

This parameter is used to determine the action taken when the refund limits for the terminal are exceeded. The valid values are as follows:

- 0 = Continue processing
- 1 = Decline
- 2 = Refer

Param Default: 0

POS-APDH-AUTH-DUP-TIMEOUT

This parameter specifies the time (in seconds) that may elapse before an authorisation message may not be deemed to be a duplicate. The Device Handler module validates if an authorisation request is in a duplicate time window by comparing transaction time plus the timeout value with the current time.

Param Default: 10

POS-APDH-STALE-REQ-TIMEOUT

This parameter specifies the time (in seconds) after which a transaction message is deemed to be *stale*. The Device Handler module determines the length of time a request has been queued and ensures that the time is less than the timeout value. A value of 45 is recommended. The valid values are from 10 –180.

Param Default: 45

POS-APDH-PRE-FLOOR-LIMIT

This parameter specifies the precommunications floor limit. Transactions with amounts less than or equal to this value are authorised offline regardless of system availability. It can be returned in EFT and Authorisation responses. The corresponding field in the PTDF record can override this value.

Param Default: 010

POS-APDH-USE-PRE-FL-LIMIT

This parameter specifies whether the value in the POS-APDH-PRE-FLOOR-LIMIT param is included in transaction response messages to the device. This value is used only if the corresponding field in the PTDF record is set to a value of 0. The valid values are as follows:

Y = Yes, include value in messages

N = No, do not include value in
messages

Param Default: Y

POS-APDH-POST-FLOOR-LIMIT

This parameter specifies the postcommunications floor limit. Transactions with amounts less than or equal to this value are authorised offline if the acquiring system is unavailable. It may be returned in EFT responses. The corresponding field in the PTDF record can override this value.

Param Default: 020

POS-APDH-USE-POST-FL-LIMIT

This parameter specifies whether the value in the POS-APDH-POST-FLOOR-LIMIT param is included in transaction response messages to the Device. This value is used only if the corresponding field in the PTDF record is set to a value of 0. The valid values are as follows:

Y = Yes, include value in messages

N = No, do not include value in
messages

Param Default: Y

POS-APDH-CONFIRM-FLAG

This parameter specifies whether confirmation messages are required from the device. A value of Y results in the CONFIRMATION-REQUEST field in authorisation response messages sent to the device being set to a value of 1. The valid values are as follows:

Y = Yes, request a confirmation message

N = No, do not request a confirmation
message

Param Default: N

POS-APDH-CONFIRM-TIMEOUT

This parameter specifies the time (in seconds) that the Device Handler module waits for a Confirmation message from the terminal.

Param Default: 15

POS-APDH-HOLD-MESSAGE-PREFIX

This parameter is used in conjunction with the POS-APDH-EXTEND-RESP-TIMEOUT to allow for card issuers who might want to use human intervention in an extended authorisation process. BASE24 issues a hold message to the terminal whilst an additional transaction timer is running for the selected card prefix or prefixes.

The valid values are up to 20 card prefixes delimited by a semicolon (;).

Param Default: None

POS-APDH-EXTEND-RESP-TIMEOUT

This parameter is used in conjunction with the POS-APDH-HOLD-MESSAGE-PREFIX, and specifies the length (in seconds) of the extended timeout period required to support these transactions.

Param Default: 120

POS-APDH-REFERRAL-PREFIX

This parameter specifies whether referral codes sent by Card Issuers are returned to the terminal in addition to the Call Card Issuer Message. BASE24 includes the referral code in response message messages for the selected card prefix or prefixes.

The valid values are up to 20 card prefixes delimited by a semicolon (;).

Param Default: None

POS-APDH-LOG-APACS-DATA

This parameter is used to allow additional APACS data to be logged to the PTLF. If set to a value of Y, the Message Header, Confirmation Code, Balance Code and Descriptive Data fields in the APACS 70 EFT Request Message, if present, are appended to the user data field at the end of the PSTM for subsequent logging to the PTLF. The valid values are as follows:

Y = Yes, log additional data

N = No, do not log additional
data

Param Default: N

POS-APDH-DISCONNECT-ON-ERROR

This parameter determines whether the APACS 70 POS Device Handler module disconnects the terminal communications under certain error conditions (e.g., when a PTDF record cannot be located). If the parameter is missing or invalid (not Y or N), the default value of Y is used and the APACS 70 POS Device Handler module issues disconnects under appropriate error conditions. The valid values are as follows:

Y = Yes, issue a
disconnect

N = No, do not disconnect

Param Default: Y

3.2.2 POS Terminal Data File (PTDF)

The POS Terminal Data File (PTDF) contains one record for each POS terminal in the network, including terminals that conform to the APACS 70 POS Device Handler module. The APACS 70 POS Device Handler module uses the TERMINAL ID field contained in the standard message header of all request messages to access the appropriate record in the PTDF. You can maintain the PTDF using the PTD screens in the BASE24 CRT access system.

The PTD fields listed in this section are required by BASE24-pos to initialise and process the BASE24-pos APACS 70 Device Handler module.

For more information on PTD screens, refer to the *BASE24-pos Files Maintenance Manual*.

TERMINAL TYPE (PTD screen 1)

This field specifies the type of terminal defined in the PTDF record. The valid values are as follows:

70 = APACS 70 (Auth Only)

71 = APACS 70

Default Value: No default value

EMV TERMINAL CAPABILITIES (PTD screen 17)

This field is defined by EMV, with each bit position representing a different capability of the EMV terminal. Each bit position is represented by a flag, which can be set to either N or Y.

Default Value: N

PRE-COMM FLOOR LIMIT (PTD screen 19)

The precommunications floor limit contains the maximum value of a transaction that can be authorised by the terminal without attempting to communicate with the host. The limit must be specified in whole currency units, in the currency specified by the CRNCY-CDE field (i.e., the primary terminal currency). The valid values are 000–999.

Default Value: 000

POST-COMM FLOOR LIMIT (PTD screen 19)

The postcommunications floor limit contains the maximum value of a transaction that can be authorised by the terminal when an attempt to communicate with the host has failed. The limit must be specified in whole currency units, in the currency specified by the CRNCY-CDE field (i.e., the primary terminal currency). The valid values are 000–999.

Default Value: 000

PRE-COMM LIMIT FLAG (PTD screen 19)

The precommunications floor limit flag indicates whether the value in the PRE-COMM FLOOR LIMIT field is included in response messages sent to the terminal. For the APACS 70 POS Device Handler module, a value of 0 indicates that the corresponding flag in the LCONF should be used. The valid values are as follows:

- 0 = Do not use the PRE-COMM FLOOR LIMIT value for this device type. Use the value of the corresponding flag in the Logical Network Configuration File (LCONF).
- 1 = Send the PRE-COMM FLOOR LIMIT value to the device.
- 2 = Do not send the PRE-COMM FLOOR LIMIT value to the device.

Default Value: 0

POST-COMMS LIMIT FLAG (PTD screen 19)

The postcommunications floor limit flag indicates whether the value in the POST COMM FLOOR LIMIT FIELD is included in response messages sent to the terminal. For the APACS 70 POS Device Handler module, a value of 0 indicates that the corresponding flag in the LCONF should be used. The valid values are as follows:

- 0 = Do not use the PRE-COMM FLOOR LIMIT value for this device type. Use the value of the corresponding flag in the Logical Network Configuration File (LCONF).
- 1 = Send the PRE-COMM FLOOR LIMIT value to the device.
- 2 = Do not send the PRE-COMM FLOOR LIMIT value to the device.

Default Value: 0

3.2.3 POS Retailer Definition File (PRDF)

The POS Retailer Definition File (PRDF) contains one record for each retailer in the network. The APACS 70 POS Device Handler module uses the PRDF in processing reconciliation transactions. The APACS 70 POS Device Handler module uses the retailer cutover window and posting date when determining whether a reconciliation transaction should cause the terminal to be cutover to the next business day.

For more information on the PRDF, refer to the *BASE24-pos Files Maintenance Manual*.

3.2.4 Token File (TKN)

To log or extract tokens, the Token File (TKN) must be configured appropriately. The token parameters can be set on the TKN screen using the AFT facility.

The following tokens used by the APACS 70 POS Device Handler module can be logged or extracted:

- POS 5.0 token
- POS 5.1 token (which includes the electronic commerce flag)
- Point of Service Data token
- POS Data1 token
- Authentication Data token
- Trans Stain XID token
- Station ID token
- Reversal Date and Time token.

For more details, refer to the *BASE24 Tokens Manual*.

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4. STARTUP AND CONTROL

This section describes the processing performed by the APACS 70 POS Device Handler module that is not related to transaction processing. It covers initialisation processing, memory management, and the interface with operators of the BASE24 system.

4.1 INITIALISATION

When the Device Handler/Router/Authorisation process is initialized, the Router module begins initialisation first. Part of the initialisation process required by the Device Handler module is within the Router module. The steps described in this section assume that the Router module has already initialised.

The APACS 70 POS Device Handler module performs the following steps to initialise:

1. Initialises the logging process with the name of the specific APACS 70 POS Device Handler module being initialised and reads the Log Message Configuration File (LMCF).
2. Retrieves assigns and params from the Logical Network Configuration File (LCONF). Refer to section 3.2 for further details.
3. Opens the PTDF and sets the file mode so any process attempting to lock or read a PTDF record while the APACS 70 POS Device Handler module has the record locked receives an error 73. This error indicates the file or record is locked. If the APACS 70 POS Device Handler module is unable to open the PTDF, the APACS 70 POS Device Handler module abends.
4. Allocates extended memory space for the number of PTDF records it needs at one time (see section 4.2 below). The number of records that can be in memory at one time is specified in the POS-DH-MAX-TDFS param.

When the APACS 70 POS Device Handler module receives a WARMBOOT command, it performs the same steps described in the above initialisation procedures with one exception.

The step to initialise the logging process with the name of the specific APACS 70 POS Device Handler module is not performed when a WARMBOOT command is received. Instead, when a WARMBOOT command is received, the first step taken by the APACS 70 POS Device Handler module is to read the Log Message Configuration File (LMCF). An internal table stored in the APACS 70 POS Device Handler module containing LMCF information requires updating if log messages have been altered using the LMCF. The APACS 70 POS Device Handler module requires warmbooting in order for any LMCF changes to take effect. The logging facility itself need not be reinitialised because it was already started during initialisation.

The remaining steps described in the initialisation procedures above are followed when the APACS 70 POS Device Handler module receives a WARMBOOT command.

4.2 PTDF MEMORY TABLE

The POS Terminal Data File (PTDF) is an Enscribe relative file, with an alternate key of Terminal ID. Each APACS Device Handler process holds a PTDF extended memory table to avoid accessing the PTDF on disk using the Terminal ID alternate key. Each entry in this table holds the PTDF Terminal ID together with the record's relative disk address.

When a transaction is received from a terminal, the Device Handler module attempts to locate the appropriate entry in the memory table. If an entry is found, the Device Handler module uses the relative record number to access the PTDF on disc. If an entry is not found the Device Handler module uses the Terminal ID to access the PTDF on disc by the alternate key, adding an entry to the memory table if the record is retrieved successfully. To accommodate the PTDF table, each Device Handler module allocates a fixed amount of extended memory at initialisation, based on the LCONF parameter POS-DH-MAX-TDFS.

4.3 NCPCOM COMMANDS

Network text commands are used to send information from a network control facility to a process. These commands can be entered by an operator using a network control facility and serve as a tool that operators can use to perform network management functions such as changing certain processing parameters, warmbooting a specific process, monitoring a specific process and resolving problems encountered by a specific process. The following text commands can be used with the APACS 70 POS Device Handler module.

See the *BASE24 Text Command Reference Manual* for detailed descriptions and the syntax of all text commands.

4.3.1 Process Warmboot

It is possible to instruct a Device Handler module to reinitialise itself, an action known as *warmbooting*. This has the effect of the Device Handler module reallocating its internal memory structures and rebuilding its internal memory tables.

The text of the NCPCOM command sent to the Device Handler module is as follows:

9501****

4.3.2 Terminal Status Change

The DEACTIVATE-FLG field in the PTDF record indicates the current status of the APACS terminal. The status of the terminal can be changed from the NCS screen.

The text of the NCPCOM command sent to the Device Handler module is as follows:

9525 *terminal-id* *n*

where:

n = 0 the terminal status is set to activated.

n = 1 the terminal status is set to deactivated or pending deactivation, depending on the contents of the totals fields in the PTDF record. If the totals fields are set to zero, then the terminal status is set to a value of 1 (deactivated); otherwise, the terminal status is set to a value of 2 (pending deactivation).

4.3.3 Reset Transaction

A Reset transaction is required if the terminal and BASE24 have become out of synchronisation during Message Authentication Code (MAC) processing of EFT transactions.

A command is initiated from the NCS terminal to reset the MAC transaction keys to the MAC Master Key. At the same time, the retailer must reinitialise the APACS 70 terminal.

When the reset transaction is performed for the PTDF, BASE24 moves the MAC Master Key value to the current, previous and next transaction key fields.

The text of the NCPCOM command sent to the Device Handler module is as follows:

9521MACR<terminal-id>

4.3.4 Terminal Management

The LOADFLAGON command allows the APACS 70 POS Device Handler module to inform an APACS terminal to contact its terminal management system on completion of a transaction. The LOADFLAGOFF command allows the APACS 70 POS Device Handler module to cancel this request (i.e., inform an APACS terminal not to contact its terminal management system).

The NCPCOM commands cause a flag to be set in the PTDF device-dependent data, and is used to set the appropriate value in the next response message sent to the terminal.

The text of the LOADFLAGON NCPCOM command sent to the Device Handler module is as follows:

9500 *terminal-id* LOADFLAGON

The text of the LOADFLAGOFF NCPCOM command sent to the Device Handler module is as follows:

9500 *terminal-id* LOADFLAGOFF

4.4 LOG MESSAGE GENERATION

The BASE24-pos APACS 70 Device Handler module generates log messages to assist network operators in performing their daily tasks. Log messages are unsolicited messages generated by BASE24 processes that provide information regarding the internal condition of the system. They are a network operator's primary link to what is actually occurring in the system. Log messages provide operators with information ranging from system status, such as the completion of a procedure, to more serious circumstances requiring operator intervention.

The log messages generated by the APACS 70 POS Device Handler module are documented in appendix C. For a complete explanation of BASE24 log message documentation, refer to the ***BASE24 Event Message Reference Manual***.

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5. MESSAGE PROCESSING

The APACS 70 POS Device Handler module is the primary interface between a POS device and BASE24-pos. Each type of POS device has its own message format and method of processing transactions. The APACS 70 POS Device Handler module supports POS terminals that conform to APACS Standard 70.

This section describes the different types of message supported by the APACS 70 POS Device Handler module.

The message formats supported by the Device Handler module are defined in Book 2 of the APACS 70 Standard. The following basic message layouts are supported:

- Authorisation Request/Reversal
- Authorisation Response
- Confirmation of Authorisation Response
- Data Capture (EFT) Request/Reversal/Advice
- Data Capture (EFT) Response
- Offline Repeat of Data Capture (EFT) Request
- Reconciliation Request/Advice
- Reconciliation Response

The Device Handler module does not support immediate confirmation messages for Data Capture Requests; confirmation is implicit based on data in the next transaction.

This section describes the normal flow of the request or confirmation messages sent to BASE24 from the APACS 70 terminal, how the message is processed by BASE24 and the response message that is returned to the terminal.

Exception conditions are also documented in this section, together with additional information on response message processing.

For more details of message layouts, refer to APACS Standard 70, Book 2.

5.1 APACS 70 MESSAGES

Standard 70 messages comprise a message header followed by the message data. Full details of the data elements supported in APACS 70 messages are provided in Book 2 of Standard 70.

5.1.1 Message Header

The APACS 70 request message header comprises five fields:

- Dial Indicator
- Terminal Identity
- Message Number
- Terminal Type
- Message Type

The APACS 70 response message header comprises six fields:

- Dial Indicator
- Terminal Identity
- Message Number
- Message Type
- Acquirer Response Code
- Confirmation Request

These fields are described below.

5.1.1.1 Dial Indicator

The Dial Indicator field indicates the number of dial attempts made by the terminal to connect to BASE24-pos. The value may indicate the first, second or third dial or “no dial” (e.g., a host-to-host connection).

The value sent in the request message is echoed in the response message.

5.1.1.2 Terminal Identity

The Terminal Identity field contains the hardwired Terminal ID, which is unique to given APACS 70 terminal.

The value sent in the request message is echoed in the response message.

5.1.1.3 Message Number

The Message Number field contains a sequence number, which is incremented for every cardholder transaction or reconciliation that is successfully completed.

The value sent in the request message is echoed in the response message.

5.1.1.4 Terminal Type

The Terminal Type field indicates the capabilities and attributes of the terminal:

- The first character identifies the card entry capabilities of the terminal
- The second character identifies the receipt printing capabilities of the terminal
- The third character identifies the optional response message fields supported by the terminal
- The fourth character identifies additional features and attributes of the terminal

5.1.1.5 Message Type

The Message Type field is a two character field, identifying:

- Format (i.e., authorisation only, EFT)
- Message type (i.e., request, response, reversal, confirmation)
- Message category (i.e., first try, retry, post-event advice)
- Card details entry (i.e., card swiped, card number keyed, no cardholder)
- Transaction type (e.g., sale, refund)
- Transaction attributes (e.g., cardholder not present, e-commerce)

5.1.1.6 Acquirer Response Code

The Acquirer Response Code field indicates the result of the transaction.

5.1.1.7 Confirmation Request

The Confirmation Request field is a one character field, identifying:

- Whether an immediate confirmation message is required
- Whether a subsequent post-event advice message is required following a referral override
- Whether the terminal should call its terminal management system

5.1.2 Message Data

For cardholder transactions, the data in APACS 70 messages includes elements such as the cardholder's primary account number (PAN) and the transaction amount

For reconciliation transactions, the data in APACS 70 messages includes elements such as the terminal totals.

5.2 MESSAGE FLOWS

All messages are subject to some common processing by the APACS 70 POS Device Handler module, until the type of message has been identified. The common processing and the message-specific processing are described below.

5.2.1 Common Request/Reversal Message Processing

1. When initiating a transaction, the Retailer enters the transaction at the terminal. The request message is formatted by the terminal and, provided that communications can be established, is sent to BASE24. When cancelling a transaction (e.g., because the cardholder has left the shop) the retailer presses the reversal key on the terminal. A reversal message is formatted by the terminal and sent to BASE24, providing that communications can be established.
2. The message is routed by the XPNET process to the Router module, which, in turn, sends the message to the Device Handler module. If the Device Handler module detects a stale request then the message is dropped, since it is assumed that the terminal already has timed out. A stale message is defined as one that has been queued in the XPNET process for longer than the value specified in the LCONF parameter POS-APDH-STALE-REQ-TIMEOUT. Stale messages should occur only if there are problems with the configuration or performance of the BASE24 system. If stale messages are encountered, contact HELP24.
3. The Device Handler module takes the terminal ID (from the message) and reads the relevant PTDF record into memory. If a PTDF record cannot be found, then the terminal is not supported by BASE24, so the Device Handler module checks the value of the LCONF parameter POS-APDH-DISCONNECT-ON-ERROR. If the parameter is set to a value of Y, the Device Handler module disconnects the line to the terminal; otherwise, the Device Handler module drops the message and allows the terminal to time out.
4. The Device Handler module checks the status of the terminal on the PTDF, using the value of the TERM STATUS field on PTD screen 1. If a BASE24 operator has deactivated the terminal (i.e., the status is not “Active”), then the transaction is processed as described in section 5.3.2.
5. The Device Handler module uses the Header Parsing Table (see appendix A.1) to validate the request message header. If the header is invalid, a message is output to the EMS log and an ERROR HOLD response (see section 5.3.1) is sent to the terminal. Provided the message header is valid, the Device Handler module uses the Message Type field in the header message to locate the appropriate entry in the Message Type Descriptor table (see appendix A.1). This entry indicates the type of request message being sent, and the Device Handler module stores this information in the device-dependent data of the PTDF for message processing.

6. The Device Handler module uses the Authorisation Request Message Parsing table (see appendix A.2) to verify the message. If the message is invalid, the Device Handler module formats an ERROR HOLD response (see section 5.3.1) and sends this to the terminal for display.
7. The Device Handler module compares the current message number with the previous message number received from the terminal (held in the PTDF). If they match, then the Device Handler module checks the contents of the message and uses the information in the device-dependent data of the PTDF to identify which the following conditions has occurred:
 - A duplicate message (see section 5.3.3)
 - A retry message, where the original message has already been processed (see section 5.3.4)
 - A post-event advice message, where the original message has already been processed (see section 5.3.6)
 - The previous response message sent to the terminal was not processed successfully

The Device Handler module then continues message processing based on the message type, as discussed below.

5.2.2 Authorisation Request/Response Messages

1. If the Device Handler module has determined that the previous response message was not processed correctly by the terminal (i.e., the request message contains the same message number as the previous authorisation message, but with a different card number or amount), then a flag is set to indicate that an implicit reversal is required (see section 5.3.6 below).
2. The Device Handler module checks to ensure that the PTLF is not full. If the PTLF is full, then a SYSTEM MALFUNCTION ERROR HOLD response is sent to the terminal.
3. If the previous transaction is to be reversed (i.e., an *implicit* reversal) then the Device Handler module formats a 0420 PSTM, and forwards the reversal message to the Router module.
4. The Device Handler module initialises the device-dependent data in the PTDF for the current authorisation message.
5. The Device Handler module checks the transactions allowed table in the PTDF, to ensure that the message contains a transaction type that is valid for the terminal. If the transaction type is not valid, then the transaction is declined, and a response is sent indicating that the transaction is not allowed at the terminal.
6. The Device Handler module ensures that the Currency Code received in the message with is supported (e.g., it must match the value of the CURRENCY CODE field on PTD screen 2 for single-currency systems). If the currency code is not valid, then the transaction is declined, and a response is sent indicating that the currency is not supported.
7. The Device Handler module formats a 0200 message from the information held in the request and from the PTDF. During this process, data is added to token fields for subsequent recording on the PTLF. If an error is encountered in adding the tokens, then a SYSTEM MALFUNCTION ERROR HOLD response is sent to the terminal.
8. The Device Handler module compares the merchant number received in the message with the value of the value of the Retailer ID field on PTD screen 1. If the values do not match, then the transaction is referred, and a response is sent indicating that the merchant number is invalid.
9. The Device Handler module forwards the 0200 message to the Router module and sets a timer to await the response to the message.

10. On receipt of the response, the Device Handler module cross-references the authorisation response code with the Device Response Code table to locate the response message to be displayed at the terminal. The Device Handler module formats the response message as specified by the Authorisation Response/Hold Message Parsing table from information in the PSTM, PTDF, and Device Response Code table. The response is sent to the terminal. The PTDF record is updated and written back to disk.
11. The terminal processes the authorisation response message.
12. If immediate confirmations are required, a confirmation of authorisation response is sent to BASE24-pos.

5.2.3 Confirmation of Authorisation Response

1. The Device Handler module initialises the device-dependent data in the PTDF for the current authorisation message.
2. If the value of the confirmation code is zero, the confirmation is valid (the data received at the terminal was accepted). If the value of the confirmation code is nonzero, then the terminal failed to process the previous response message correctly. The Device Handler module formats a 0420 PSTM, and forwards the message to the Router module as an implicit reversal (see section 5.3.6 below).

5.2.4 Authorisation Reversal Messages

1. The Device Handler module checks to ensure that the PTLF is not full. If the PTLF is full, then a SYSTEM MALFUNCTION ERROR HOLD response is sent to the terminal.
2. The Device Handler module initialises the device-dependent data in the PTDF for the current authorisation message.
3. The Device Handler module checks the transactions allowed table in the PTDF, to ensure that the message contains a transaction type that is valid for the terminal. If the transaction type is not valid, then the transaction is rejected, and a response is sent indicating that the transaction is not allowed at the terminal.
4. The Device Handler module ensures that the currency code received in the message with is supported (e.g., it must match the value in the CURRENCY CODE field on PTD screen 1 for single-currency systems). If the currency code is not valid, then the transaction is rejected, and a response is sent indicating that the currency is not supported.
5. The Device Handler module formats a 0420 message from the information held in the reversal and from the PTDF. During this process, data is added to token fields for subsequent recording on the PTLF. If an error is encountered in adding the tokens, then a SYSTEM MALFUNCTION ERROR HOLD response is sent to the terminal.
6. The Device Handler module compares the merchant number received in the message with the value in the RETAILER ID field on PTD screen 1. If the values do not match, then the transaction is rejected, and a response is sent indicating that the merchant number is invalid.
7. The Device Handler module forwards the 0420 message to the Router module.
8. The Device Handler module formats the response message as specified in the Authorisation Response Message Parsing table from information in the PTDF. The message is sent to the terminal. The PTDF record is updated and written back to disk.
9. The terminal processes the authorisation response message.
10. If immediate confirmations are required, a confirmation of authorisation response is sent to BASE24-pos.

5.2.5 Data Capture Request/Response Messages

1. The Device Handler module verifies the message authentication code (MAC), having derived the appropriate transaction MAC key. If the MAC is invalid, then a BAD MAC ERROR HOLD message is sent to the terminal for display. If the MAC key cannot be derived, then a SYSTEM MALFUNCTION ERROR HOLD message is sent to the terminal for display. If the MAC is successfully verified using the transaction key that should have been valid for the previous transaction from the terminal, then a flag is set to indicate that an implicit reversal is required (see section 5.3.6 below).
2. The Device Handler module checks to ensure that the PTLF is not full. If the PTLF is full, then a SYSTEM MALFUNCTION ERROR HOLD response is sent to the terminal.
3. If the previous transaction is to be reversed (i.e., an implicit reversal) then the Device Handler module formats a 0420 message, and forwards the reversal message to the Router module.
4. The Device Handler module initialises the device-dependent data in the PTDF for the current EFT message.
5. The Device Handler module checks the transactions allowed table in the PTDF, to ensure that the message contains a transaction type that is valid for the terminal. If the transaction type is not valid, then the transaction is declined, and a response is sent indicating that the transaction is not allowed at the terminal.
6. The Device Handler module ensures that the Currency Code received in the message with is supported (e.g., it must match the PTDF Currency Code for single-currency systems). If the currency code is not valid, then the transaction is declined, and a response is sent indicating that the currency is not supported.
7. The Device Handler module formats a 0200 PSTM from the information held in the request and from the PTDF. During this process, data is added to token fields for subsequent recording on the PTLF. If an error is encountered in adding the tokens, then a SYSTEM MALFUNCTION ERROR HOLD response is sent to the terminal.
8. The Device Handler module compares the Merchant Number received in the message with the Retailer ID specified in the PTDF record. If the values do not match, then the transaction is referred, and a response is sent indicating that the merchant number is invalid.
9. The Device Handler module forwards the PSTM to the Router module and sets a timer to await the PSTM response to the message.

10. On receipt of the response, the Device Handler module cross-references the AUTH response code with the Device Response Code Table to locate the response message to be displayed at the terminal. The Device Handler module formats the response message as specified by the EFT Response Message Parsing Table from information in the PSTM, PTDF and Device Response Code Table. The response MAC is generated, and the message is sent to the terminal. The PTDF record is updated and written back to disk.
11. The terminal processes the EFT Response Message.

5.2.6 Data Capture Post-Event Advice Messages

1. The Device Handler module verifies the MAC, having derived the appropriate transaction MAC key. If the MAC is invalid, then a BAD MAC ERROR HOLD message is sent to the terminal for display. If the MAC key cannot be derived, then a SYSTEM MALFUNCTION ERROR HOLD message is sent to the terminal for display. If the MAC is successfully verified using the transaction key that should have been valid for the previous transaction from the terminal, then a flag is set to indicate that an implicit reversal is required (see section 5.3.6 below).
2. The Device Handler module checks to ensure that the PTLF is not full. If the PTLF is full, then a SYSTEM MALFUNCTION ERROR HOLD response is sent to the terminal.
3. If the previous transaction is to be reversed (i.e., either the transaction details in the previous request message correspond to those in the current advice message (see section 5.3.4 below) or an implicit reversal is required), then the Device Handler module formats a 0420 message, and forwards the reversal message to the Router module.
4. The Device Handler module initialises the device-dependent data in the PTDF for the current EFT message.
5. The Device Handler module checks the transactions allowed table in the PTDF, to ensure that the message contains a transaction type that is valid for the terminal. If the transaction type is not valid, then the transaction is rejected, and a response is sent indicating that the transaction is not allowed at the terminal.
6. The Device Handler module ensures that the currency code received in the message with is supported (e.g., it must match the value in the CURRENCY CODE field on PTD screen 1 for single-currency systems). If the currency code is not valid, then the transaction is rejected, and a response is sent indicating that the currency is not supported.
7. The Device Handler module formats a 0220 message from the information held in the request and from the PTDF. During this process, data is added to token fields for subsequent recording on the PTLF. If an error is encountered in adding the tokens, then a SYSTEM MALFUNCTION ERROR HOLD response is sent to the terminal. The Device Handler module forwards the 0220 message to the Router module.
8. The Device Handler module formats the response message as specified in the EFT Response Message Parsing table from information in the PTDF. The response MAC is generated, and the message is sent to the terminal. The PTDF record is updated and written back to disk.
9. The terminal processes the EFT Response Message.

5.2.7 Data Capture Reversal Messages

1. The Device Handler module verifies the MAC, having derived the appropriate transaction MAC key. If the MAC is invalid, then a BAD MAC ERROR HOLD message is sent to the terminal for display. If the MAC key cannot be derived, then a SYSTEM MALFUNCTION ERROR HOLD message is sent to the terminal for display.
2. The Device Handler module checks to ensure that the PTLF is not full. If the PTLF is full, then a SYSTEM MALFUNCTION ERROR HOLD response is sent to the terminal.
3. The Device Handler module initialises the device-dependent data in the PTDF for the current EFT message.
4. The Device Handler module checks the transactions allowed table in the PTDF, to ensure that the message contains a transaction type that is valid for the terminal. If the transaction type is not valid, then the transaction is rejected, and a response is sent indicating that the transaction is not allowed at the terminal.
5. The Device Handler module ensures that the Currency Code received in the message with is supported (e.g., it must match the value of the CURRENCY CODE field on PTD screen 1 for single-currency systems). If the currency code is not valid, then the transaction is rejected, and a response is sent indicating that the currency is not supported
6. The Device Handler module formats a 0420 message from the information held in the request and from the PTDF. During this process, data is added to token fields for subsequent recording on the PTLF. If an error is encountered in adding the tokens, then a SYSTEM MALFUNCTION ERROR HOLD response is sent to the terminal.
7. The Device Handler module compares the merchant number received in the message with the value in the RETAILER ID field on PTDF screen 1. If the values do not match, then the transaction is rejected, and a response is sent indicating that the merchant number is invalid.
8. The Device Handler module forwards the 0420 message to the Router module.
9. The Device Handler module formats the response message as specified in the EFT Response Message Parsing table from information in the PTDF. The response MAC is generated, and the message is sent to the terminal. The PTDF record is updated and written back to disk.
10. The terminal processes the EFT Response Message.

5.2.8 Reconciliation Request/Response/Advice Messages

1. The Device Handler module verifies the MAC, having derived the appropriate transaction MAC key. If the MAC is invalid, then a BAD MAC ERROR HOLD message is sent to the terminal for display. If the MAC key cannot be derived, then a SYSTEM MALFUNCTION ERROR HOLD message is sent to the terminal for display.
2. The Device Handler module checks to ensure that the PTLF is not full. If the PTLF is full, then a SYSTEM MALFUNCTION ERROR HOLD response is sent to the terminal.
3. If the previous transaction is to be reversed (i.e., either the transaction details in the previous request message correspond to those in the current repeat message (see section 5.3.4 below) or an implicit reversal is required), then the Device Handler module formats a 0420 message, and forwards the reversal message to the Router module.
4. The Device Handler module initialises the device-dependent data in the PTDF for the current EFT message.
5. The Device Handler module reads the POS Retailer Definition File (PRDF) to check whether the reconciliation transaction is being performed within the retailer cutover and balancing window, and, if not, whether the terminal was forced cutover in the last balancing period. As a result of these checks, the Device Handler module performs either a balance transaction or a balance and cutover transaction.
6. The Device Handler module formats a network totals PTLF record from the information sent in the request, and a services totals PTLF record from the information held in the PTDF. If the totals sent in the message do not match the network totals accumulated in the PTDF, then the Device Handler module formats an additional Settlement PTLF record (network totals) from the information held in the PTDF.
7. If the message is not an advice, then the Device Handler module compares the merchant number received in the message with the value of the RETAILER ID field on PTD screen 1. If the values do not match, then the transaction is referred, and a response is sent indicating that the merchant number is invalid.
8. The Device Handler module forwards the PTLF record (or records) to the Router module, if configured on the PTDF.
9. The Device Handler module formats the response message as specified in the Reconciliation Response Message Parsing table from information in the PTDF. The response MAC is generated, and the message is sent to the terminal. The PTDF record is updated and written back to disk.
10. The terminal processes the EFT response message.

5.3 EXCEPTION HANDLING

Further details of the message exception processing performed by the APACS 70 POS Device Handler module are provided below.

5.3.1 Error Hold Messages

The Device Handler module performs a number of validation checks on messages sent by APACS 70 terminals. If a message format is invalid or the terminal is not recognised, the Device Handler module responds with an ERROR HOLD message. An ERROR HOLD message is a HOLD type response, but because the response code is set to a nonzero value the terminal disconnects on receiving the message and attempts to redial if appropriate.

5.3.2 Deactivated Terminals

If the DEACTIVATE-FLG in the PTDF record is set to “Deactivated” or “Pending Deactivation,” but the terminal sends a request message to BASE24, the Device Handler module acts on the message as described below.

- DEACTIVATED

The Device Handler module formats a TERM DEACTIVATED ERROR HOLD message that is sent to the terminal for display.

- PENDING DEACTIVATION

If the batch/shift/day/network totals are nonzero, then the terminal accepts only reconciliation transactions (until the totals are zero). All other message types are processed as if the terminal is fully deactivated. When the totals are zero, the Device Handler module changes the terminal status to Deactivated.

5.3.3 Duplicate Transactions

If the terminal sends a duplicate message to BASE24, the message is ignored. The Device Handler module determines that a message is a duplicate in the following ways.

- An Authorisation Request or Confirmation message is classified as a duplicate message if it contains the same message number, dial indicator, card number, transaction amount and is within the same time window as the previously received Authorisation message, using the LCONF param POS-APDH-AUTH-DUP-TIMEOUT.
- An EFT Request is classified as a duplicate if it contains the same message number and dial indicator as the previously received EFT message.

5.3.4 Retry Message Sent After First Try Message Received

The Device Handler module determines the number of times a terminal has attempted to send a message to BASE24 by referencing the request message type against the message Tries in the Message Type Descriptor Table.

If a First Try message was sent by the terminal and received by BASE24 for Data Capture and/or Authorisation, but the terminal then sends a Retry message (determined by the message number and an incremented dial indicator), the Device Handler module acts on the Retry message according to when it is received.

- If the Retry is received before the Device Handler module has sent a response message to the terminal, the Device Handler module responds to the Retry message when a response is received for the request. In this situation, the Retry overrides the First Try message.
- If the Retry is received after the Device Handler module has sent the response message to the terminal, it is assumed that the terminal did not receive the response. In this situation, the Device Handler module returns the response message formatted for the First Try request.

5.3.5 Advice Message After Approved Response

If a Post-Event Advice message for an EFT transaction is received by BASE24 when the terminal has previously sent a First Try or Retry request message, the Device Handler module reverses the original transaction, if approved (PSTM type 0420), and sends a Force Post transaction (PSTM type 0220) to Authorisation for logging to the PTLF.

5.3.6 Implicit Reversals

If the Device Handler module detects that the previous transaction did not complete successfully at the terminal, then a reversal is generated to back out the effect of that transaction. If the previous message received from the terminal was a reversal, then a force post message is generated to back out the effect of the reversal.

The Device Handler module uses data from the previous transaction stored in the PTDF record to format the PSTM reversal message. The Device Handler module determines that the previous Authorisation or EFT transaction performed at the terminal did not complete successfully as described below

5.3.6.1 Authorisation Messages

An authorisation message has not completed successfully if one of the following events has occurred:

- An authorisation request message is received with the same message number as the previous authorisation message, but with a different card number or amount.
- An authorisation confirmation message is received with a nonzero confirmation code.
- The Device Handler module times out waiting for an authorisation confirmation message, using the LCONF param POS-APDH-CONFIRM-TIMEOUT.

5.3.6.2 EFT Messages

An EFT message has not completed successfully if the MAC is verified using the previous transaction key, instead of the current transaction key, (i.e., the terminal has not updated its key registers).

5.3.7 Transaction Timeout

When a transaction request message is sent to BASE24, following verification of the message content and if applicable the MAC, the Device Handler module formats a PSTM message for internal processing by BASE24. The Device Handler module sends the message to the Router module, sets a transaction timer using the value of the LCONF param POS-DH-AUTH-RESP-TIMEOUT, and awaits the PSTM response to the message. If the PSTM response is not received within the timer period, the Device Handler module formats an UNABLE TO AUTH response and sends the message to the terminal. The retailer can reenter the transaction request details at the terminal.

If an approved PSTM response is received following the transaction timeout, then the Device Handler module generates a reversal transaction in PSTM format and passes this to the Router module.

If the UNABLE TO AUTH messages persist, contact HELP24. The institution may decide to increase its transaction timer limits to prevent timeout occurring.

5.4 RESPONSE MESSAGE FORMATTING

Having processed a transaction message, the Device Handler module formats and sends a response message to the terminal.

5.4.1 Terminal Display

For cardholder requests sent to the BASE24-pos Router/Authorisation modules for authorisation, the Device Handler module cross-references the response code returned by the Authorisation process with its own Response Code table, to obtain a device response table entry (see appendix B.1). If the response code is unrecognised, then the transaction is referred, and a response is sent indicating that the transaction could not be authorised. For all other messages, the Device Handler module sets the device response table entry itself.

Except for a response to a post-event advice message (for which no message is displayed), the Device Handler module then uses the Device Response table to select the response message to be displayed at the terminal (see appendix B.2):

- The response message displayed at the terminal for an approved Refund transaction is REFUND ACCEPTED.
- The response message displayed at the terminal for an approved Reversal transaction is REVERSAL ACCEPTED.
- The response message displayed at the terminal for an approved Refund Reversal transaction is REFUND REVERSAL ACCEPTED.
- The response message displayed at the terminal for an approved Balance Enquiry is BALANCE SZZZ,ZZZ,ZZ9.99.
- The response message displayed at the terminal for an approved Reconciliation Request is COMPLETE.

For other approved transaction requests, the response message displayed at the terminal includes the authorisation code.

5.4.2 Terminal Floor Limits

Book 2 of the APACS 70 Standard allows a host to update the terminal's communications limits in response messages if the terminal supports it (see section 6.10.2).

5.4.2.1 Precommunications Floor Limit

For Authorisation and EFT messages, the APACS 70 POS Device Handler module checks the value of the PRE-COMMS LIMIT FLAG in the PTDF record.

- If the field contains a value of 2, then no precommunications floor limit is included in response messages to the terminal.
- If the field contains a value of 1, then the Device Handler module returns the value of the PTDF field PRE-COMMS FLOOR LIMIT in response messages to the terminal.
- If the field contains 0, then the APACS 70 POS Device Handler module checks the value of the LCONF param POS-APDH-USE-PRE-FL-LIMIT.
 - If the param is set to a value of Y, then the Device Handler module returns the value of the LCONF param POS-APDH-PRE-FLOOR-LIMIT in response messages to the terminal.
 - If the param is set to a value of N, then no precommunications floor limit is included in response messages to the terminal.

5.4.2.2 Postcommunications Floor Limit

For EFT messages only, the APACS 70 POS Device Handler module checks the value of the POST-COMMS LIMIT FLAG in the PTDF record.

- If the field contains a value of 2, then no postcommunications floor limit is included in response messages to the terminal.
- If the field contains a value of 1, then the Device Handler module returns the value of the PTDF field POST-COMMS FLOOR LIMIT in response messages to the terminal.

- If the field contains a value of 0, then the APACS 70 POS Device Handler module checks the value of the LCONF param POS-APDH-USE-POST-FL-LIMIT.
 - If the param is set to a value of Y, then the Device Handler module returns the value of the LCONF param POS-APDH-POST-FLOOR-LIMIT in response messages to the terminal.
 - If the param is set to a value of N, then no precommunications floor limit is included in response messages to the terminal.

5.4.3 Explicit Confirmations

If immediate confirmations are required for authorisation responses (i.e., the LCONF param POS-APDH-CONFIRM-FLAG is set to a value of Y), then the Device Handler module sets the CONFIRMATION REQUEST field in the response message header to an appropriate value.

This results in the terminal generating an authorisation confirmation message after processing the response message.

5.4.4 Terminal Management Trigger

The Device Handler module uses the REMOTE-CONF field in the PTDF record to determine whether to inform the terminal to contact its terminal management system. The REMOTE-CONF field can be set by either a LOADFLAGON or a LOADFLGOFF NCPCOM command (see section 4.3.4).

Based on the value of the REMOTE-CONF field, the Device Handler module sets the CONFIRMATION REQUEST field in the response message header to an appropriate value.

If the REMOTE-CONF flag is set to a value of 1, then the terminal calls its terminal management system on completion of the transaction. The REMOTE-CONF flag is reset to a value of 0 the next time a message is received from the terminal, which indicates the successful completion of the previous transaction.

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6. CARDHOLDER TRANSACTIONS

This section describes the different types of cardholder transaction supported by the APACS 70 POS Device Handler module.

6.1 TRANSACTION OVERVIEW

The following transactions are permissible at APACS 70 (Auth Only) and APACS 70 terminals:

- CHEQUE (Card Verification)
- SALE (Normal Purchase)
- CASH (Cash Advance)
- SALE WITH CASH (Purchase with Cash Back)
- REFUND (Merchandise Return)
- ACCOUNT FUNDING (Account Funding)

The following transaction is permissible at APACS 70 terminals only:

- CARDHOLDER BALANCE ENQUIRY (Balance Inquiry)

The following transactions are no longer defined in Standard 70, but may be supported by some “legacy” APACS 70 terminals:

- PRE-AUTHORISATION SALE (Preauthorisation Purchase)
- PRE-AUTH SALE COMPLETION (Preauthorisation Completion)

6.1.1 Cheque

The cheque transaction type allows a cardholder to obtain goods, services or cash against a personal cheque supported by a cheque guarantee card without data capture taking place.

The APACS 70 POS Device Handler module converts cheque requests to BASE24-pos card verification messages (PSTM.TRAN.TRAN-CDE.TC = 16). The PT-SRV-COND-CDE field in the PSTM is set to a value of 00 (normal presentment).

6.1.2 Sale

The sale transaction type allows a sale of goods or services to be authorised.

The APACS 70 POS Device Handler module converts sale requests to either BASE24-pos normal purchase messages (PSTM.TRAN.TRAN-CDE.TC = 10) or BASE24-pos mail/telephone order messages (PSTM.TRAN.TRAN-CDE.TC = 13), based on the attributes identified by the message type (see section 6.2 below). The PT-SRV-COND-CDE field in the PSTM is also based on the attributes identified by the message type.

6.1.3 Cash

The cash transaction type allows a cash advance to be authorised.

The APACS 70 POS Device Handler module converts cash requests to BASE24-pos cash advance messages (PSTM.TRAN.TRAN-CDE.TC = 15). The PT-SRV-COND-CDE field in the PSTM is set to a value of 00 (normal presentment).

6.1.4 Sale With Cash

The sale with cash transaction type is designed to permit the return of cash to the customer in response to a sale.

The APACS 70 POS Device Handler module converts sale with cash requests to BASE24-pos purchase with cash back messages (PSTM.TRAN.TRAN-CDE.TC = 18). The PT-SRV-COND-CDE field in the PSTM is set to a value of 00 (normal presentment).

6.1.5 Refund

The refund transaction type allows a sale of refund (credit to a cardholder and debit to a retailer) to be authorised.

The APACS 70 POS Device Handler module converts refund requests to BASE24-pos merchandise return messages (PSTM.TRAN.TRAN-CDE.TC = 14). The PT-SRV-COND-CDE field in the PSTM is based on the attributes identified by the message type (see section 6.2 below).

The LCONF parameter POS-DH-LMT-EXCEED-DISP can be configured such that the APACS 70 POS Device Handler module declines or refers a refund transaction should the value of the transaction exceed the maximum returns limit set for the terminal.

The parameter values and corresponding actions are shown in the table below:

Value	Action
0	Continue Processing (default)
1	Decline
2	Refer

6.1.6 Account Funding

The account funding transaction type is designed to allow a cardholder to obtain funds in advance of their use to pay for goods or services where funds are loaded into another account or wallet for example for micro-payments or brokerage.

The APACS 70 POS Device Handler module converts account funding requests to either a BASE24-pos normal purchase message (PSTM.TRAN.TRAN-CDE.TC = 10) or a BASE24-pos cash advance message (PSTM.TRAN.TRAN-CDE.TC = 15) depending on the value of the LCONF param FUND-TRAN-TO-CASH-ADV. If FUND-TRAN-TO-CASH-ADV is set to “Y”, account funding transactions are processed as cash advance transactions. Otherwise, account funding transactions are processed as a normal purchase transactions. For account funding requests where the cardholder is not present and the param FUND-TRAN-TO-CASH-ADV is set to “Y”, the request is converted into a BASE24-pos mail/phone order message (PSTM.TRAN.TRAN-CDE.TC = 13). The PT-SRV-COND-CDE field in the PSTM is also based on the attributes identified by the message type.

6.1.7 Cardholder Balance Enquiry

The cardholder balance enquiry transaction type allows the cardholder's account balance to be viewed. A balance enquiry transaction messages contain a MAC.

For purposes of generating the authorization parameter, balance enquiry transactions (message type '70', '72', '73' and '74') fall under 'credit' type of transactions.

The APACS 70 POS Device Handler module converts cardholder balance enquiry requests to BASE24-pos balance inquiry messages (PSTM.TRAN.TRAN-CDE.TC = 17). The PT-SRV-COND-CDE field in the PSTM is set to a value of 00 (normal presentment).

6.1.8 Preauthorisation Sale

The preauthorisation transaction type is designed to allow goods or services to be authorised without data capture taking place, and to notify the acquirer that this preauthorised amount may be subsequently adjusted so should be placed on hold in the customer's account.

The APACS 70 POS Device Handler module converts preauthorisation sale requests to BASE24-pos preauthorisation purchase messages (PSTM.TRAN.TRAN-CDE.TC = 11) with the PRE-AUTH-SEQ-NUM field in the PSTM set to a value of APPROVALCODE. The PT-SRV-COND-CDE field in the PSTM is set to a value of 00 (normal presentment).

6.1.9 Preauthorisation Sale Completion

The preauthorisation sale completion transaction type is designed to allow a preauthorisation for one amount to be completed by the Acquirer for a different amount. The hold entered in the preauthorisation is released and data capture is takes place for the transaction using the final transaction amount

The APACS 70 POS Device Handler module converts preauthorisation completion requests to BASE24-pos preauthorisation completion messages (PSTM.TRAN.TRAN-CDE.TC = 12) with the PRE-AUTH-SEQ-NUM field in the PSTM set to the authorisation code used for the preauthorisation request. The PT-SRV-COND-CDE field in the PSTM is set to a value of 06 (preauthorised request).

6.2 TRANSACTION ATTRIBUTES

For sale and refund transactions, different message types are used to identify specific transaction attributes.

6.2.1 Cardholder Present

If the message type from the terminal indicates that the cardholder is present at the point of sale, then the BASE24-pos transaction code is set to Normal Purchase (except for refunds), and the PT-SRV-COND-CDE field in the PSTM is set to a value of 00 (normal presentment).

The Standard Industry Classification (SIC) code or Merchant Category Code (MCC) indicates the retailer's line of business, and is defined at either the terminal or retailer level. The APACS 70 POS Device Handler module sets the RETL-SIC-CDE field in the PSTM using the value in the TERM SIC CODE field on PTD screen 1. If the TERM SIC CODE field on PTD screen 1 contains zeros or blanks, then the Router/Authorisation module sets the RETL-SIC-CDE field in the PSTM using the value in the SIC CODE field on PRDF screen 1.

6.2.2 Cardholder Not Present

If the message type from the terminal indicates that the cardholder is not present at the point of sale, then the BASE24-pos transaction code is set to Mail/Phone Order (except for refunds), and the PT-SRV-COND-CDE field in the PSTM is set to a value of 08 (mail or telephone order).

The SIC code or MCC indicates the retailer's line of business and is defined at either the terminal or retailer level. The APACS 70 POS Device Handler module sets the RETL-SIC-CDE field in the PSTM using the value in the MAIL/PHONE SIC CODE field on PTD screen 1. If the MAIL/PHONE SIC CODE field on PTD screen 1 contains zeros or blanks, then the Router/Authorisation module sets the RETL-SIC-CDE field in the PSTM using the MAIL/PHONE SIC CODE field on PRDF screen 1.

6.2.3 Continuous Authority

The APACS 70 POS Device Handler module supports the message types used for continuous authority transactions. These transactions are treated in the same way as “cardholder not present” sale or refund transactions, e.g. the PT-SRV-COND-CDE field in the PSTM is set to a value of 08 (mail or telephone order).

In addition, BASE24-pos Release 5.1 token fields and POS Data1 token fields are set to indicate recurring payment transaction.

For a complete list of the tokens used by the BASE24-pos APACS 70 Device Handler module, refer to section 6.6.

6.2.4 e-Commerce Transactions

The APACS 70 POS Device Handler module supports the message types used for Electronic Commerce (e-commerce) transactions. These transactions are treated in the same way as normal purchase or refund transactions, except that the PT-SRV-COND-CDE field in the PSTM is set to a value of 15 (customer/home terminal).

In addition, BASE24-pos Release 5.1 token fields and Point of Service Data token fields are set to indicate an e-commerce transaction.

For a complete list of the tokens used by the BASE24-pos APACS 70 Device Handler module, refer to section 6.6.

6.3 EMV SUPPORT

The BASE24-pos EMV add-on product is an additional licensable module, built on top of the existing APACS 70 POS Device Handler module.

The standard APACS 70 POS Device Handler source module includes a number of product-specific “hooks” or user exits. These provide an interface between the standard APACS 70 POS Device Handler module and the BASE24-pos EMV add-on product. The BASE24-pos EMV add-on product is bound together with the standard APACS 70 POS Device Handler module to provide support for EMV transactions, in addition to non-EMV transactions.

The BASE24-pos APACS 70 POS Device Handler module EMV routines support the following EMV functions:

- A function to identify EMV request messages
- A function to validate and store EMV-specific data sent in request messages
- A function to accept and process card details read from a chip
- A function to determine whether a MAC is present in an EMV request message
- A function to verify the MAC in an EMV request message
- A function to set EMV-specific values in the PSTM (e.g., a point-of-service entry mode of 05) and to determine which EMV-specific tokens should be added to the PSTM
- A function to format the EMV Status token and to add the token to the PSTM
- A function to format the EMV Request token and to add the token to the PSTM
- A function to format the EMV Discretionary Data token and to add the token to the PSTM
- A function to retrieve the EMV Response token from the PSTM
- A function to retrieve the EMV Script token from the PSTM
- A function to include EMV-specific data in response messages
- A function to determine whether a MAC is required in an EMV response message
- A function to generate the MAC in an EMV response message

For further details of the BASE24-pos EMV add-on product, refer to the *BASE24-pos EMV Support Manual*.

6.4 MULTIPLE CURRENCY SUPPORT

The BASE24-pos Multiple Currency add-on product is an additional licensable module, built on top of the existing APACS 70 POS Device Handler module.

The standard APACS 70 POS Device Handler source module includes a number of product-specific “hooks” or user exits. These provide an interface between the standard APACS 70 POS Device Handler module and the BASE24-pos Multiple Currency add-on. The BASE24-pos Multiple Currency add-on product is bound together with the standard APACS 70 POS Device Handler module to provide support for transactions in more than one currency received from a single APACS 70 terminal.

The BASE24-pos APACS 70 POS Device Handler module multiple currency routines support the following multiple currency functions:

- A function to verify that the transaction currency code is valid. The POS Device Handler Multiple Currency Library procedure (MCU^DH^VERIFY^CRNCY) that verifies a currency code is invoked to support this function.
- A function to convert the transaction amount to the primary terminal currency to allow the transaction amount to be checked against PTDF limits. The POS Device Handler Multiple Currency Library procedure (MCU^DH^CONVERT^AMTS) that converts up to two amounts from one currency code to another is invoked to support this function.
- A function to convert the transaction amount to the primary terminal currency to allow the resulting value to be added to the Multiple Currency token. The POS Device Handler Multiple Currency Library procedure (MCU^DH^CNVRT^PSTM^CRNCY) that ensures that the PSTM amount fields are represented in the primary terminal currency is invoked to support this function.
- A function to convert the PTDF limits (stored in the PSTM) to the transaction currency, and to ensure that the transaction currency code is included in the PSTM instead of the primary terminal currency code. The POS Device Handler Multiple Currency Library procedure (MCU^DH^SET^PSTM^CRNCY) that ensures that PSTM amount fields are represented in the transaction currency is invoked to support this function.

Note that the BASE24-pos APACS 70 Device Handler module can detect if it is operating in a Multiple Currency environment (i.e., whether the BASE24-pos Multiple Currency add-on product is present).

The BASE24-pos APACS 70 POS Device Handler Multiple Currency routines provide access to the BASE24-pos Multiple Currency POS Device Handler Libraries, as shown below.

Notes:

1. The APACS 70 POS Device Handler module does not support Terminal Currency Migration (i.e., conversion of terminal totals from one currency to another).
2. For APACS 70 terminals supporting multiple currencies, reconciliation totals take the form of hash totals of the transaction amounts. This is in conformance with Book 2 of Standard 70.

6.4.1 mcu^dcb^check^crncy

Parameter	Description	Action
TXN^CRNCY	Transaction Currency Code	Invoke the POS Device Handler Library routine MCU^DH^VERIFY^CRNCY using TXN^CRNCY as the only parameter. If this returns PSE^OK^L then return PSE^OK^L (true), otherwise return PSE^NOT^OK^L (false).

6.4.2 mcu^dcbe^convert^amts

Parameter	Description	Action
INPUT^CRNCY	Input Currency	Invoke the POS Device Handler Library routine MCU^DH^CONVERT^AMTS using the four mandatory parameters supplied as input to this procedure (do not pass any of the optional params, that is, input amount2, output amount2 or rate). If this returns PSE^OK^L then return PSE^OK^L (true), otherwise return PSE^NOT^OK^L (false).
INPUT^AMT1	Input Transaction Amount	
OUTPUT^CRNCY	Output Currency	
OUTPUT^AMT1	Output Transaction Amount	

6.4.3 mcu^dcb^set^pstm^crncy

Parameter	Description	Action
INPUT^CRNCY	Terminal Currency	Invoke the POS Device Handler Library routine MCU^DH^CNVRT^PSTM^CRNCY with parameters INPUT^CRNCY and PSTM. This sets up the terminal currency in the Multiple Currency token array. If this returns PSE^NOT^OK^L then return PSE^NOT^OK^L (false). Invoke the POS Device Handler Library routine MCU^DH^SET^PSTM^CRNCY using the three mandatory parameters supplied as input to this procedure. If this returns PSE^OK^L then return PSE^OK^L (true), otherwise return PSE^NOT^OK^L (false).
OUTPUT^CRNCY	Transaction Currency	
PSTM	PSTM	

6.5 TERMINAL CAPABILITIES

The APACS 70 POS Device Handler module supports the extended terminal type information in request messages from the APACS terminal.

The APACS 70 POS Device Handler module recognises the values in the first and fourth character of the Terminal Type field, and sets the Point of Service Condition Code and Point of Service Entry Mode fields in the PSTM to appropriate values.

It also uses this information to set some of the subfields in the Point of Service Data token (see section 6.6.4 below).

The third character of the Terminal Type field is used to determine if the terminal can be sent additional information in a response message (see section 6.10).

The fourth character of the Terminal Type field is used to indicate additional features and capabilities. The APACS 70 POS Device Handler module checks the fourth character to identify e-commerce capable terminals.

6.5.1 Point of Service Condition Code

If the PT-SRV-COND-CDE field in the PSTM has not been set to a nonzero value because of the transaction type and/or transaction attributes indicated by the message type (see sections 6.1 and 6.2 above), then the APACS 70 POS Device Handler module checks the fourth character of the terminal type in the transaction request message:

- If the fourth character of the terminal type is set to a value of 1 or 3, then the PT-SRV-COND-CDE field is set to a value of 27 (Unattended term unable to retain card).
- If the fourth character of the terminal type is set to a value of 5 or 7, then the PT-SRV-COND-CDE field is set to a value of 02 (Unattended terminal able to retain card).

6.5.2 Point of Service Entry Mode

The first two digits of the PT-SRV-ENTRY-MDE field in the PSTM are set based on how the card details are entered at the point of sale. For the third digit, the APACS 70 POS Device Handler module checks the fourth character of the terminal type in the transaction request message:

- If the fourth character of the terminal type is set to a value of 2, 3, 6 or 7, then the third byte of the PT-SRV-ENTRY-MDE field is set to a value of 1 (PIN entry capability).
- If the fourth character of the terminal type is set to 1, 4, 5 or 8, then the third byte of the PT-SRV-ENTRY-MDE field is set to a value of 2 (No PIN entry capability).

6.6 TOKEN PROCESSING

The APACS 70 Device Handler module supports the processing of BASE24 tokens. For definitions of the tokens supported and for further details on BASE24 token processing, refer to the *BASE24 Tokens Manual*.

On receipt of a transaction or authorisation response, the Device Handler module stores token data in the PTDF. This data is added to the PSTM if a reversal message is subsequently generated for the transaction.

6.6.1 BASE24-pos Release 5.0 Token (Token ID = 04)

On receipt of a transaction or authorisation request from an APACS device, the Device Handler module appends the BASE24-pos Release 5.0 token to the PSTM. The APACS 70 POS Device Handler module can set the following fields in the BASE24-pos Release 5.0 token:

- RTE-GRP (a field used in routing not-on-us transactions).
- CRD-VRFY-FLG (a field used to indicate the result of card verification).
- CITY-EXT (a field used to store bytes 14-18 of the name of the city where the terminal is located).
- COMPLETE-TRACK2-DATA (a field indicating whether the terminal has the ability to capture and transmit complete Track 2 data on card-swiped transactions).

6.6.2 Reversal Date and Time Token (Token ID = BH)

On receipt of a reversal message (explicit reversal) from an APACS device or when an implicit reversal is generated due to a timeout, the Device Handler module adds the Reversal Date and Time token to the PSTM.

For explicit reversals, the TXN-DAT and TXN-TIM fields of the token are set from the Date and Time field in the request message. For implicit reversals, the TXN-DAT and TXN-TIM fields of the token are set from the system time.

6.6.3 BASE24-pos Release 5.1 Token (Token ID = C0)

On receipt of a transaction or authorisation request from an APACS device, the Device Handler module appends the BASE24-pos Release 5.1 token to the PSTM. The APACS 70 POS Device Handler module can set the following fields in the BASE24-pos Release 5.1 token:

Field Name	Description	
CVD-FLD	This is set from the CV2 value in the Descriptive Data field in the request message.	
E-COM-FLG	The data in the Auxiliary Data record type 01 in the request message is mapped to the E-COM-FLG.	
	Additional Transaction Security Data, position 3 and 4	E-COM-FLG
	41 or C1 - VbV where authentication was successful	5 - Secure electronic transaction with cardholder authentication
	42 or C2 – VbV with evidence of attempted authentication	6 - Encrypted electronic transaction where the merchant is capable of authenticating the cardholder, but was unable to complete the authentication (e.g., because the issuer or cardholder does not participate in the appropriate authentication program)
	C4 – VbV where the system was unable to verify	7 - Encrypted electronic commerce transaction
	44 – VbV/SSL where the system was unable to verify	8 - Non-secure electronic commerce transaction. (This is the default value)
MOTO-FLG	This is set to “2” for a continuous authority transaction, “1” for a mail/telephone order transaction, “0” otherwise.	
CVD-FLD-PRESENT	This is set to “1” if CVD-FLD has been populated, “0” otherwise.	
SAF-OR-FORCE-POST	This is set to “S” for a post-event advice transaction.	

Field Name	Description	
AUTHN-COLL-IND	The data in the Auxiliary data record type 01 in the request message is mapped to the AUTHN-COLL-IND.	
	Additional Transaction Security Data	AUTHN-COLL-IND
	For merchants that attempted SecureCode (Additional Transaction Security Data, position 3 = “1” or “9”) and CAV is present.	2 – Universal Cardholder Authentication field (UCAF) data collection is supported by the merchant, and UCAF data was populated.
	For merchants that support SecureCode (Additional Transaction Security Data, position 1 = “1”, “3”, “5”, “7”, “9”, “B”, “D” or “F”) and CAV not present.	1 – Universal Cardholder Authentication field (UCAF) data collection is supported by the merchant, but UCAF data was not populated.
	If neither of the above apply, default AUTHN-COLL-IND to 0.	0 – Universal Cardholder Authentication field (UCAF) data collection is not supported at the merchant’s Web site. (This is the default value)

6.6.4 Point of Service Data Token (Token ID = C4)

On receipt of a transaction or authorisation request from an APACS device, the Device Handler module appends the Point of Service Data token to the PSTM. The APACS 70 POS Device Handler module can set the following fields in the Point of Service Data token:

Field Name	Description
TERM-ATTEND-IND	If this is an e-commerce transaction, the field is set to a value of 2 (No terminal used). If the fourth character of the terminal type field in the request message is set to 1, 3, 5 or 7, the field is set to a value of 1 (Unattended terminal). Otherwise, the TERM-ATTEND-IND field is set to a value of 0 (Attended terminal).
TERM-OPER-IND	Reserved for future use. This field is zero filled.
TERM-LOC-IND	If this is an e-commerce transaction, the field is set to a value of 3 (On premises of cardholder—Home PC). Otherwise, the TERM-LOC-IND field is set to a value of 0 (On premises of card acceptor facility).
CRDHLD-PRESENT-IND	If this is a continuous authority request, the field is set to a value of 4 (Cardholder not present, standing order/recurring transaction). If this is an e-commerce transaction, the field is set to a value of 5 (Electronic order—Home PC, Internet). If this is a mail order transaction, the field is set to a value of 1 (Cardholder not present, unspecified). Otherwise, the CRDHLD-PRESENT-IND field is set to a value of 0 (Cardholder present).
CRD-PRESENT-IND	If this is a continuous authority, e-commerce or mail order transaction, the field is set to a value of 1 (Card not present). Otherwise, the CRD-PRESENT-IND field is set to a value of 0 (Card present).
CRD-CAPTR-IND	If the fourth character of the terminal type field in the request message is set to 4, 5, 6 or 7, the field is set to a value of 1 (Card capture capability). Otherwise, the CRD-CAPTR-IND field is set to a value of 0 (No card capture capability).
TXN-STAT-IND	If this is a preauthorisation request, the field is set to a value of 4 (Preauthorised request). Otherwise, the TXN-STAT-IND field is set to a value of 0 (Normal request).

Field Name	Description
TXN-SEC-IND	The TXN-SEC-IND field is set to a value of 0 (No security concern).
TXN-RTN-IND	Reserved for future use. This field is zero filled.
CRDHLDR-ACTVT-TERM-IND	If this is an e-commerce transaction, the field is set to a value of 6 (Electronic commerce). Otherwise, the CRDHLDR-ACTVT field is set to a value of 0 (Not a CAT transaction).
TERM-INPUT-CAP-IND	If this is an e-commerce transaction, the field is set to a value of 1 (No terminal used). The first character of the terminal type field in the request message and the card details entry method are checked to set the field.
	1st Digit of Terminal Type; Card Details Entry Method
	TERM-INPUT-CAP-IND
	1 = Integrated Circuit Reader Only
	9 = EMV Compatible ICC Reader
	2 = Magnetic Stripe Reader Only; this is not a keyed transaction
	2 = Magnetic Stripe Reader
	2 = Magnetic Stripe Reader Only; this is a keyed transaction
	7 = Magnetic Stripe Reader and Key Entry
	3 = Integrated Circuit and Magnetic Stripe Reader; this is not a keyed transaction
	5 = Magnetic Stripe Reader and EMV Compatible ICC Reader.
	3 = Integrated Circuit and Magnetic Stripe Reader; this is a keyed transaction
	8 = Magnetic Stripe Reader, Key Entry and EMV Compatible ICC Reader
	4 = No Card Reader
	6 = Key Entry Only

6.6.5 POS Data1 Token (Token ID = CH)

On receipt of a recurring payment transaction from an APACS device, the Device Handler module appends the POS Data1 token to the PSTM. The APACS 70 POS Device Handler module sets the RECUR-PMNT-IND field in the POS Data1 token to “R”.

6.6.6 Station ID Token (Token ID = C1)

The APACS 70 POS Device Handler module can track station or port usage. Port usage tracking gives an institution the ability to analyse, control, and balance network service usage. The APACS 70 POS Device Handler module retrieves the station ID from the network message header and places it in the Station ID token. The token is appended to the PSTM, and can be logged to the POS Transaction Log File (PTLF).

6.6.7 Trans Stain XID Token (Token ID = C6)

On receipt of a Visa VbV transaction from an APACS device, the Device Handler module appends the Trans Stain XID token to the PSTM. The APACS 70 POS Device Handler module sets the TRANS-STAIN field in the Trans Stain XID token.

6.6.8 Authentication Data Token (Token ID = CE)

On receipt of a MasterCard SecureCode transaction from an APACS device, the Device Handler module appends the Authentication Data token to the PSTM. The APACS 70 POS Device Handler module can set the following fields in the Authentication Data token:

- AUTHN-IND-FLG (set to 01 if the cardholder authentication value is present)
- AUTHN-IND-DATA (contains the cardholder authentication value)

6.7 LOGGING ADDITIONAL DATA

The LCONF parameter POS-APDH-LOG-APACS-DATA is used to allow additional APACS data to be logged to the PTLF.

If the parameter is set to a value of Y for yes, for EFT request messages, the following data fields are appended to the user data field at the end of the PSTM, (if present). Subsequently, this data is logged to the PTLF.

- Dial Indicator
- Message Type
- Transaction Date and Time
- Reason Online Code
- Confirmation Code (for previous transaction)
- Balance Code (for previous transaction)
- Descriptive Data

6.8 CV2/AVS SUPPORT

The APACS 70 POS Device Handler module supports manually-entered Card verification Digits (CV2), and the U.K.'s Address Verification Service (AVS).

6.8.1 Request Processing

The CV2/AVS data will be presented in the APACS 70 Auth and EFT requests as Descriptive Data and will be in the following format:

Descriptive Data					
CV2		Post Code		Address	
3 or 4 numeric digits	US	Up to 5 numeric digits from Post Code	US	Up to 14 numeric digits from Address	FS

It is assumed that both unit separators will always be present in the field, even if one or more of the sub fields is absent.

When parsing Auth or EFT request messages, the three optional variable length, all numeric, sub fields (CV2 Value, Post Code, and Address) are extracted from field 11 and placed in the PS50 token (token ID = 04), PS51 token (token ID = C0) or the PSTM as follows:

Data Item	Max Size	Field	Comments
CV2	4	PS51-TKN. CVD-FLD	PS51-TKN.CVD-FLD-PRESENT will be set to "1" if CV2 is present PS50-TKN.CRD-VRFY-FLG will be set to "0" (zero) if CV2 is present.
Post Code	5	PSTM. ZIP-CODE	PSTM.ADDR-TYP will be set to "98" and
Address	5	PSTM. ADDR-FLDS. ADDR	PSTM.ADDR-FLDS.ADDR-VRFY-STAT will be set to "0" (zero) if one or both of Post Code or Address are present. It is possible to receive a CV2/AVS Address of up to 14 characters in length. Only the first five characters will be moved into the PSTM

6.8.2 Response Processing

If the terminal is capable of receiving and processing the Response Additional Data (see section 6.10.3), and one or more of the Descriptive Data CV2/AVS sub fields were included in the request, then Response Additional Data should be included in the response. It has the following format:

	Byte 1 CV2				Byte 2 Post Code				Byte 3 Address			
Description	8	4	2	1	8	4	2	1	8	4	2	1
Not Checked				X				X				X
Matched			X				X				X	
Not Matched		X				X				X		

When generating the response to the terminal the CV2 Result Code held in PS50-TKN.CRD-VRFY-FLG will be mapped to Position 1 CV2 value in the Response Additional Data field as follows:

CV2 Result Code	Description	Response Additional Data Byte 1
“P”, “O”	CV2 Check Not Performed	0
“N”, “0”, “L”, “J”, “K”, “U”, “ ”	CV2 Not Processed	1
“Y”	CV2 Match	2
“C”, “D”, “R”	CV2 No Match	4

PSTM.ADDR-FLDS.ADDR-VRFY-STAT will be mapped to Position 2 (Post Code Result) and Position 3 (Address Result) in the Response Additional Data field as follows:

BASE24 AVS Result Codes	Description	Response Additional Data	
		Byte 2	Byte 3
A	ADDRESS-Addresses matched, but ZIP codes did not match.	4	2
E*	ERROR-The transaction was not eligible for address verification or an editing error occurred while attempting to process the message.	1	1
N	NO-Addresses did not match and ZIP codes did not match.	4	4
R*	RETRY-Primary and secondary authorizers were unavailable or declined the transaction and address verification was not performed on BASE24-pos.	1	1
S*	SERVICE NOT SUPPORTED-BASE24-pos authorized the transaction, but did not have the add-on Address Verification module.	1	1
U*	UNAVAILABLE-Address information was not available to the processor performing address verification.	1	1
W	WHOLE ZIP-Nine-digit ZIP codes matched, but addresses did not match.	2	4
X	EXACT-Addresses and nine-digit ZIP codes matched.	2	2
Y	YES-Addresses and five-digit ZIP codes matched.	2	2
Z	ZIP-Five-digit ZIP codes matched, but addresses did not match.	2	4
“ “* “ “*	Address verification information was not included in this transaction	0	0
0*	Address verification information was included in this transaction, but was not verified. A transaction to be verified by a host or interchange carries this code. A transaction to be verified on BASE24-pos, but declined before address verification could be performed also carries this code.	1	1

* Identifies a reason that a comparison was not made. Each interchange specifies the value it uses to identify this reason. Interchange Interfaces substitute the interchange-specific value before sending the message from BASE24-pos to the interchange. Refer to the ***BASE24-pos Address Verification Manual*** for a listing of interchange specific values.

Response Additional Data is a fixed format 6 bytes field, but only the first three are used for the CV2/AVS service. Only values 0, 1, 2, 4 or 8 are valid in bytes 1 to 3. For any sub field that was not present in the request, the value for that item in the response should be set to Zero.

Byte 4 (Authorising Entity) is set to indicate the authoriser of the transaction, e.g. acquirer or issuer.

Bytes 5 and 6 are reserved.

6.9 EXTENDING TRANSACTION TIMEOUT

Book 2 of the APACS 70 Standard allows for the host to send a hold message to the device to request more time in which to respond to a pending authorisation. This facility is required for some card issuers (e.g., AMEX) where there may be human intervention in the Authorisation process.

When a request message sent to the Authorisation module times out in the Device Handler module, a hold message may be returned to the terminal, and the authorisation response timer restored. This processing occurs if the card prefix is specified in the LCONF parameter POS-APDH-HOLD-MESSAGE-PREFIX, when the length of the restarted response timer is set to the value of the LCONF parameter POS-APDH-EXTEND-RESP-TIMEOUT. Only one hold message per transaction may be generated.

This processing occurs only if the terminal is capable of receiving and processing Hold Messages. If the value in the third byte of Terminal Type field in the request message is one of the following values: “2”, “3”, “6”, “7”, “A”, “B”, “E” or “F” then the terminal is capable of receiving and processing Hold Messages.

6.10 OPTIONAL RESPONSE MESSAGE FIELDS

The 3rd Character of the Terminal Type field in the request message is used to determine if the terminal can handle Referral Telephone number, Floor limits and date and Response Additional Data. This is used to ensure that the relevant data is only included in responses sent to the Terminals that can use the information.

6.10.1 Referral Telephone Number

If the value in the third byte of terminal type is one of the following values: “1”, “3”, “5”, “7”, “9”, “B”, “D” or “F” the terminal is capable of receiving and processing the Referral Telephone number.

6.10.2 Terminal Floor Limits

If the value in the third bytes of the terminal type is one of the following values: “4”, “5”, “6”, “7”, “C”, “D”, “E” or “F” then terminal is capable of receiving and processing the floor limits and date.

6.10.3 Response Additional Data

If the value in the third byte of the terminal type is one of the following values: “8”, “9”, “A”, “B”, “C”, “D”, “E” or “F” or if the Terminal is Level 15 then the terminal is capable of receiving and processing the Response Additional Data.

6.11 REFERRAL PROCESSING

The authorisation of a transaction by either an issuer or BASE24 may result in a referral response being returned to the terminal. The message displayed on the terminal asks the retailer to call for authorisation. The Device Handler module includes the referral telephone number in the response message, if available, and if supported by the terminal (see section 6.10.1)

On EFT responses, if the card prefix is specified in the LCONF parameter POS-APDH-REFERRAL-PREFIX, then a referral code returned by the authoriser also is displayed on the terminal. Some card issuers (e.g., AMEX) require the display of the referral code.

Following a referred EFT transaction, the retailer may enter the result of the referral (i.e., approved, declined or not acted upon) into the terminal. This information is then sent to BASE24 as a post-event advice message. Fields in the PSTM are set to indicate that the message is a referral override, and the response code field is set to indicate the outcome of the referral.

6.12 SECURE INTERNET VALIDATION

The APACS 70 POS Device Handler module supports Secure Internet Validation (SIV / 3D Secure). Both Visa and MasterCard methodologies are supported. Visa's method is known as Verified by Visa (VbV). MasterCard's method is SecureCode. SIV support requires the APACS device handler to support e-commerce Auxiliary data records. The APACS device handler will map the data from the Auxiliary data record (if it exists) to the appropriate BASE24 token. If the transaction is determined to be VbV, the Auxiliary data will be mapped to the Trans Stain XID token. If the transaction is determined to be SecureCode, the Auxiliary data will be mapped to the Authentication Data token.

6.13 CONTACTLESS TRANSACTIONS

The APACS 70 POS Device Handler module allows for identifying contactless transactions from terminals. Contactless transactions are transactions initiated without physical contact between the card and terminal. They can include contactless magnetic stripe transactions and contactless chip card transactions.

Contactless transactions are identified by the data in Auxiliary data record type 06, terminal attributes used field. If the TERM-ATTR-USED is a “2”, the PSTM.PT-SRV-ENTRY-MDE is set to a “07” (contactless chip card transaction). If the TERM-ATTR-USED is a “1”, the PSTM.PT-SRV-ENTRY-MDE is set to a “91” (contactless magnetic stripe transaction).

6.13.1 Contactless Chip

Contactless chip transactions are processed in the same way as contact EMV transactions (see section 6.3).

6.13.2 Contactless Magnetic Stripe

Contactless magnetic stripe transactions contain the same data elements in the terminal message as contact magnetic stripe transactions. However, the Track 2 data sent in the message may contain additional information, used to provide a more secure card authentication method, known as Dynamic Card Verification.

The APACS 70 POS Device Handler module is compatible with BASE24 Dynamic Card Verification, a type of card verification performed by BASE24 on contactless magnetic stripe transactions. BASE24 Card Verification supports MasterCard’s CVC3 card verification value (used for PayPass card verification) and Visa’s dCVV card verification value (used for payWave card verification).

Note: Dynamic Card Verification is a BASE24 add-on product that is used exclusively for contactless magnetic stripe transactions. It cannot be used with contactless chip card transactions, which are handled through EMV verification. Dynamic Card Verification is also the only type of BASE24 card verification supported for contactless magnetic stripe transactions. For more information on Dynamic Card Verification, refer to the ***BASE24 Transaction Security Services Processing Guide***.

6.13.2.1 Terminal Requirements

For contactless magnetic stripe transactions, the applicable card verification values for Dynamic Card Verification are included in the Issuer Discretionary Data fields of the Track 1 and Track 2 data. To support Dynamic Card Verification, the terminal must include the following in the track information it sends:

- Card verification value (the CVC3 or dCVV)
- Application Transaction Counter (ATC) – a transaction counter maintained by the application on the integrated chip card.
- Unpredictable Number (UN) – a number generated by the terminal and used to create uniqueness in the transaction (required by MasterCard only).

In addition, the track location and length of this card verification information must match that defined to BASE24 in the Card Prefix File.

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7. RECONCILIATION AND BALANCING

Terminal balancing and reconciliation is closely linked with BASE24-pos settlement, providing the institution with the opportunity to ensure that the BASE24-pos totals agree with the actual terminal totals at the retailer outlet.

The EFT (APACS 70) Payment Terminal is capable of performing interactive administrative functions. BASE24-pos supports a range of interactive administrative functions for balancing terminals by batch, shift and day, which enables it to support all APACS 70 balancing facilities.

For more details on Settlement and Reporting procedures, refer to the ***BASE24-pos Settlement and Reporting Manual***.

Note: Whilst the BASE24-pos APACS 70 Device Handler module supports the APACS 70 function that allows the input of a session number in reconciliation requests, no processing is performed on this field.

7.1 APACS 70 TERMINAL BALANCING

APACS 70 terminals support the following reconciliation transactions that cause totals to be logged to the PTLF. Screen 3 of the PTDF specifies which totals are recorded for the terminal in the file (all default to N) and a totals record is written only if the corresponding flag in the PTDF is set.

Where a device is located in a different time zone to the HP NonStop processor, the records in the PTLF are set to local time for the device. This is achieved by taking the HP NonStop System Time and adjusting by the amount in the TIME OFFSET held on the PTDF.

7.1.1 Reconciliation Transaction Within Retailer Window

When an end-of-day reconciliation transaction is received from the APACS 70 terminal within the Retailer Cutover and Balancing Window, BASE24 performs a Balance and Cutover transaction and sets the BALANCE FLAG in the POS Terminal Data File (PTDF) accordingly. To perform this transaction, BASE24 compares the terminal totals with the PTDF totals.

If the totals match, BASE24:

- Sets the new posting date.
- Formats a Network Settlement type PTLF record with a Balance Flag value of Balanced from the PTDF terminal totals (if configured on the PTDF) and sends this to the Authorisation module for writing to the PTLF.
- Formats a Services Reconciliation settlement type PTLF record with a Balance Flag value of Balanced from the service group totals (if configured on the PTDF) and sends this to the Authorisation module for writing to the PTLF.
- When processing the next transaction from the terminal, moves the current NET totals to the previous NET totals and clears the following totals in the PTDF:

SERVICE GROUP DAILY	
BATCH	DAILY-DC
BATCH-DC	CURRENT NET
SHIFT	CURRENT NET-DC
SHIFT-DC	

If the totals do not match, BASE24:

- Sets the new posting date.
- Formats a Network Settlement type PTLF record with a Balance Flag value of Not Balanced from the real terminal totals (if configured on the PTDF) and sends this to the Authorisation module for writing to the PTLF.
- Formats a Network Settlement type PTLF record with a Balance Flag of *Not Balanced* from the PTDF terminal totals (if configured on the PTDF) and sends this to Authorisation for writing to the PTLF.
- Formats a Services Reconciliation settlement type PTLF record with a Balance Flag of *Not Balanced* from the service group totals (if configured on the PTDF) and sends this to Authorisation for writing to the PTLF.
- When processing the next transaction from the terminal, moves the current NET totals to the previous NET totals and clears the following totals in the PTDF:

SERVICE GROUP DAILY
BATCH DAILY-DC
BATCH-DC CURRENT NET
SHIFT CURRENT NET-DC
SHIFT-DC

7.1.2 Reconciliation Transaction Outside Retailer Window

Reconciliation transactions can be performed at any time of the day from an APACS 70 terminal. If a transaction is initiated by the Retailer outside the Cutover and Balancing Window, BASE24 compares the terminal totals with the totals in the PTDF.

If the totals match, BASE24:

- Formats a Terminal Reconciliation settlement type PTLF record with a Balance Flag of *Balanced* from the PTDF terminal totals (if configured on the PTDF) and sends this to Authorisation for writing to the PTLF.
- Formats a Services Reconciliation settlement type PTLF record with a Balance Flag of *Balanced* from the service group totals (if configured on the PTDF) and sends this to Authorisation for writing to the PTLF.
- Clears the following totals in the PTDF:

SERVICE GROUP
BATCH
BATCH-DC
SHIFT
SHIFT-DC
DAILY
DAILY-DC

If the totals do not match, BASE24:

- Formats a Terminal Reconciliation settlement type PTLF record with a Balance Flag of *Not Balanced* from the real terminal totals (if configured on the PTDF) and sends this to Authorisation for writing to the PTLF.
- Formats a Terminal Reconciliation settlement type PTLF record with a Balance Flag of *Not Balanced* from the PTDF terminal totals (if configured on the PTDF) and sends this to Authorisation for writing to the PTLF.
- Formats a Services Reconciliation settlement type PTLF record with a Balance Flag of *Not Balanced* from the service group totals (if configured on the PTDF) and sends this to Authorisation for writing to the PTLF.
- Clears the following totals in the PTDF:

SERVICE GROUP
BATCH
BATCH-DC
SHIFT
SHIFT-DC
DAILY
DAILY-DC

7.1.3 Duplicate Reconciliation Transaction Processing

A merchant can re-request terminal balancing totals in the event of a communication failure, thus ensuring that the terminal receives the correct balancing information.

With the enhanced processing of administrative transactions, the PTDF is flagged as “Balancing Request, Awaiting Confirmation” and the system does not clear the PTDF totals prior to sending the balance request response to the terminal. The PTDF now clears the terminal totals on receipt of the next transaction, which is not a duplicate. This allows the merchant to re-request and receive accurate balancing totals.

7.2 FORCED CUTOVER

The cutover of APACS 70 terminals can be tied to the BASE24-pos balancing functionality and reconciliation transactions, as described above. The cutover of APACS 70 (Auth Only) terminals is “forced” each day by the BASE24-pos system (because APACS 70 (Auth Only) terminals do not support reconciliation messages), and is not tied to terminal balancing. Tying cutover to the balancing function is a convenient way to close the terminal reporting day at the same time for both the terminal and the BASE24-pos system. Note that the cutover of APACS 70 terminals is also forced if the retailer does not perform a reconciliation transaction within the retailer window.

Terminals can be forced cutover by either of the following methods:

- Device Handler process force cutover with the first financial transaction received within the Device Handler cutover window
- Secondary Settlement process force cutover at the end of the retailer cutover window

7.2.1 Forced Cutover by the Device Handler Module

The Device Handler module forces the terminal to cut over when it receives a financial transaction during the Device Handler cutover window. The transaction that causes the Device Handler module to cut over the terminal is then logged to the next day's PTLF. Only the first transaction within the Device Handler cutover window causes the Device Handler module to cut the terminal over. Subsequent transactions within the Device Handler cutover window do not cause the terminal to cut over.

If the TERMINAL CUTOVER flag on screen 3 of the PTDF is set to 4 (Network forced cutover by the Device Handler module), the Device Handler module can cutover the terminal. This occurs if the Router /Authorization module returns a response with a new posting date in the PSTM, and the Device Handler module performs the following steps:

- Formats a Network Settlement type PTLF record with a Balance Flag of "Force Cutover" from the PTDF terminal totals (if configured on the PTDF) and writes this to the PTLF.
- Formats a Services Reconciliation settlement type PTLF record with a Balance Flag of "Force Cutover" from the service group totals (if configured on the PTDF) and writes this to the PTLF.
- Moves the current NET totals to the previous NET totals and clears the following totals in the PTDF:

SERVICE GROUP
BATCH
SHIFT
SHIFT-DC
DAILY
DAILY-DC
CURRENT NET
CURRENT NET-DC

- Sets the new posting date.

Note: The transaction that causes the Device Handler module to perform force cutover is applied to the PTDF record after the settlement function has been completed. The BATCH-DC totals are cleared. There is no indication at the terminal that a BASE24 forced cutover has occurred.

7.2.2 Forced Cutover by Settlement

The Secondary Settlement process forces the terminal to cut over at the time indicated in the RETAILER CUTOVER/BALANCE END field on screen 1 of the PRDF.

If an end-of-day reconciliation transaction is not received from the APACS 70 terminal during the Retailer Cutover and Balancing Window, and the Device Handler module has not forced the terminal cutover, BASE24-pos Settlement performs a forced cutover.

To perform this transaction the BASE24-pos Settlement Initialisation process:

1. Sets the new posting date.
2. Formats a Network Settlement type PTLF record with a Balance Flag of *Force Cutover* from the PTDF terminal totals (if configured on the PTDF) and writes this to the PTLF.
3. Formats a Services Reconciliation settlement type PTLF record with a Balance Flag of *Force Cutover* from the service group totals (if configured on the PTDF) and writes this to the PTLF.
4. Moves the current NET totals to the previous NET totals and clears the following totals in the PTDF:

SERVICE GROUP
BATCH
SHIFT
SHIFT-DC
DAILY
DAILY-DC
CURRENT NET
CURRENT NET-DC

5. Sets the BALANCE FLAG in the PTDF to FORCE CUTOVER.

Note: The BATCH-DC totals are not cleared. They will only be cleared on the next Balance Transaction from the terminal, regardless of whether this is inside or outside the Retailer Cutover window. There is no indication at the terminal that a BASE24 forced cutover has occurred.

A. DEVICE HANDLER INTERNAL TABLES

Messages received from and sent to the APACS terminals are processed using internal tables to authorise or capture transaction information.

Messages sent by APACS terminals conform to the message format defined in Book 2 of the APACS 70 Standard. They are routed by the XPNET process to the BASE24-pos APACS 70 Device Handler module where the message header details are verified against the Message Header Parsing table. This table determines the terminal ID and type of message being sent (data capture, authorisation or reconciliation).

Provided that the terminal and header details are valid, and, in the case of data capture or reconciliation messages, that the MAC is verified, the message is processed using the Message Type Descriptor table.

Following processing, the message is reformatted using the relevant Response Message Parsing table and device dependent data stored in the POS Terminal Data File (PTDF), and is sent back to the terminal.

A.1 MESSAGE PARSING TABLES

The Header Parsing table is applicable to all message formats and describes each field in the request message header.

There is one Message Parsing table per input message format:

- Authorisation
- Authorisation confirmation
- Data capture
- Reconciliation

Each table contains the following information:

- DEFINITION of the field
- MIN/MAX (the minimum and maximum length of the field in bytes)
- FIX/VAR or F/V (whether the field is fixed length or variable length)
- VALID VALUES or A/N (whether the field requires alphanumeric or numeric data).

A.2 MESSAGE TYPE DESCRIPTOR TABLE

The Message Type Descriptor table contains all the possible message types that are sent by the APACS terminals to the Device Handler module. The table contains one entry for each message type. The message type in the input message is used to locate the entry in the table and the entry is copied to the device dependent data of the PTDF. The contents of each table entry are used to drive the processing of the message.

Each table contains the following information:

Content	Description
Request Message Type (Rqst Msg Type)	Two hexadecimal characters to indicate the request message sent.
Description	Description of the request message type, the transaction attributes, and whether the card information was keyed into the terminal or card initiated. Valid values for the request message type are as follows: • BALANCE ENQ (Balance Enquiry) • SALE (Cardholder Purchase) • CASH (Cash Advance) • CHQ (Card Verification) • PWCB (Purchase With Cash Back) • PRE-AUTH (Pre-Authorisation Request) • COMPLETION (Pre-Authorisation Completion) • REFUND (Refund) • REVERSAL (Reversal of last transaction type) • REFUND REVERSAL (Reversal of refund) • CONFIRMATION (Confirmation) • RECONCILIATION (Reconciliation) • ACCOUNT FUNDING (Account Funding) Valid values for the transaction attributes are as follows: • CONT. AUTH (Continuous Authority) • E.COM (E-Commerce) Valid values for card details entry are as follows: • Keyed (operator entered transaction details manually) • Card (transaction initiated by swiping card through card reader) • No c/h (no cardholder)

Content	Description																				
Message Format (Msg Format)	Identifies whether the message is for authorisation purposes only or authorisation and data capture: • AUTH (authorisation/verification) • EFT (authorisation and data capture) • CONF (authorisation confirmation) • REC (reconciliation)																				
Tries	Indicates the type of request message sent: • 1st Try—generated the first time the terminal attempts to send a message to the Device Handler. • Retry—if the terminal times out waiting for a response from the Device Handler, subsequent attempts to send the message are classified as RETRIES. Two Retry attempts are permitted per transaction request. • Advice—refers to request messages stored in the terminal because they fall below the terminal floor limits or were approved referrals. Such messages are batched and sent to BASE24 at intervals as post-event advices. For example, every fifth transaction may be sent to BASE24 for authorisation and/or data capture; at the same time, the four previous approved EFT transactions stored at the terminal is sent to BASE24 (before the fifth transaction) for force-posting to cardholder accounts. If a transaction exceeds the terminal floor limit parameters, it and any preceding transactions in that particular batch are sent to BASE24 for force-posting and authorisation/data capture.																				
Authorisation Transaction Code (Auth Tran Code)	The authorisation transaction code associated with the particular type of request message: <table> <tr><td>10</td><td>Sales</td></tr> <tr><td>11</td><td>Preauthorisation</td></tr> <tr><td>12</td><td>Preauthorisation completion</td></tr> <tr><td>13</td><td>Mail order</td></tr> <tr><td>14</td><td>Refund</td></tr> <tr><td>15</td><td>Cash advance</td></tr> <tr><td>16</td><td>Card verification</td></tr> <tr><td>17</td><td>Balance enquiry</td></tr> <tr><td>18</td><td>Purchase with cash back</td></tr> <tr><td>50</td><td>Reconciliation request</td></tr> </table>	10	Sales	11	Preauthorisation	12	Preauthorisation completion	13	Mail order	14	Refund	15	Cash advance	16	Card verification	17	Balance enquiry	18	Purchase with cash back	50	Reconciliation request
10	Sales																				
11	Preauthorisation																				
12	Preauthorisation completion																				
13	Mail order																				
14	Refund																				
15	Cash advance																				
16	Card verification																				
17	Balance enquiry																				
18	Purchase with cash back																				
50	Reconciliation request																				

Content	Description
Response Message Type (Resp Msg Type)	Specifies the response code the APACS 70 terminal expects to receive from the Device Handler module for the particular type of request message sent: 12 Debit (authorisation/verification) 13 Debit (authorisation and data capture) 41 Credit (refund/reversal) 62 Administrative 71 Balance enquiry 81 Hold
Credit or Debit (Cr Db)	Indicates whether the transaction is a debit from or credit to the cardholder account: • Credit • Debit • Admin
PSTM Type	The POS Standard Internal Message (PSTM) type for the request message: 0200 Request 0220 Forced-post 0420 Reversal

Rqst Msg Type	Description	Msg Format	Tries	Auth Tran Code	Resp Msg Type	Cr Db	PSTM Type
00	Not Implemented	-	-	00	00	-	-
01	CHQ, card	AUTH	1st Try	16	12	Debit	0200
02	CHQ, keyed	AUTH	1st Try	16	12	Debit	0200
03	PWCB, card	AUTH	1st Try	18	12	Debit	0200
04	PWCB, keyed	AUTH	1st Try	18	12	Debit	0200
05	PWCB, card	AUTH	Retry	18	12	Debit	0200
06	PWCB, keyed	AUTH	Retry	18	12	Debit	0200
07	CHQ, card	AUTH	Retry	16	12	Debit	0200
08	CHQ, keyed	AUTH	Retry	16	12	Debit	0200
09	SALE, no c/h	AUTH	1st Try	13	12	Debit	0200
10	SALE, card	AUTH	1st Try	10	12	Debit	0200
11	SALE, card	EFT	1st Try	10	13	Debit	0200
12	Response Message Type	-	-	00	00	-	-
13	Response Message Type	-	-	00	00	-	-
14	REVERSAL	EFT	1st Try	10 *	41	Credit	0420
15	REVERSAL	EFT	Retry	10 *	41	Credit	0420
16	SALE, card	AUTH	Retry	10	12	Debit	0200
17	SALE, card	EFT	Retry	10	13	Debit	0200
18	REVERSAL	EFT	Advice	10 *	41	Credit	0420
19	SALE, card	EFT	Advice	10	13	Debit	0220
20	SALE, keyed	AUTH	1st Try	10	12	Debit	0200
21	SALE, keyed	EFT	1st Try	10	13	Debit	0200
22	SALE, no c/h	EFT	1st Try	13	13	Debit	0200
23	SALE, no c/h	EFT	Retry	13	13	Debit	0200
24	SALE, no c/h	EFT	Advice	13	13	Debit	0220
25	REVERSAL	AUTH	1st Try	10 *	12	Debit	0420
26	SALE, keyed	AUTH	Retry	10	12	Debit	0200
27	SALE, keyed	EFT	Retry	10	13	Debit	0200
28	REVERSAL	AUTH	Retry	10 *	12	Debit	0420
29	SALE, keyed	EFT	Advice	10	13	Debit	0220

Rqst Msg Type	Description	Msg Format	Tries	Auth Tran Code	Resp Msg Type	Cr Db	PSTM Type
30	CASH, card	AUTH	1st Try	15	12	Debit	0200
31	CASH, keyed	AUTH	1st Try	15	12	Debit	0200
32	CASH, card	EFT	1st Try	15	13	Debit	0200
33	CASH, keyed	EFT	1st Try	15	13	Debit	0200
34	CASH, card	EFT	Retry	15	13	Debit	0200
35	CASH, keyed	EFT	Retry	15	13	Debit	0200
36	CASH, card	AUTH	Retry	15	12	Debit	0200
37	CASH, keyed	AUTH	Retry	15	12	Debit	0200
38	CASH, card	EFT	Advice	15	13	Debit	0220
39	CASH, keyed	EFT	Advice	15	13	Debit	0220
40	REFUND, card	EFT	1st Try	14	41	Credit	0200
41	Response Message Type	-	-	00	00	-	-
42	PRE-AUTH, card/keyed	AUTH	1st Try	11	12	Debit	0200
43	PRE-AUTH, card/keyed	AUTH	Retry	11	12	Debit	0200
44	COMPLETION, card/keyed	EFT	1st Try	12	13	Debit	0200
45	SALE, no c/h	AUTH	Retry	13	12	Debit	0200
46	REFUND, card	EFT	Retry	14	41	Credit	0200
47	REFUND, no c/h	AUTH	1st Try	14	12	Credit	0200
48	REFUND, card	EFT	Advice	14	41	Credit	0220
49	REFUND, no c/h	AUTH	Retry	14	12	Credit	0200
50	REFUND, no c/h	EFT	1st Try	14	41	Credit	0200
51	COMPLETION, card/keyed	EFT	Retry	12	13	Debit	0200
52	REFUND, no c/h	EFT	Retry	14	41	Credit	0200
53	COMPLETION, card/keyed	EFT	Advice	12	13	Debit	0220
54	REFUND, no c/h	EFT	Advice	14	41	Credit	0220
55	REFUND, keyed	EFT	1st Try	14	41	Credit	0200
56	REFUND, keyed	EFT	Retry	14	41	Credit	0200
57	Not Implemented	-	-	00	00	-	-
58	REFUND, card	AUTH	1st Try	14	12	Credit	0200
59	REFUND, keyed	EFT	Advice	14	41	Credit	0220

Rqst Msg Type	Description	Msg Format	Tries	Auth Tran Code	Resp Msg Type	Cr Db	PSTM Type
60	RECONCILIATION	REC	1st Try	50	62	Admin	-
61	REFUND, keyed	AUTH	1st Try	14	12	Credit	0200
62	Response Message Type	-	-	00	00	-	-
63	REFUND, card	AUTH	Retry	14	12	Credit	0200
64	Not Supported	-	-	00	00	-	-
65	Not Supported	-	-	00	00	-	-
66	Not Implemented	-	-	00	00	-	-
67	REFUND, keyed	AUTH	Retry	14	12	Credit	0200
68	RECONCILIATION	REC	Retry	50	62	Admin	-
69	RECONCILIATION	REC	Advice	50	62	Admin	-
70	BALANCE ENQ, card	EFT	1st Try	17	71	-	0200
71	Response Message Type	-	-	00	00	-	-
72	BALANCE ENQ, keyed	EFT	1st Try	17	71	-	0200
73	BALANCE ENQ, card	EFT	Retry	17	71	-	0200
74	BALANCE ENQ, keyed	EFT	Retry	17	71	-	0200
75	PWCB, card	EFT	1st Try	18	13	Debit	0200
76	PWCB, keyed	EFT	1st Try	18	13	Debit	0200
77	PWCB, card	EFT	Retry	18	13	Debit	0200
78	PWCB, keyed	EFT	Retry	18	13	Debit	0200
79	PWCB, card	EFT	Advice	18	13	Debit	0220
80	CONFIRMATION	CONF	-	00	00	-	-
81	Response Message Type	-	-	00	00	-	-
82	Not Supported	-	-	00	00	-	-
83	REFUND REVERSAL	EFT	1st Try	14	13	Debit	0420
84	PWCB, keyed	EFT	Advice	18	13	Debit	0220
85	REFUND REVERSAL	EFT	Retry	14	13	Debit	0420
86	REFUND REVERSAL	AUTH	1st Try	14	12	Debit	0420
87	REFUND REVERSAL	EFT	Advice	14	13	Debit	0420

Rqst Msg Type	Description	Msg Format	Tries	Auth Tran Code	Resp Msg Type	Cr Db	PSTM Type
88	REFUND REVERSAL	AUTH	Retry	14	12	Debit	0420
89	Not Implemented	-	-	00	00	-	-
90	Not Implemented	-	-	00	00	-	-
91	Not Implemented	-	-	00	00	-	-
92	Not Implemented	-	-	00	00	-	-
93	Not Implemented	-	-	00	00	-	-
94	Not Implemented	-	-	00	00	-	-
95	Not Implemented	-	-	00	00	-	-
96	Not Implemented	-	-	00	00	-	-
97	Not Implemented	-	-	00	00	-	-
98	Not Implemented	-	-	00	00	-	-
99	Not Implemented	-	-	00	00	-	-
A0	SALE (CONT. AUTH), no c/h	AUTH	1st Try	13	12	Debit	0200
A1	SALE (CONT. AUTH), no c/h	AUTH	Retry	13	12	Debit	0200
A2	SALE (CONT. AUTH), no c/h	EFT	1st Try	13	13	Debit	0200
A3	SALE (CONT. AUTH), no c/h	EFT	Retry	13	13	Debit	0200
A4	REFUND (CONT. AUTH), no c/h	EFT	1st Try	14	41	Credit	0200
A5	REFUND (CONT. AUTH), no c/h	EFT	Retry	14	41	Credit	0200
A6	SALE (CONT. AUTH), no c/h	EFT	Advice	13	13	Debit	0220
A7	REFUND (CONT. AUTH), no c/h	EFT	Advice	14	41	Credit	0220
A8	SALE (E.COM), card	AUTH	1st Try	10	12	Debit	0200
A9	SALE (E.COM), card	AUTH	Retry	10	12	Debit	0200
AA	REFUND (E.COM), card	AUTH	1st Try	14	12	Credit	0200
AB	REFUND (E.COM), card	AUTH	Retry	14	12	Credit	0200
AC	SALE (E.COM), card	EFT	1st Try	10	13	Debit	0200
AD	SALE (E.COM), card	EFT	Retry	10	13	Debit	0200

Rqst Msg Type	Description	Msg Format	Tries	Auth Tran Code	Resp Msg Type	Cr Db	PSTM Type
AE	REFUND (E.COM), card	EFT	1st Try	14	41	Credit	0200
AF	REFUND (E.COM), card	EFT	Retry	14	41	Credit	0200
B0	SALE (E.COM), card	EFT	Advice	10	13	Debit	0220
B1	REFUND (E.COM), card	EFT	Advice	14	41	Credit	0220
B2	SALE (E.COM), keyed	AUTH	1st Try	10	12	Debit	0200
B3	SALE (E.COM), keyed	AUTH	Retry	10	12	Debit	0200
B4	REFUND (E.COM), keyed	AUTH	1st Try	14	12	Credit	0200
B5	REFUND (E.COM), keyed	AUTH	Retry	14	12	Credit	0200
B6	SALE (E.COM), keyed	EFT	1st Try	10	13	Debit	0200
B7	SALE (E.COM), keyed	EFT	Retry	10	13	Debit	0200
B8	REFUND (E.COM), keyed	EFT	1st Try	14	41	Credit	0200
B9	REFUND (E.COM), keyed	EFT	Retry	14	41	Credit	0200
BA	SALE (E.COM), keyed	EFT	Advice	10	13	Debit	0220
BB	REFUND (E.COM), keyed	EFT	Advice	14	41	Credit	0220
C0	ACCOUNT FUNDING, card	AUTH	1st Try	10	12	Debit	0200
C1	ACCOUNT FUNDING, card	AUTH	Retry	10	12	Debit	0200
C2	ACCOUNT FUNDING, keyed	AUTH	1st Try	10	12	Debit	0200
C3	ACCOUNT FUNDING, keyed	AUTH	Retry	10	12	Debit	0200
C4	ACCOUNT FUNDING, no c/h	AUTH	1st Try	13	12	Debit	0200
C5	ACCOUNT FUNDING, no c/h	AUTH	Retry	13	12	Debit	0200

Rqst Msg Type	Description	Msg Format	Tries	Auth Tran Code	Resp Msg Type	Cr Db	PSTM Type
C6	ACCOUNT FUNDING, card	EFT	1st Try	10	13	Debit	0200
C7	ACCOUNT FUNDING, card	EFT	Retry	10	13	Debit	0200
C8	ACCOUNT FUNDING, card	EFT	Advice	10	13	Debit	0220
C9	ACCOUNT FUNDING, keyed	EFT	1st Try	10	13	Debit	0200
CA	ACCOUNT FUNDING, keyed	EFT	Retry	10	13	Debit	0200
CB	ACCOUNT FUNDING, keyed	EFT	Advice	10	13	Debit	0220
CC	ACCOUNT FUNDING, no c/h	EFT	1st Try	13	13	Debit	0200
CD	ACCOUNT FUNDING, no c/h	EFT	Retry	13	13	Debit	0200
CE	ACCOUNT FUNDING, no c/h	EFT	Advice	13	13	Debit	0220

* These message types are also used for cash, account funding and PWCB reversals. BASE24 interprets the request appropriately depending on the last transaction type received by the Device Handler module.

B. DEVICE HANDLER EXTERNAL TABLES

An EDIT file (RESPTBL) contains the Authorisation Response table and the Device Handler Response table.

The edit file is compiled with the APACS 70 POS Device Handler module, and therefore provides customers with some flexibility in configuring the response data returned to a POS terminal.

The Authorisation Response table documents all possible response codes from the Authorisation process and an associated pointer to the Device Response table. The device response code and text displayed at the terminal is found at the offset, starting at zero, in the Device Response table.

The Device Response table contains the response codes that are sent by the Device Handler module to be displayed by the APACS 70 terminal at the retailer outlet or bank branch.

For example, a response code of 150 (Merchant Not On File) is returned to the Device Handler module from the Authorisation module. The device response code associated with 150 in the Authorisation Response table is 03. Therefore, INVALID MERCHANT (device response 03) is displayed at the terminal.

B.1 AUTHORISATION RESPONSE TABLE

The device response code is the only part of this table that should be modified.

AUTH RESPONSE CODE	DEVICE RESPONSE TABLE ENTRY	DESCRIPTION
***** Approval Codes *****		
000	00	00 Approved balances
001	00	01 Approved no balances
002	02	02 Approved country club
003	01	03 Approved maybe (more ID)
004	01	04 Approved pending ID (sign draft)
005	00	05 Approved blind
006	02	06 Approved VIP
007	00	07 Approved administrative transaction
008	00	08 Approved NNEG hit OK
009	00	09 Approved commercial
***** Decline Codes *****		
050	15	10 Declined
051	14	11 Expired card
052	15	12 PIN tries exceeded
053	15	13 No sharing with card user
054	15	14 No ATALLA box
055	16	15 Invalid transaction
056	16	16 Not supported
057	19	17 Lost or stolen card
058	16	18 Invalid card status
059	15	19 Restricted card
060	16	20 No accounts
061	16	21 No PBF
062	16	22 PBF update error
063	16	23 Invalid authorisation type
064	11	24 Bad track 2
065	10	25 Adjustment not allowed

AUTH RESPONSE CODE	DEVICE RESPONSE TABLE ENTRY	DESCRIPTION
066	12	26 Invalid credit card advance increment
067	16	27 Invalid transaction date
068	16	28 TLF error
069	16	29 Invalid message format
070	16	30 No IDF
071	16	31 Invalid route authorisation
072	15	32 Card on National Negative File
073	16	33 Invalid route service
074	15	34 Unable to authorise
075	11	35 Invalid PAN length
076	15	36 Low on funds
077	16	37 Pre-authorisation table full
078	16	38 Duplicate transaction
079	15	39 Maximum on-us refund reached
080	15	40 Maximum not-on-us refund reached
081	15	41 Maximum credit per refund
082	15	42 Maximum number times used
083	15	43 Maximum refund credit reached
084	15	44 Customer selected negative reason
085	10	45 Enquiries not allowed
086	15	46 Over floor limit
087	15	47 Maximum number refund credits
088	15	48 Place call
089	15	49 CAF status 0 or 9
090	16	50 Referral file full
091	16	51 NEG problem
092	12	52 Cash advance less than minimum
093	15	53 Delinquent
094	15	54 Over limit table
095	16	55 Amount exceeds maximum
096	16	56 PIN required
097	11	57 MOD 10 check
099	15	58 Bad PBF status

AUTH RESPONSE CODE	DEVICE RESPONSE TABLE ENTRY	DESCRIPTION	
***** Referral Codes *****			
100	04	59	Unable to process transaction
101	04	60	Unable to authorise, issue call
102	04	61	Call
103	04	62	NEG problem
104	04	63	CAF problem
105	04	64	Card not supported
106	04	65	Amount exceeds maximum
107	04	66	Over daily limit
108	04	67	CAPF not found
109	04	68	Cash advance less than minimum
110	04	69	Number times used
111	04	70	Delinquent
112	04	71	Over limit table
113	04	72	Timeout
115	04	73	PTLF full
120	04	74	UAF problem
121	04	75	Admin File Problem
122	04	76	Unable to Validate PIN
***** EMV Referral Codes *****			
130	04	77	ARQC failure Referral
131	04	78	CVR Referral
132	04	79	TVR Referral
133	04	80	Reason Online Code Referral
134	04	81	Fallback referral
134	02	Fallback referral	
***** Service Codes *****			
150	13	82	Merchant not on File
***** Transaction Error Codes *****			
200	16	83	Invalid account
201	08	84	Incorrect PIN
202	12	85	Cash advance less than minimum
203	16	86	Administrative card needed
204	12	87	Enter lesser amount

AUTH RESPONSE CODE	DEVICE RESPONSE TABLE ENTRY	DESCRIPTION	
205	12	88	Invalid credit card advance amount
206	16	89	CAF not found
207	16	90	Invalid transaction date
208	16	91	Invalid expiry date
251	15	92	Error in cash amount
252	15	93	Debit card not supported
***** EMV DECLINE CODES *****			
400	17	94	ARQC failure decline
401	16	95	HSM Parameter Failure
402	16	96	HSM Failure
403	16	97	KEYI Record not found
404	18	98	ATC Check Failure
405	18	99	CVR decline
406	18	100	TVR decline
407	18	101	Reason online code decline
408	18	102	Fallback decline

AUTH RESPONSE CODE	DEVICE RESPONSE TABLE ENTRY	DESCRIPTION
***** Capture Codes *****		
900	09	103 PIN tries exceeded
901	09	104 Expired card
902	09	105 NEG capture code
903	09	106 CAF status 3
904	09	107 Cash advance less than minimum
905	09	108 Number times used
906	09	109 Delinquent
907	09	110 Over limit table
908	09	111 Amount over maximum
909	09	112 Capture!
***** EMV Capture Codes *****		
910	09	112ARQC Failure Capture
911	09	113 CVR Capture
912	09	114 TVR Capture

B.2 DEVICE RESPONSE TABLE

The text of any or all of the display messages can be modified, but the length of each must remain the same. Also, display entries must not be added, deleted or re-arranged.

B.2.1 Response Types Generated by Authorisation Process

Response Type	Resp Code	Terminal Display Text	Description	Device Response Table Entry
Standard	00	AUTH CODE : XXXXXX	Approval; XXXXX is the approval number	00
Standard	08	AUTH CODE : XXXXXX	Approval; XXXXX is the approval number	01
Standard	11	AUTH CODE : XXXXXX	Approval; XXXXX is the approval number	02
Standard	00	RESERVED	Not used	03
Standard	01	REFERRAL	Referral; over limit	04
Standard	02	REFERRAL	Referral; restricted card	05
Standard	60	REFERRAL	Referral; restricted card	06
Standard	00	RESERVED	Not used	07
Standard	55	PIN ERROR	Invalid PIN format; re-enter	08
Standard	04	RETAIN CARD	Capture Card	09
Standard	05	REQUEST INVALID	Invalid request	10
Standard	05	INVALID TRACK2	Invalid card details	11
Standard	05	INVALID AMOUNT	Invalid credit card advance; re-enter	12
Standard	05	INVALID MERCHANT	Merchant Number not found in PRDF	13
Standard	05	CARD EXPIRED	Expired Card	14
Standard	05	NOT AUTHORISED	Declined	15
Standard	05	UNABLE TO AUTH	Referral; general	16
Standard	05	INVALID EMV DATA	Invalid EMV data; re-enter	17
Standard	05	EMV AUTH FAILED	EMV Authorisation Failure	18
Standard	05	LOST/STOLEN CARD	Capture Card	19

B.2.2 Error Response Types Generated by Device Handler Process

Response Type	Resp Code	Terminal Display Text	Description	Device Response Table Entry
Error Hold	21	TERM DEACTIVETED	Device deactivated	20
Error Hold	30	BAD MERCH	Invalid merchant id	21
Error Hold	30	BAD TRAN CODE	Error in transaction information	22
Error Hold	30	RESERVED	Not used	23
Error Hold	30	BAD AMOUNT	Invalid amount entered	24
Error Hold	30	BAD ACCOUNT	Invalid account number	25
Error Hold	30	BAD EXPIRY DATE	Invalid card expiry date	26
Error Hold	30	BAD TRACK2	Invalid card details	27
Error Hold	30	BAD DESCR	Invalid description data	28
Error Hold	30	BAD CONFCODE	Invalid confirmation code	29
Error Hold	30	BAD BAL CODE	Invalid balance code	30
Error Hold	30	BAD AUTHCODE	Invalid authorisation code	31
Error Hold	30	BAD FORMAT	Invalid format	32
Error Hold	30	BAD TOTALS	Invalid total	33
Error Hold	30	BAD CIPHER	Invalid cipher block	34
Error Hold	30	BAD TIME	Invalid time format	35
Error Hold	96	SYSTEM MALFUNCTN	MAC verification failure	36
Error Hold	63	BAD MAC	MAC not in sync	37
Error Hold	63	RESERVED	Not used	38
Error Hold	63	RESERVED	Not used	39

An error hold response is formatted if the request message received from the terminal cannot be processed by BASE24. The response is a hold response, but because the response code is set to a non-zero value, the terminal disconnects on receiving the message and attempts to redial if appropriate.

B.2.3 Standard Response Types Generated by Device Handler Process

Response Type	Resp Code	Terminal Display Text	Description	Device Response Table Entry
Standard	05	INVALID CURRENCY	Invalid currency	40
Standard	05	UNABLE TO AUTH	Referral; general	41
Standard	05	RESERVED	Not used	42
Standard	05	RESERVED	Not used	43
Standard	05	RESERVED	Not used	44
Standard	05	NOT AUTHORISED	Declined	45
Standard	02	REFERRAL Z	Referral; refund over limit	46
Standard	30	REQUEST INVALID	Invalid request	47
Standard	05	UNABLE TO AUTH	Referral; general	48
Standard	02	REFERRAL X	Referral; invalid response code	49
Standard	00	COMPLETED	Reconciliation completed	50
Standard	00	REVERSAL	Reversal authorised	51
Standard	00	REFUND	Refund authorised	52
Standard	00	REFUND REVERSAL	Refund reversal authorised	53
Standard	00	ACCEPTED	Accepted	54

B.2.4 Ignore and Disconnect Messages

Response Type	Resp Code	Terminal Display Text	Description	Device Response Table Entry
Ignore	98	IGNORE		55
Disconnect	99	DISCONNECT		56
Hold	00	PLEASE WAIT		57
Standard	00	RESERVED	Not used	58
Standard	00	RESERVED	Not used	59

Ignore and Disconnect Messages are not displayed at the terminal.

C. EVENT MESSAGES

Event messages provide information ranging from simple process statuses to problems of a serious nature requiring operator intervention. They are generated by the Device Handler module, based on conditions encountered, and are sent to the BASE24 EMS log.

Event messages also provide notification to operators of actions and conditions occurring in the system and as such also provide an audit trail of these actions and conditions.

C.1 FORMAT OF EVENT MESSAGES

Event messages are generated to report an error condition or information. Error messages may require corrective action before the appropriate process can function properly.

The standard format for BASE24 log messages is:

YY/MM/DD hh:mm:ss X PROCESS ID OBJECT MMMM RRRR TEXT

where: YY/MM/DD specifies the date the message was sent.

hh:mm:ss specifies the time the message was sent.

X specifies the severity code where:

0 or I	=	Status Information
1 or W	=	Warning
2 or E	=	Error
3 or C	=	Critical

PROCESS ID specifies the symbolic process name.

OBJECT specifies a 16-byte string defined and passed by the process.

MMMM specifies the message number.

RRRR specifies the routing code.

TEXT specifies the text of the message.

For example:

01/12/10 09:09:48 (2) P1A^IBDRA DEVICE HANDLER 7070 A Failed message was received from 02400005
--

To avoid the repetition of showing the date, time, process identifier, object name and routing code for each message; only the message number, severity code and the text of the message are documented. Therefore, the display format of log messages adopted for documentation is:

MMMM X TEXT

For example:

7070 (2) A Failed message was received from 02400005

For each event message, additional information is given:

- The cause or reason the message is displayed.
- The remedy or solution for the message, if relevant.

Also, the following support or service personnel may need to be contacted:

- Operations Personnel: In-house operations staff responsible for initiating fault action for a particular problem.
- Service Personnel: In-house staff who are responsible for maintaining user data on BASE24.
- HELP24 ACI personnel who are responsible for solving customer queries and problems.

C.2 DEVICE HANDLER EVENT MESSAGES

7020	(3)	PTDF^MCB Initialisation error \e
-------------	------------	---

Where: \e specifies the error number.

Cause: The system was unable to allocate memory or initialise all of the data structures that use the PTDF due to the error identified in the message.

Remedy: Determine the correct action to take based on the error listed.

7030	(0)	Unable to send timeout to terminal \S - TDF read error
-------------	------------	---

Where: \S specifies the terminal ID.

Cause: System error. The system was unable to send the timeout to the terminal due to a problem that occurred while attempting to read the PTDF.

Remedy: Check the PTDF.

7031	(0)	Unable to send timeout to terminal \S - TDF read error
-------------	------------	---

Where: \S specifies the terminal ID.

Cause: System error. The system was unable to send the timeout to the terminal due to a problem that occurred while attempting to read the PTDF.

Remedy: Check the PTDF.

7032	(0)	Unable to send timeout to terminal \S - TDF read error
-------------	------------	---

Where: \S specifies the terminal ID.

Cause: System error. The system was unable to send the timeout to the terminal due to a problem that occurred while attempting to read the PTDF.

Remedy: Check the PTDF.

7040	(0)	Unable to process transaction request for terminal \S - TDF read error
-------------	------------	---

Where: \S specifies the terminal ID.

Cause: System error. The system was unable to process the transaction request due to a problem that occurred while attempting to read the PTDF.

Remedy: Check the PTDF.

7050	(0)	Transaction reversed due to TDF read error on terminal \S
-------------	------------	--

Where: \S specifies the terminal ID.

Cause: System error. The system was unable to reverse the transaction due to a problem that occurred while attempting to read the PTDF.

Remedy: Check the PTDF.

7060	(2)	Invalid msg received from @ - *****
-------------	------------	---

Where: @ specifies the message source.
* specifies the message text.

Cause: The BASE24 APACS Device Handler module received an invalid message.

Remedy: If the problem persists, check the source sending the message.

7070	(2)	A Failed message was received from \S
-------------	------------	--

Where: \S specifies the terminal ID.

Cause: The BASE24 APACS Device Handler module prepared a response for the identified terminal, but the XPNET process was unable to deliver the message.

Remedy: Check the line and station associated with the identified terminal.

7080	(0)	Terminal \S has been deactivated
-------------	------------	---

Where: \S specifies the terminal ID.

Cause: The terminal has been deactivated in response to a command from the user.

Remedy: No action required.

7090 (0) Terminal \S is pending deactivation

Where: \S specifies the terminal ID.

Cause: A deactivate command was issued by the terminal did not meet the conditions to be deactivated. The terminal is deactivated when the conditions are met.

Remedy: No action required.

7100 (2) Bad merchant number, PTDF = \S1 while \S2 received from terminal \S3

Where: \S1 specifies the retailer ID in the PTDF.
\S2 specifies the retailer ID received.
\S3 specifies the terminal ID.

Cause: The retailer ID specified in the terminal message does not match the retailer ID specified in the PTDF.

Remedy: Check the retailer IDs in the stored in the terminal and in the PTDF.

7110 (2) Field no #, message too short, terminal \S

Where: # specifies the field number.
\S specifies the terminal ID.

Cause: The terminal message is too short and does not contain the required field shown in the message.

Remedy: Contact the terminal vendor.

7115 (2) Field no #, message too short, terminal \S

Where: # specifies the field number.
\S specifies the terminal ID.

Cause: The terminal message is too short and does not contain the required field shown in the message.

Remedy: Contact the terminal vendor.

7120	(2)	Field no #, field too long, terminal \S
-------------	------------	--

Where: # specifies the field number.
 \S specifies the terminal ID.

Cause: The terminal message contains a field that is not of the correct length.

Remedy: Contact the terminal vendor.

7125	(2)	Field no # not numeric, terminal \S
-------------	------------	--

Where: # specifies the field number.
 \S specifies the terminal ID.

Cause: The terminal message contains a field that is required to be numeric, but contains non-numeric characters.

Remedy: Contact the terminal vendor.

7130	(2)	Field no # not numeric, terminal \S
-------------	------------	--

Where: # specifies the field number.
 \S specifies the terminal ID.

Cause: The terminal message contains a field which is not of the correct length.

Remedy: Contact the terminal vendor.

7135	(2)	Field no #, message too short, terminal \S
-------------	------------	---

Where: # specifies the field number.
 \S specifies the terminal ID.

Cause: The terminal message is too short and does not contain the required field shown in the message.

Remedy: Contact the terminal vendor.

7140	(2)	Field no #, field too long, terminal \S
-------------	------------	--

Where: # specifies the field number.
 \S specifies the terminal ID.

Cause: The terminal message contains a field that is not of the correct length.

Remedy: Contact the terminal vendor.

7145 (2) Field no # not numeric, terminal \S

Where: # specifies the field number.
\S specifies the terminal ID.

Cause: The terminal message contains a field which is not of the correct length.

Remedy: Contact the terminal vendor.

7150 (2) Number of fields in header incorrect, terminal \S

Where: \S specifies the terminal ID.

Cause: The header part of the terminal message contains data that is not expected by BASE24.

Remedy: Contact the terminal vendor.

7155 (2) Field no #, message too short, terminal \S

Where: # specifies the field number.
\S specifies the terminal ID.

Cause: The terminal message is too short and does not contain the required field shown in the message.

Remedy: Contact the terminal vendor.

7160 (2) No. fields in auth req incorrect, terminal \S

Where: \S specifies the terminal ID.

Cause: The terminal auth request message contains fields that are not expected by BASE24.

Remedy: Contact the terminal vendor.

7170 (2) Confirmation received but not expected, terminal \S

Where: \S specifies the terminal ID.

Cause: The terminal has sent an authorisation confirmation message to BASE24 but BASE24 was not expecting the message.

Remedy: Determine why the terminal sent the confirmation message and contact the terminal vendor.

7171	(2)	Confirmation received but not expected, terminal \S
-------------	------------	--

Where: \S specifies the terminal ID.

Cause: The terminal has sent an authorisation confirmation message to BASE24 but BASE24 was not expecting the message.

Remedy: Determine why the terminal sent the confirmation message and contact the terminal vendor.

7180	(2)	Illegal auth conf code \, terminal \S
-------------	------------	--

Where: \ specifies the confirmation code field sent in the terminal message.
\S specifies the terminal ID.

Cause: The terminal has sent an authorisation confirmation message that contains an invalid value in the confirmation code field.

Remedy: Determine why the terminal sent an invalid confirmation code and contact the terminal vendor.

7190	(2)	No. fields in auth conf incorrect, terminal \S
-------------	------------	---

Where: \S specifies the terminal ID.

Cause: The terminal has sent an authorisation confirmation message that contains more data than BASE24 is expecting.

Remedy: Determine why the terminal sent extra fields and contact the terminal vendor.

7191	(2)	Invalid Additional Transaction Security Data from terminal \S is \?
-------------	------------	--

Where: \S specifies the terminal ID.
\? specifies the Additional Security Data field.

Cause: The terminal has sent invalid data in the Additional transaction security data field in Auxiliary data record type 01.

Remedy: Determine why the terminal sent the invalid data in the Additional transaction security data field and contact the terminal vendor.

7192 (2) Auxiliary rec type \ not supported, terminal \S

- Where: \ specifies the Auxiliary data record type.
\S specifies the terminal ID.
- Cause: The terminal has sent an Auxiliary data record type that is not supported by BASE24.
- Remedy: Determine why the terminal sent the Auxiliary data record type not supported and contact the terminal vendor.

7200 (2) Invalid cipher field \, terminal \S

- Where: \ specifies the cipher field sent in the terminal message.
\S specifies the terminal ID.
- Cause: The terminal message contained an invalid cipher field that BASE24 cannot recognise.
- Remedy: Determine why the terminal sent the invalid cipher field and contact the terminal vendor.

7205 (2) Invalid cipher value \, terminal \S

- Where: \ specifies the cipher value sent in the terminal message.
\S specifies the terminal ID.
- Cause: The terminal message contained an invalid cipher value that BASE24 cannot recognise.
- Remedy: Determine why the terminal sent the invalid cipher value code and contact the terminal vendor.

7210 (2) No. fields in EFT conf incorrect, terminal \S

- Where: \S specifies the terminal ID.
- Cause: The terminal has sent an EFT confirmation message that contains more data than BASE24 is expecting.
- Remedy: Determine why the terminal sent extra fields and contact the terminal vendor.

7220	(2)	\S - MAC VERIFICATION FAILURE # - CONFIRMATION DROPPED. SEQUENCE NUMBER ##
-------------	------------	---

Where: \S specifies the terminal ID.
 # specifies the Racal error number.
 ## specifies the sequence number.

Cause: The confirmation message sent by the terminal contained a MAC that failed verification.

Remedy: Contact Systems Manager.

7221	(2)	\S - MAC VERIFICATION FAILURE # - CONFIRMATION DROPPED. SEQUENCE NUMBER ##
-------------	------------	---

Where: \S specifies the terminal ID.
 # specifies the Racal error number.
 ## specifies the sequence number.

Cause: The confirmation message sent by the terminal contained a MAC that failed verification.

Remedy: Contact Systems Manager.

7230	(2)	***** Bad MAC *****, terminal \S
-------------	------------	---

Where: \S specifies the terminal ID.

Cause: The message sent by the terminal contained a MAC that failed verification.

Remedy: Contact the Systems Manager.

7240	(2)	No. fields in rec req incorrect, terminal \S
-------------	------------	---

Where: \S specifies the terminal ID.

Cause: The terminal sent a reconciliation request that contains more data than BASE24 is expecting.

Remedy: Determine why the terminal sent extra fields and contact the terminal vendor.

7250	(2)	Invalid message format from terminal \S
-------------	------------	--

Where: \S specifies the terminal ID.

Cause: The terminal has sent a message that BASE24 does not recognise.

Remedy: Contact the terminal vendor.

7251 (2) Invalid message length from terminal \S

Where: \S specifies the terminal ID.
Cause: The terminal has sent a message with invalid message length.
Remedy: Contact the terminal vendor.

7261 (2) Bad conversion of transaction type, terminal \S

Where: \S specifies the terminal ID.
Cause: The terminal message contained an invalid transaction code that BASE24 cannot recognise.
Remedy: Determine why the terminal sent the invalid transaction code and contact the terminal vendor.

7270 (2) Duplicate message received from terminal \S

Where: \S specifies the terminal ID.
Cause: A duplicate message was received from the specified terminal.
Remedy: If problem persists, contact the terminal vendor.

7271 (2) Duplicate message received from terminal \S

Where: \S specifies the terminal ID.
Cause: A duplicate message was received from the specified terminal.
Remedy: If problem persists, contact the terminal vendor.

7280 (2) \S - UNABLE TO CONVERSE WITH THE SECURITY MODULE - ERROR #

Where: \S specifies the terminal ID.
specifies the security module error number.
Cause: The system was unable to check the MAC due to the error listed.
Remedy: Take the action indicated by the error.

7281	(2)	\S - UNABLE TO CONVERSE WITH THE SECURITY MODULE - ERROR #
-------------	------------	---

Where: \S specifies the terminal ID.
 # specifies the security module error number.

Cause: The system was unable to check the MAC due to the error listed.

Remedy: Take the action indicated by the error.

7290	(2)	***** Failed MAC 1st time try with previous/updated key terminal \S, conf code # *****
-------------	------------	---

Where: \S specifies the terminal ID.
 # specifies the terminal confirmation code.

Cause: The MAC verification process failed with the new updated key so the system tries with the previous key.

Remedy: If this happens frequently, contact the terminal vendor.

7300	(2)	Unable to reverse TDF totals - invalid service \\ in STM
-------------	------------	---

Where: \\ specifies the card service identifier.

Cause: An attempt to reverse totals in the PTFDF was unsuccessful due to an invalid card service in the PSTM.

Remedy: Note that the totals are incorrect. In addition, determine why an invalid card service was entered in the PSTM.

7301	(2)	Unable to reverse TDF totals - invalid service \\ in STM
-------------	------------	---

Where: \\ specifies the card service identifier.

Cause: An attempt to reverse totals in the PTFDF was unsuccessful due to an invalid card service in the PSTM.

Remedy: Note that the totals are incorrect. In addition, determine why an invalid card service was entered in the PSTM.

7309	(2)	Invalid format for key-entered transaction, terminal \S
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Where: \S specifies the terminal ID.

Cause: The terminal message was in an invalid format.

Remedy: Determine why the terminal formatted the message incorrectly and contact the terminal vendor.

7310 (2) Wrong length account number from terminal \S

Where: \S specifies the terminal ID.
Cause: The terminal message contained an account number that was either too short or too long.
Remedy: Determine why the terminal sent the invalid account number and contact the terminal vendor.

7320 (2) Non-numeric account number from terminal \S

Where: \S specifies the terminal ID.
Cause: The terminal message contained an account number that was not numeric.
Remedy: Determine why the terminal sent the invalid account number and contact the terminal vendor.

7330 (2) A Member Number greater than 3 digits received from terminal \S

Where: \S specifies the terminal ID.
Cause: The terminal message contained a member number which was longer than three characters.
Remedy: Determine why the terminal sent the invalid member number and contact the terminal vendor.

7340 (2) Non numeric key-entered Member No # from terminal \S

Where: # specifies the member number in the terminal request message.
\S specifies the terminal ID.
Cause: The terminal message contained a member number that was not numeric.
Remedy: Determine why the terminal sent the invalid member number and contact the terminal vendor.

7350 (2) Wrong length expiry date from terminal \S

Where: \S specifies the terminal ID.
Cause: The terminal message contained an expiry date that was not the correct length.
Remedy: Determine why the terminal sent the invalid expiry date and contact the terminal vendor.

7360	(2)	Non-numeric expiry date from terminal \S
-------------	------------	---

Where: \S specifies the terminal ID.

Cause: The terminal message contained an expiry date that was not numeric.

Remedy: Determine why the terminal sent the invalid expiry date and contact the terminal vendor.

7370	(2)	Invalid track2 from terminal \S is \?
-------------	------------	--

Where: \S specifies the terminal ID.
\? specifies the track 2 data sent in the terminal message.

Cause: The terminal message contained a track 2 data that was invalid.

Remedy: Determine why the terminal sent the invalid track 2 data and contact the terminal vendor.

7380	(2)	Invalid amount from terminal \S
-------------	------------	--

Where: \S specifies the terminal ID.

Cause: The terminal message contained an amount field that was invalid.

Remedy: Determine why the terminal sent the invalid amount field and contact the terminal vendor.

7390	(2)	Bad conversion of confirmation code, terminal \S
-------------	------------	---

Where: \S specifies the terminal ID.

Cause: The confirmation code in the terminal message could not be converted into a numeric format because it contained invalid characters.

Remedy: Determine why the terminal sent the confirmation code and contact the terminal vendor.

7400	(2)	Balance code cannot be converted to integer \S
-------------	------------	---

Where: \S specifies the terminal ID.

Cause: The balance code in the terminal message could not be converted into numeric format because it contained an invalid value.

Remedy: Determine why the terminal sent the invalid balance code and contact the terminal vendor.

7410 (2) More than 7 PIN tries so far on terminal \S

Where: \S specifies the terminal ID.
Cause: More than seven PIN tries were received by the specified terminal.
Remedy: No action required.

7420 (2) # PIN tries so far on terminal \S

Where: # specifies the number of PIN tries.
\S specifies the terminal ID.
Cause: Fewer than seven PIN tries were received by the specified terminal.
Remedy: No action required.

7430 (0) Terminal out of balance for first time \S

Where: \S specifies the terminal ID.
Cause: The specified terminal is out of balance for the first time.
Remedy: Contact Systems Manager. If problem persists contact the terminal vendor and HELP24.

7440 (0) Terminal still out of balance \S

Where: \S specifies the terminal ID.
Cause: Once the terminal has become out of balance, this message is displayed each time a new transaction is performed.
Remedy: Contact Systems Manager. If problem persists contact the terminal vendor and HELP24.

7450 (0) Balance confirmed and reset to zero \S

Where: \S specifies the terminal ID.
Cause: The terminal and BASE24 totals have been reconciled and the totals agree.
Remedy: No action required.

7460	(0)	Balance not confirmed but reset to zero \S
-------------	------------	---

Where: \S specifies the terminal ID.

Cause: The terminal and BASE24 totals have been reconciled, but the totals do not agree.

Remedy: Contact Systems Manager.

7470	(2)	Invalid balance code # received from terminal \S
-------------	------------	---

Where: # specifies the balance code number.
\S specifies the terminal ID.

Cause: The terminal message contained an invalid balance code.

Remedy: Determine why the terminal sent the invalid balance code and contact the terminal vendor.

7480	(2)	Read error # on PRDF, not in window assumed. Terminal \S
-------------	------------	---

Where: # specifies the Guardian file error.
\S specifies the terminal ID.

Cause: The system was unable to check if reconciliation occurred during the balancing window due to a read error in the POS Retailer Data File (PTDF).

Remedy: Check the PRDF.

7485	(3)	Error while attempting to get token \
-------------	------------	--

Where: \ specifies the BASE24-pos Release 5.0 token ID.

Cause: A fatal error was encountered whilst attempting to read the token indicated. The process abended.

Remedy: Contact Systems Manager.

7490	(1)	\S Unable to add POS 5.0 token \, token already exists
-------------	------------	---

Where: \S specifies the terminal ID.
\ specifies the BASE24-pos Release 5.0 token ID.

Cause: The APACS Device Handler module attempted to add the BASE24-pos Release 5.0 token that already existed. The transaction was declined.

Remedy: Contact Systems Manager.

7500 (1) \S Unable to add POS 5.0 token \, no room available

Where: \S specifies the terminal ID.
\ specifies the BASE24-pos Release 5.0 token ID.

Cause: The APACS Device Handler module attempted to add the BASE24-pos Release 5.0 token, but there was not enough room. The transaction was declined.

Remedy: Contact Systems Manager.

**7510 (3) \S - Error while attempting to add token **

Where: \S specifies the terminal ID.
\ specifies the BASE24-pos Release 5.0 token ID.

Cause: A fatal error was encountered whilst attempting to add the token indicated. The process abended.

Remedy: Contact Systems Manager.

7520 (1) \S - Unable to add Station ID token \, token already exists

Where: \S specifies the terminal ID.
\ specifies the station token ID.

Cause: The system was unable to add the station ID token but it already exists.

Remedy: Contact Systems Manager.

7530 (1) \S - Unable to add Station ID token \, no room available

Where: \S specifies the terminal ID.
\ specifies the station token ID.

Cause: The system was unable to add the station ID token because no room was available.

Remedy: Contact Systems Manager.

**7540 (3) \S - Error while attempting to add token **

Where: \S specifies the terminal ID.
\ specifies the station token ID.

Cause: An error occurred while trying to add the token indicated.

Remedy: Contact Systems Manager.

7580	(0)	Refund cnt and/or amt put device \\\\\\\\\\\\\\\ over refund limits
-------------	------------	--

Where: \\\\\\\\\\\\\\\ specifies the terminal ID.

Cause: A refund performed at the specified terminal has put the terminal over the refund limit in the POS Retailer Data File (PTDF).

Remedy: No action required.

7585	(2)	Currency conversion failed in dcbe^check^tdf^limits
-------------	------------	--

Cause: Currency conversion of amount fields failed.

Remedy: Contact HELP24.

7590	(1)	Tran number \\\\\ for \S received - no longer expected
-------------	------------	---

Where: \\\\\ specifies the transaction sequence number.
 \S specifies the terminal ID.

Cause: The system received a response message from a terminal that was no longer expected.

Remedy: No action required unless the message appears frequently, then contact Systems Manager.

7591	(1)	Tran number \\\\\ for \S received - no longer expected
-------------	------------	---

Where: \\\\\ specifies the transaction sequence number.
 \S specifies the terminal ID.

Cause: The system received a response message from a terminal that was no longer expected.

Remedy: No action required unless the message appears frequently, then contact Systems Manager.

7600	(2)	Invalid response code \\\ received from AUTH
-------------	------------	---

Where: \\\ specifies the transaction response code.

Cause: An invalid response code was received from the Authorisation process. This is a program error.

Remedy: Contact HELP24.

**7610 (1) Token caused token buffer to exceed maximum allowable limit
- trn \?, card **

Where: \? specifies the sequence number.
\\\ specifies the PAN.

Cause: There was more than 500 bytes of token data to be stored in the POS Terminal Data File (PTDF). Transaction processing continued.

Remedy: Contact Systems Manager.

**7620 (3) Error logging Token - trn \?, card **

Where: \? specifies the sequence number.
\\\ specifies the PAN.

Cause: There was more than 500 bytes of token data to be stored in the POS Terminal Data File (PTDF). Transaction processing continued.

Remedy: Contact Systems Manager.

7630 (0) \S - RESET OF MAC KEYS SUCCESSFUL

Where: \S specifies the terminal ID.

Cause: The operator command to reset the MAC keys has been successfully processed.

Remedy: No action required.

7640 (1) \S - RESET OF MAC KEYS FAILED

Where: \S specifies the terminal ID.

Cause: The operator command to reset the PTDF MAC key failed because of an invalid value entered in the PTDF MAC master key.

Remedy: Check the POS Retailer Data File (PTDF).

7650	(0)	Invalid command received from @ - *****
-------------	------------	---

Where: @ specifies the message source.
***** specifies the message text.

Cause: An invalid 9500 command received. The valid values are:

9500	=	LOADFLAGON/LOADFLAGOFF
9501	=	Warmboot
9521	=	Reset MAC keys
9525	=	ACTIVATE/DEACTIVATE PTDF status

Remedy: No action required.

7660	(0)	Warmboot of Device Handler completed
-------------	------------	---

Cause: The operator's warmboot request was completed.

Remedy: No action required.

7670	(2)	Invalid 9525 command format
-------------	------------	------------------------------------

Cause: An invalid 9525 deactivate command has been entered by an operator.

Remedy: No action required.

7680	(2)	Invalid 9525 command format
-------------	------------	------------------------------------

Cause: An invalid 9525 deactivate command has been entered by an operator.

Remedy: No action required.

7690	(0)	Terminal \S - Deactivated
-------------	------------	----------------------------------

Where: \S specifies the terminal ID.

Cause: The system has received a deactivate command which has been processed successfully.

Remedy: No action required.

7700 (0) Terminal \S - Pending Deactivation

Where: \S specifies the terminal ID.
Cause: The system has received a deactivate command which has been processed successfully and the terminal has been placed in a pending deactivation state until it is reconciled.
Remedy: No action required.

7710 (0) Terminal \S - Activated

Where: \S specifies the terminal ID.
Cause: The system has received an activate command which has been processed successfully.
Remedy: No action required.

7720 (0) Unable to Activate/Deactivate terminal \S - TDF read error

Where: \S specifies the terminal ID.
Cause: The system was unable to update the deactivate flag in the PTDF due to a read error.
Remedy: Check the PTDF.

7750 (2) Invalid case index # in DUTL^AC^GET^ASSIGNS

Where: # specifies the case index.
Cause: A programming error has occurred.
Remedy: Contact HELP24.

7760 (3) LCONF read error \e, Param Value: \S

Where \e specifies the GUARDIAN file error.
\S specifies the terminal ID.
Cause: Read error on LCONF.
Remedy: Contact Systems Manager.

7770	(1)	Param POS-DH-MAX-TDFS invalid in \F; defaulting to 20 TDFS
-------------	------------	---

Where: \F specifies the LCONF file name.

Cause: The value in the LCONF parameter POS-DH-MAX-TDFS is invalid.

Remedy: Check the LCONF parameter.

7780	(1)	Param POS-DH-MAX-TDFS too small in \F; defaulting to 20 TDFS
-------------	------------	---

Where: \F specifies the LCONF file name.

Cause: The value in the LCONF parameter POS-DH-MAX-TDFS is too small.

Remedy: Check the LCONF parameter.

7790	(1)	Param POS-DH-AUTH-RESP-TIMEOUT invalid in \F defaulting to 30 sec.
-------------	------------	---

Where: \F specifies the LCONF file name.

Cause: The value in the LCONF parameter POS-DH-AUTH-RESP-TIMEOUT (authorisation response timeout) is invalid.

Remedy: Check the LCONF parameter.

7800	(1)	Param POS-APDH-AUTH-DUP-TIMEOUT invalid in \F; defaulting to 10 sec.
-------------	------------	---

Where: \F specifies the LCONF file name.

Cause: The value in the LCONF parameter POS-APDH-AUTH-DUP-TIMEOUT is invalid.

Remedy: Check the LCONF parameter.

7810	(1)	Param POS-APDH-STALE-REQ-TIMEOUT invalid in \F defaulting to 45 seconds.
-------------	------------	---

Where: \F specifies the LCONF name.

Cause: The value in the LCONF parameter POS-APDH-STALE-REQ-TIMEOUT is invalid.

Remedy: Check the LCONF parameter.

7820	(1)	Param POS-APDH-PRE-FLOOR-LIMIT not present or invalid in \F, defaulting to 10
-------------	------------	--

Where: \F specifies the LCONF file name.

Cause: The LCONF parameter POS-APDH-PRE-FLOOR-LIMIT is either not present or contains an invalid value.

Remedy: Check the LCONF parameter.

7830	(1)	Param POS-APDH-USE-PRE-FL-LIMIT not present or invalid in \F, defaulting to Y
-------------	------------	--

Where: \F specifies the LCONF file name.

Cause: The LCONF parameter POS-APDH-USE-PRE-FL-LIMIT is either not present or contains an invalid value.

Remedy: Check the LCONF parameter.

7840	(1)	Param POS-APDH-USE-POST-FL-LIMIT not present or invalid in \F, defaulting to Y
-------------	------------	---

Where: \F specifies the LCONF file name.

Cause: The LCONF parameter POS-APDH-USE-POST-FL-LIMIT is either not present or contains an invalid value.

Remedy: Check the LCONF parameter.

7850	(1)	Param POS-APDH-POST-FLOOR-LIMIT not present or invalid in \F, defaulting to 20
-------------	------------	---

Where: \F specifies the LCONF file name.

Cause: The LCONF parameter POS-APDH-POST-FLOOR--LIMIT is either not present or contains an invalid value.

Remedy: Check the LCONF parameter.

7860	(1)	Param POS-APDH-CONFIRM-FLAG not present or invalid in \F, defaulting to N
-------------	------------	--

Where: \F specifies the LCONF file name.

Cause: The LCONF parameter POS-APDH-CONFIRM-FLAG is either not present or contains an invalid value.

Remedy: Check the LCONF parameter.

7870	(1)	Param POS-APDH-CONFIRM-TIMEOUT not present or invalid in \F, defaulting to \$ seconds
-------------	------------	--

Where: \F specifies the LCONF file name and \$ = 15.

Cause: The LCONF parameter POS-APDH-CONFIRM-TIMEOUT is either not present or contains an invalid value.

Remedy: Check the LCONF parameter.

7910	(1)	More than 20 HOLD MESSAGE PREFIXES defined, only the first 20 will be used.
-------------	------------	--

Cause: The LCONF parameter POS-APDH-HOLD-MESSAGE-PREFIXES contains more than the maximum 20 prefixes.

Remedy: Check the LCONF parameter.

7920	(0)	Param POS-APDH-HOLD-MESSAGE-PREFIX not present, no hold messages sent
-------------	------------	--

Cause: The parameter POS-APDH-HOLD-MESSAGE-PREFIX was not present in the LCONF and so no messages were sent.

Remedy: Check the LCONF parameter.

7930	(1)	More than 20 REFERRAL PREFIXES defined, only the first 20 will be used.
-------------	------------	--

Cause: The LCONF parameter POS-APDH-REFERRAL-PREFIXES contains more than the maximum 20 prefixes.

Remedy: Check the LCONF parameter.

7940	(0)	Param POS-APDH-REFERRAL-PREFIX not present, no referral codes sent to device
-------------	------------	---

Cause: The parameter POS-APDH-REFERRAL-PREFIX was not present in the LCONF and so no referral codes were sent to the device.

Remedy: Check the LCONF parameter.

7950 (1) Param POS-APDH-EXTEND-RESP-TIMEOUT not present or invalid, defaulting to 120 seconds

Cause: The LCONF parameter POS-APDH-EXTEND-RESP-TIMEOUT is either not present or contains an invalid value.

Remedy: Check the LCONF parameter.

7960 (0) Param POS-APDH-LOG-APACS-DATA not present or invalid in \F, defaulting to N

Where: \F specifies the LCONF filename.

Cause: The LCONF parameter POS-APDH-LOG-APACS-DATA was either not present or invalid and so defaulted to N.

Remedy: Check the LCONF parameter.

7970 (0) Param POS-DH-LMT-EXCEED-DISP not present or invalid in \F, defaulting to 0

Where: \F specifies the LCONF filename.

Cause: The LCONF parameter POS-DH-LMT-EXCEED-DISP was either not present or invalid and so defaulted to 0.

Remedy: Check the LCONF parameter.

7980 (2) Invalid case (j) encountered in DUTL^AD^GET^PARAMS

Cause: A programming error has occurred.

Remedy: Contact HELP24.

7990 (1) Unable to get posting date from PRDF for \\\\\\\\\\\\\\\\\ - #

Where: \\\\\\\\\\\\\\\\\ specifies the retailer ID.

specifies the Guardian file error.

Cause: The system was unable to read the PTDF because of the specified Guardian error.

Remedy: Determine the correct action to take based on the error listed.

8000	(2)	Terminal \S - PTDF table full \e - increase LCONF param MAX-TDFS
-------------	------------	---

Where: \S specifies the terminal ID.
 \e specifies the Guardian file error.

Cause: The PTDF table in extended memory is full.

Remedy: Increase the LCONF parameter POS-DH-MAX-TDFS.

8010	(2)	Terminal \S - PTDF read error \e
-------------	------------	---

Where: \S specifies the terminal ID.
 \e specifies the Guardian file error.

Cause: The system was unable to add the PTDF record for the terminal to the PTDF table in extended memory.

Remedy: Determine the correct action to take based on the error listed.

8011	(2)	Terminal \S - PTDF read error \e
-------------	------------	---

Where: \S specifies the terminal ID.
 \e specifies the Guardian file error.

Cause: The system was unable to add the PTDF record for the terminal to the PTDF table in extended memory.

Remedy: Determine the correct action to take based on the error listed.

8020	(2)	Terminal \S - Invalid terminal type \\ in PTDF
-------------	------------	---

Where: \S specifies the terminal ID.
 \\ specifies the terminal type in the PTDF record

Cause: The APACS Device Handler module received a message from a terminal that it does not support.

Remedy: Contact Systems Manager.

8040	(2)	Terminal \S - unable to decrypt MAC master key
-------------	------------	---

Where: \S specifies the terminal ID.

Cause: The system was unable to decrypt the MAC master key.

Remedy: If problem persists, reinitialise both the terminal and PTDF record.

8050 (2) Terminal \S - Invalid MAC key in PTFD

Where:	\S specifies the terminal ID.
Cause:	The decrypted data in the MAC key field in the PTDF does not correspond to hexadecimal characters.
Remedy:	If problem persists, re-initialise both the terminal and PTDF record.

**8060 (2) Open error # ON **

Where:	# specifies the Guardian file error. \\ specifies the file name.
Cause:	The system was unable to open the specified file due to the specified Guardian error.
Remedy:	Check the specified Guardian file error.

8070 (0) Terminal \S - Deactivated

Where:	\S specifies the terminal ID.
Cause:	A terminal pending deactivation has been deactivated following a terminal cutover.
Remedy:	No action required.

8080 (2) Unable to reverse TDF totals - invalid service \\ in STM

Where:	\\ specifies the card service identifier.
Cause:	An attempt to reverse the totals in the PTDF was unsuccessful due to an invalid card service in the PSTM.
Remedy:	Note that the totals are incorrect. In addition, determine why an invalid card service was in the PSTM.

8100 (1) Invalid language code # in PTDF for \S, defaulting to 0

Where:	# specifies the language code. \S specifies the terminal ID.
Cause:	The language code entered in the PTDF is invalid.
Remedy:	Check the PTDF language code.

8110	(2)	Invalid response type # for terminal \S, no response sent
-------------	------------	--

Where: # specifies the response type.

\S specifies the terminal ID.

Cause: Error in program logic.

Remedy: Contact HELP24.

8120	(3)	PROGRAM LOGIC ERROR: Missing parameter in proc duti^al^send^reversal
-------------	------------	---

Cause: A program logic error has occurred. The process abends.

Remedy: Contact HELP24.

8130	(3)	Error while attempting to retrieve all token data
-------------	------------	--

Cause: A fatal error occurred while attempting to retrieve the token data from the PTDF for reversal processing.

Remedy: Contact Systems Manager.

8140	(2)	RDF locate error # for \\\\\\\\\\\ during term cutover
-------------	------------	---

Where: # specifies the Guardian file error.

\\\\\\\\\\\\ specifies the retailer ID.

Cause: The system was unable to read the POS Retailer Data File (PRDF) during terminal cutover.

Remedy: Check the PRDF.

8150	(2)	Unable to update TDF totals - invalid service \\ in STM
-------------	------------	--

Where: \\ specifies the service (card type) in the PSTM.

Cause: An invalid service has been returned to the APACS Device Handler module from the Authorisation process.

Remedy: Possible program error. Contact HELP24.

8160 (2) PTDF write error \e for terminal \s

Where: \e specifies the Guardian file error.
\s specifies the terminal ID.

Cause: The system was unable to update the PRDF because of the specified Guardian error.

Remedy: Determine the correct action to take based on the error listed.

8161 (2) PTDF write error \e for terminal \s

Where: \e specifies the Guardian file error.
\s specifies the terminal ID.

Cause: The system was unable to update the PRDF because of the specified Guardian error.

Remedy: Determine the correct action to take based on the error listed.

8166 (2) Invalid hex digit '\'

Where: '\' specifies the hex digit.

Cause: An invalid hex digit was received. A conversion is done on the terminal type to check whether the terminal type is > 7.

Remedy: Determine why the terminal sent the invalid hex digit for terminal type and contact the terminal vendor.

8170 (2) No decoupling key for this req type, terminal \S

Where: \S specifies the terminal ID.

Cause: MAC processing cannot be performed for the type of message received.

Remedy: Contact HELP24.

8200 (1) Param POS-APDH-DISCONNECT-ON-ERROR not present or invalid - defaulting to Y

Cause: The LCONF param POS-APDH-DISCONNECT-ON-ERROR param was either not present or invalid and so defaulted to Y.

Remedy: Check the LCONF parameter.

8202	(1)	Param FUND-TRAN-TO-CASH-ADV not present or invalid - defaulting to N
-------------	------------	---

Cause: The LCONF param FUND-TRAN-TO-CASH-ADV param was either not present or invalid and so defaulted to N.

Remedy: Check the LCONF parameter.

8210	(0)	Unable to process transaction request for terminal \S
-------------	------------	--

Where: \S specifies the terminal ID.

Cause: The transaction message has been dropped instead of the station being disconnected.

Remedy: Check that a record for the terminal ID exists in the PTDF.

8220	(2)	\S - Unable to add POS 5.1 token C0, no room available
-------------	------------	---

Where: \S specifies the terminal ID.

Cause: The BASE24-pos Release 5.1 token could not be added to the PSTM due to lack of space.

Remedy: The APACS Device Handler module attempted to add the BASE24-pos Release 5.1 token but there was not enough room. The transaction was declined.

8230	(2)	\S - Unable to add POS 5.1 token C0, token already exists
-------------	------------	--

Where: \S specifies the terminal ID.

Cause: An attempt was made to add the BASE24-pos Release 5.1 token to the PSTM, but it already exists.

Remedy: The APACS Device Handler module attempted to add the BASE24-pos Release 5.1 token that already existed. The transaction was declined.

8240	(3)	\S - Error while attempting to add token C0
-------------	------------	--

Where: \S specifies the terminal ID.

Cause: The BASE24-pos Release 5.1 token could not be added to the PSTM. The Device Handler module abends after this message.

Remedy: Contact Systems Manager.

8250 (2) Unable to update PTDF totals - invalid service \\ in PSTM

Where: \\ specifies the card service.

Cause: The PSTM service does not match any of the PTDF services when an attempt is made to update the PTDF totals.

Remedy: Ensure that the service is configured on the PTDF record.

8251 (2) Unable to update PTDF totals - invalid service \\ in PSTM

Where: \\ specifies the card service.

Cause: The PSTM service does not match any of the PTDF services when an attempt is made to update the PTDF totals.

Remedy: Ensure that the service is configured on the PTDF record.

8261 (1) Terminal \S received reversal, original txn not reversible

Where: \S specifies the terminal ID.

Cause: A reversal message was received but the original transaction was not approved. An UNABLE TO AUTH response is sent to the device and the transaction is dropped.

Remedy: No action required, warning only.

8270 (2) Currency Code \\ not supported for terminal \S

Where: \\ specifies the currency code
 \S specifies the terminal ID.

Cause: The BASE24-pos Multiple Currency add-on product is not bound in and the currency code received in the request does not match the currency code on the PTDF.

Remedy: License the BASE24-pos Multiple Currency add-on product or ensure that the currency code in request messages matches the value on the PTDF record.

8271 (2) Currency code \\ not supported for terminal \S

Where: \\ specifies the currency code
 \S specifies the terminal ID.

Cause: The BASE24-pos Multiple Currency add-on product is bound in but the currency code received in the request does not exist in the Exchange Rate File (ERF).

Remedy: Add an exchange rate for the currency code to the ERF.

8280	(1)	EMV Data received in EFT msg from terminal \S but EMV product is not supported
-------------	------------	---

Where: \S specifies the terminal ID.

Cause: The BASE24-pos EMV add-on product is not bound in, and an EFT transaction with EMV data was received. The transaction continues if possible.

Remedy: License the BASE24-pos EMV add-on product or stop the terminal from processing EMV transactions.

8285	(2)	No. fields in eft req incorrect, terminal \S
-------------	------------	---

Where: \S specifies the terminal ID.

Cause: The terminal message contains data that is not expected by BASE24.

Remedy: Contact the terminal vendor.

8290	(1)	\S – Unable to add Point of Service Data token C4, no room available
-------------	------------	---

Where: \S specifies the terminal ID.

Cause: The Point of Service Data token could not be added to the PSTM due to lack of space.

Remedy: The APACS Device Handler module attempted to add the Point of Service Data token, but there was not enough room. The transaction was declined.

8300	(1)	\S - Unable to add Point of Service Data token C4, token already exists
-------------	------------	--

Where: \S specifies the terminal ID.

Cause: An attempt was made to add the Point of Service Data token to the PSTM, but it already exists.

Remedy: The APACS Device Handler module attempted to add the POS Data token that already existed. The transaction was declined.

8310 (3) \S - Error while attempting to add token C4

Where: \S specifies the terminal ID.

Cause: The Point of Service Data token could not be added to the PSTM. The Device Handler module abends after this message.

Remedy: Contact Systems Manager.

8320 (1) EMV Data received in Auth msg from terminal \S but EMV product is not supported

Where: \S specifies the terminal ID.

Cause: The BASE24-pos EMV add-on product is not bound in and an Authorisation transaction with EMV data was received. The transaction continues if possible.

Remedy: License the BASE24-pos EMV add-on product or stop the terminal from processing EMV transactions.

8330 (0) Invalid command received from @ - *****

Where: @ specifies the message source.
***** specifies the message text.

Cause: An invalid 9500 command received. The valid values are as follows:

9500	=	LOADFLAGON/LOADFLAGOFF
9501	=	Warmboot
9521	=	Reset MAC keys
9525	=	ACTIVATE/DEACTIVATE PTDF status

Remedy: No action required.

8340 (0) Invalid command type # returned by dcdda^edit^dct^command

Where: # specifies the command type.

Cause: An invalid 9500 command received. The valid command types are as follows:

0	=	LOADFLAGON
1	=	LOADFLAGOFF

Remedy: No action required.

8350	(0)	Invalid command received from - *****
-------------	------------	---

Where: ***** specifies the message text.

Cause: An invalid 9500 command received. The valid text values are LOADFLAGON and LOADFLAGOFF.

Remedy: No action required.

8351	(0)	Invalid command received from - *****
-------------	------------	---

Where: ***** specifies the message text.

Cause: An invalid 9500 command received. The valid text values are LOADFLAGON and LOADFLAGOFF.

Remedy: No action required.

8360	(0)	Downline load flag is turned on for term \S
-------------	------------	--

Where: \S specifies the terminal ID.

Cause: The system has received a LOADFLAGON command that has been processed successfully, and the terminal is requested to call its terminal management system.

Remedy: No action required.

8370	(0)	Downline load flag is turned on for term \S
-------------	------------	--

Where: \S specifies the terminal ID.

Cause: The system has received a LOADFLAGON command that has been processed successfully, and the terminal is requested not to call its Terminal Management System.

Remedy: No action required.

8380	(0)	Unable to set Downline load flag for term \S due to PTDF read error
-------------	------------	--

Where: \S specifies the Terminal ID.

Cause: The system was unable to update the REMOTE-CONF flag in the PTDF due to a read error.

Remedy: Check the PTDF.

8390 (2) Currency conversion failed in dcbb^format^stm^req

Cause: Currency conversion of amount fields failed.

Remedy: Contact HELP24.

8400 (2) \S - Unable to add Reversal Date and Time token BH, token already exists

Where: \S specifies the terminal ID.

Cause: An attempt was made to add the BASE24 Reversal Date and Time token to the PSTM, but it already exists.

Remedy: The APACS Device Handler module attempted to add the BASE24 Reversal Date and Time token that already existed. The transaction was declined.

8410 (2) \S - Unable to add Reversal Date and Time token BH, no room available

Where: \S specifies the terminal ID.

Cause: The BASE24 Reversal Date and Time token could not be added to the PSTM due to lack of space.

Remedy: The APACS Device Handler module attempted to add the BASE24 Reversal Date and Time token but there was not enough room. The transaction was declined.

8420 (3) \S - Error while attempting to add token BH

Where: \S specifies the terminal ID.

Cause: The BASE24 Reversal Date and Time token could not be added to the PSTM. The Device Handler module abends after this message.

Remedy: Contact Systems Manager.

8425	(2)	Terminal \S – Unable to convert CAVV from base 64 for token C6
-------------	------------	---

Where: \S specifies the terminal ID.

Cause: An error occurred when attempting to decode the cardholder authentication value received from the terminal. The Device Handler module declines the transaction.

Remedy: Determine the source of the transaction containing the invalid data and correct the problem. If this does not solve the problem, collect all relevant information and contact HELP24.

8430	(3)	Error \E while attempting to get token 04
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Where: \E specifies the Guardian Error.

Cause: The BASE24-pos Release 5.0 token could not be retrieved. The Device Handler module abends after this message.

Remedy: Contact Systems Manager.

8430	(1)	\S - Unable to add Trans Stain XID token C6, token already exists
-------------	------------	--

Where: \S specifies the terminal ID.

Cause: An attempt was made to add the BASE24-pos Trans Stain XID token to the PSTM, but it already exists.

Remedy: The APACS Device Handler module attempted to add the BASE24-pos Trans Stain XID token that already existed. The transaction was declined.

8435	(1)	\S - Unable to add Trans Stain XID token C6, no room available
-------------	------------	---

Where: \S specifies the terminal ID.

Cause: The BASE24-pos Trans Stain XID token could not be added to the PSTM due to lack of space.

Remedy: The APACS Device Handler module attempted to add the BASE24-pos Trans Stain XID token but there was not enough room. The transaction was declined.

8440 (3) \S - Error while attempting to add token C6

Where: \S specifies the terminal ID.
Cause: The BASE24-pos Trans Stain XID token could not be added to the PSTM. The Device Handler module abends after this message.
Remedy: Contact Systems Manager.

8440 (1) \S - MAC GENERATION FAILURE #

Where: \S specifies the terminal ID.
specifies the security module error number.
Cause: The system was unable to generate a MAC due to the error listed.
Remedy: Take the action indicated by the error.

8445 (1) \S - Unable to add Authentication Data token CE, token already exists

Where: \S specifies the terminal ID.
Cause: An attempt was made to add the BASE24-pos Authentication Data token to the PSTM, but it already exists.
Remedy: The APACS Device Handler module attempted to add the BASE24-pos Authentication Data token that already existed. The transaction was declined.

8450 (3) Required parameter missing in proc dcb^format^stm^req; process abending

Cause: A required parameter is missing. The process abends.
Remedy: Contact HELP24.

8450 (1) \S - Unable to add Authentication Data token CE, no room available

Where: \S specifies the terminal ID.
Cause: The BASE24-pos Authentication Data token could not be added to the PSTM due to lack of space.
Remedy: The APACS Device Handler module attempted to add the BASE24-pos Authentication Data token but there was not enough room. The transaction was declined.

8455	(3)	\S - Error while attempting to add token CE
-------------	------------	--

Where: \S specifies the terminal ID.

Cause: The BASE24-pos Authentication Data token could not be added to the PSTM. The Device Handler module abends after this message.

Remedy: Contact Systems Manager.

8460	(1)	\S - Unable to add POS Data1 token CH, token already exists
-------------	------------	--

Where: \S specifies the terminal ID.

Cause: An attempt was made to add the BASE24-pos POS Data1 token to the PSTM, but it already exists.

Remedy: The APACS Device Handler module attempted to add the BASE24-pos POS Data1 token that already existed. The transaction was declined.

8465	(1)	\S - Unable to add POS Data1 token CH, no room available
-------------	------------	---

Where: \S specifies the terminal ID.

Cause: The BASE24-pos POS Data1 token could not be added to the PSTM due to lack of space.

Remedy: The APACS Device Handler module attempted to add the BASE24-pos POS Data1 token but there was not enough room. The transaction was declined.

8470	(3)	\S - Error while attempting to add token CH
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Where: \S specifies the terminal ID.

Cause: The BASE24-pos POS Data1 token could not be added to the PSTM. The Device Handler module abends after this message.

Remedy: Contact Systems Manager.

D. TRANSACTION KEY SECURITY

This section provides an overview of the security aspects concerning the transmission of messages to and from the APACS 70 terminal and the Device Handler module. More details of the security system are given in Book 5 of the APACS 70 Standard.

D.1 TERMINOLOGY

One Way Function	This is a non-reversible cryptographic sum of two 64-bit fields D (data register) and K (key register) to produce a 64-bit result.
Key Register	The memory location in the terminal or acquirer containing a value known only to those two parties which is used in the computation of a transaction key between the terminal and acquirer. Initial key register values are provided by the manufacturer and are hard-coded into the terminal for each dataset maintained. The values are input to the POS Terminal Data File (PTDF) record where they are stored in encrypted form.
Decoupling Keys	These keys, derived either from the PAN or merchant number data depending on transaction type, are used as input to the OWF. They are moved into the Data register and used together with the current contents of the Key register to derive other keys.
MAC Processing Key	This key is used to verify and generate message authentication codes (MACs).
PIN Processing Key	This key is used to encrypt PIN blocks.
Key Register Substitute	This key is commonly referred to as KEYVAL and is used together with the card key to construct the current PIN Processing key.
Card Key	This key, derived from track 2 data, is used together with the key register substitute (KEYVAL) to construct the current PIN Processing key.
Authorisation Parameter	This parameter is a one way function of transaction data block and the transaction key acquirer or transaction key issuer.

D.2 TRANSACTION KEY MANAGEMENT

At the start of every transaction, a set of transaction keys is generated for use in all cryptographic and authentication procedures executed during the course of the transaction. The transaction keys are derived from decoupling keys and the contents of a key register, both of which are formulated to ensure that the keys selected are unique to the particular transaction.

The way in which the terminal obtains the decoupling keys depends on its mode of operation, but the selection is not predictable and does not enter into building the transaction key until the transaction is initiated. The key register is also updated in a way that cannot be predicted in advance.

The decoupling keys and key registers hold 64 bits. The terminal is equipped with a one way function (OWF) F , which has two 64 bit inputs; data register D and key register K and a single 64 bit output $F(D,K)$.

Figure D-1 is an example of the OWF, based on the Data Encryption Algorithm (DEA) in accordance with the ANSI Standard X3.92.

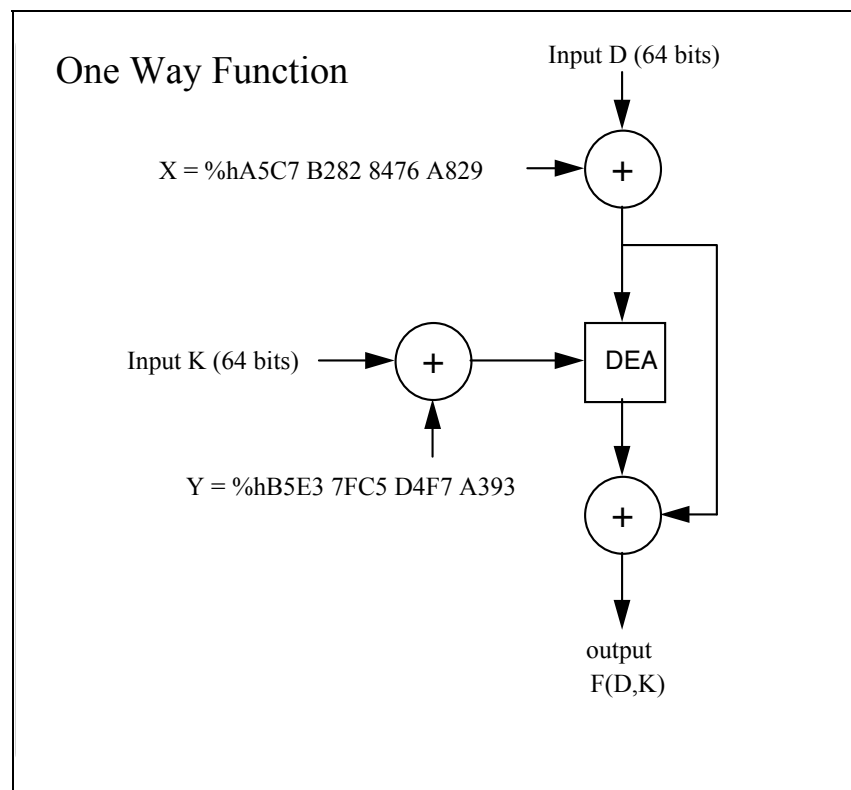


Figure D-1 Example of One Way Function (OWF)

D.3 KEY REGISTER MANAGEMENT

Figure D-2 shows the formation of transaction keys for both acquirers and issuers.

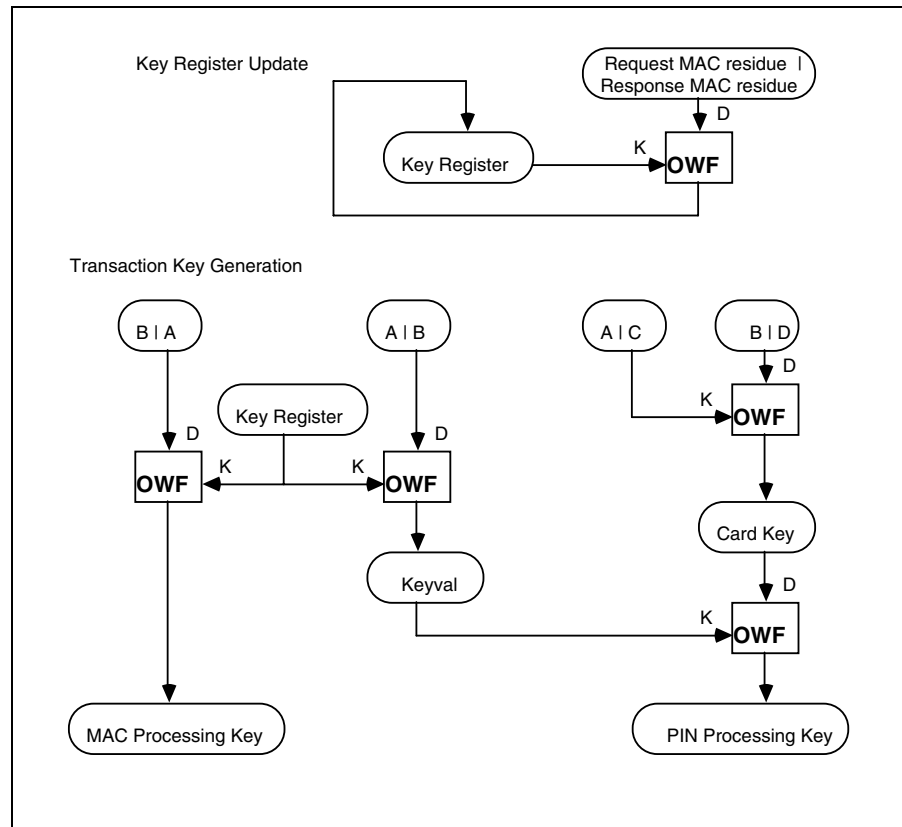


Figure D-2 Transaction Key Generation

D.3.1 Derivation of Cryptographic Keys

To build the transaction key between the terminal and the acquirer, the terminal reads an appropriate decoupling key into D and the key register contents into K. The output $F(D,K)$ of the OWF is the MAC processing key (that is, the transaction key between the terminal and the acquirer). The decoupling key (B/A) is derived from card data or merchant number, as appropriate for the transaction.

If the terminal is to communicate via the acquirer with third parties who have no knowledge of the terminal and hence the key register value, a substitute for the key register value is computed. The terminal reads an appropriate decoupling key into D and the key register contents into K. The substitute value (KEYVAL) is the output $F(D,K)$ of the OWF. The decoupling key (A/B) is derived from card data or merchant number, as appropriate for the transaction.

To construct the PIN processing key (that is, the transaction key between the terminal and the third party), a card key must first be computed. The terminal reads two further decoupling keys into D and K. The output $F(D,K)$ of the OWF is the card key. The decoupling keys (A/C and B/D) are derived from card data or merchant number, as appropriate for the transaction.

To construct the PIN processing key (that is, the transaction key between the terminal and the third party), the terminal reads the card key into D and the substitute value (KEYVAL) into K. The output $F(D,K)$ of the OWF is the PIN processing key which can be advised by the acquirer to the third party without revealing the decoupling keys or the contents of the key register.

D.3.2 Key Register Update

The key register is updated at the completion of every transaction to assure the uniqueness of the keys used in each transaction.

The new key register value is generated by passing the Request MAC residue and the Response MAC residue through a one way function (OWF) with the previous key register value. The new key register value is the output from this OWF.

The key register held at the acquirer's host, and that held in the terminal need to be identical.

D.4 MESSAGE AUTHENTICATION

To perform message authentication, an authorisation parameter is required.

The authorisation parameter is a parameter generated under the PIN processing key or the MAC processing key that is used within the response message when generating the response MAC. It is not transmitted in the response message. The use of either key is determined by agreement between the acquirer and card issuer but either is acceptable to the terminal. By using the PIN processing key when some discretionary data is suppressed and not transmitted as part of the request message, the card issuer is authenticated by the terminal and break-forward protection achieved (depending upon the amount of data suppressed).

The parameter is generated as described below.

A 64-bit transaction data block is formed as follows:

C	N	DATA	FILLER
---	---	------	--------

where: C = 8 (Debit)
= 9 (Credit, including balance enquiries)
= 7 (Admin)
N = Length of data portion
DATA = Credit or Debit amount (Not present if Admin)
FILLER = Fs

The transaction data block is written into the Data register of the OWF. The MAC processing key or PIN processing key is written into the key register of the OWF. The result of the OWF is the authorisation parameter.

D.4.1 MAC Verification of Request Message

Message authentication is used in financial request messages (EFT or reconciliation) sent by the terminal to the Device Handler module. When the Device Handler receives a message, it must be able to check for any insertions, deletions or alterations, and to confirm that the message originates from an authorised source, i.e. the terminal. To achieve this authentication, every message includes a message authentication code (MAC), constructed using the appropriate transaction key. The method used to generate MACs is also used to link all messages and to provide random data to update the key registers at the end of a transaction.

The transaction key is used to construct a 64 bit message authentication block (MAB) from the incoming message using the DES algorithm (DEA). If the first 32 bits of the MAB match the MAC on the message then the message is verified. The remaining 32 bits of the MAB form the Request MAC residue.

Figure D-3 illustrates how the request message MAC is verified.

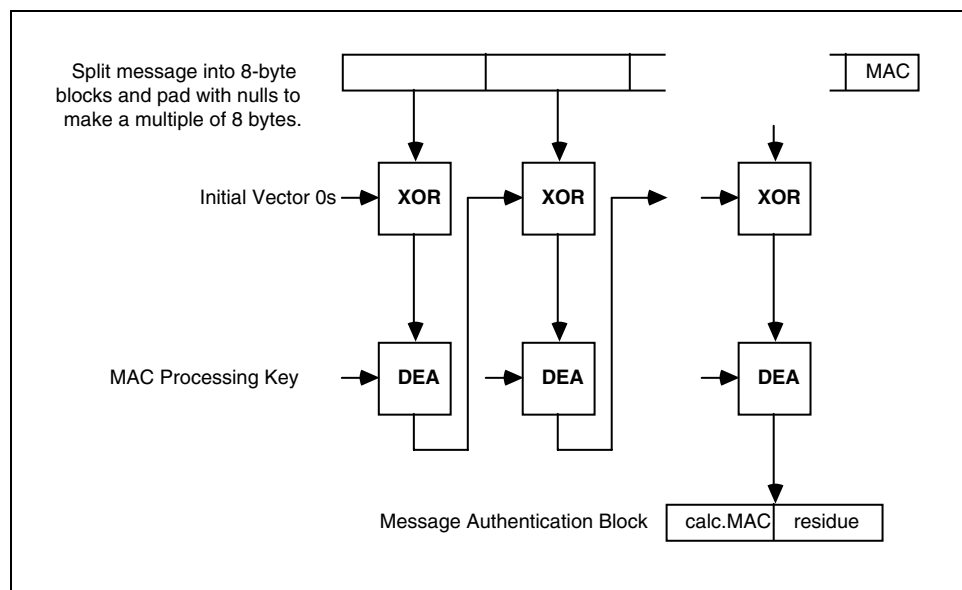


Figure D-3 Request MAC Verification

D.4.2 MAC Computation of Response Message

When a terminal receives a response message, it must be able to check that the message source is the Device Handler module with which it is communicating and not a fraudulent source. This authentication is provided by the response MAC, because only the holder of the current terminal transaction key and the request MAC can compute a valid response MAC.

Before computing the MAC for the response message, the 32-bit Request MAC residue is attached to the front of the message and the authorisation parameter appended to the end of the response message. As with the request message the data block itself must be in 8 byte blocks, padded with nulls.

The transaction key is used to generate the response MAB for this extended message. The Request MAC residue, padding and authorisation parameter bits are removed from the response message and the first 32 bits of the MAB are attached to the end of the response message as the response MAC. The remaining 32 bits of the MAB are known as the Response MAC residue.

Figure D-4 illustrates how the response message MAC is computed.

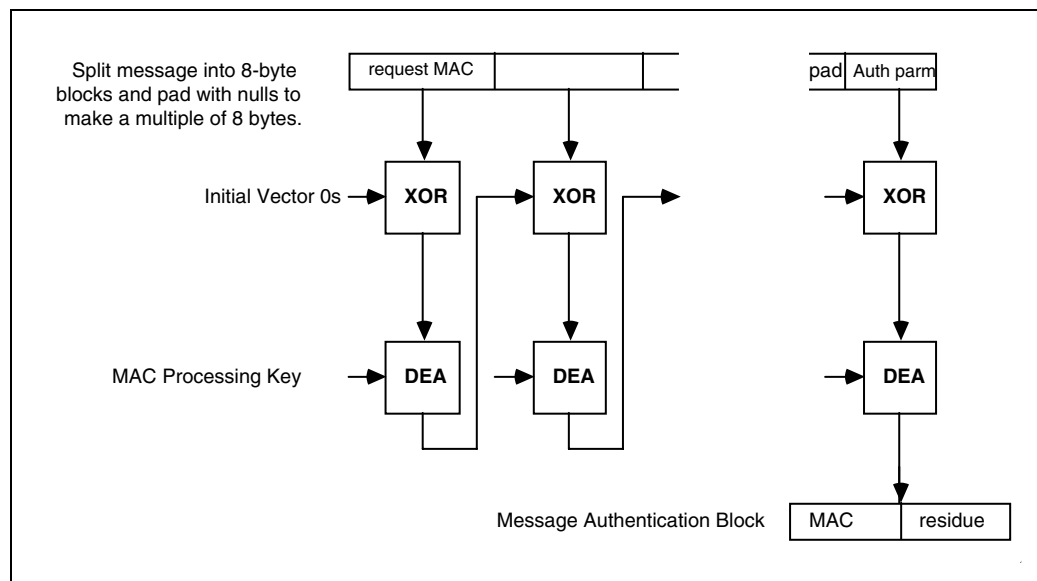


Figure D-4 Response MAC Computation

D.5 PIN BLOCK ENCRYPTION

A clear text PIN/PAN block is constructed from the PIN and the cardholder's PAN. The PIN/PAN block construction is in accordance with ANSI X9.8.

The clear text PIN/PAN block is then encrypted under the current PIN processing key, using the DES algorithm (DEA), to produce the PIN cipher block.

D.6 TRIPLE-DES PROCESSING EXTENSION

This security scheme provides optional Triple-DES encryption of PIN blocks and MACs in accordance with the Triple-DES algorithm defined in ANSI X9.52. When a terminal is configured to support Triple-DES for a particular host, it is required that both the MAC and the PIN block be Triple-DES encrypted if either value is sent to the host.

D.6.1 Key Registers

By labelling the bits in each key register byte from left to right, bits 1 to 7 are used in the DEA calculation and bit 8 is a parity bit that is set to ensure the byte has odd parity overall. Terminals and hosts supporting Single-DES maintain 64-bit key registers for key derivation (see section D.2), of which 56 bits are used by the DEA. Terminals and hosts supporting Triple-DES maintain 128-bit key registers, of which 112 bits are used by the DEA. Each Triple-DES key register is split into two halves. By labelling bits 1 to 128 from left to right, then bits 1 to 64 form Key_L (which is the 64-bit key register used for Single-DES) and bits 65 to 128 form Key_R (which is the key register extension for Triple-DES).

D.6.2 Initial Key Register Values

Terminals and hosts supporting Triple-DES use two initial key register values (also known as start keys). Both initial key register values Start Key_L and Start Key_R are derived from different unrelated secret values, as it is important that Start Key_L be mathematically unrelated to Start Key_R.

D.6.3 Derivation of Cryptographic Keys

The key derivation method described in section D.3.1 is applied separately to the two key register values (Key_L and Key_R), using the same decoupling key values (A, B, C and D blocks) as input data to produce the following four results: (MAC Processing Key)_L, (PIN Processing Key)_L, (MAC Processing Key)_R, and (PIN Processing Key)_R.

D.6.4 Transaction Key Management

The key register update processing described in section D.3.2 is applied separately to Key_L and Key_R to produce updated key register values for Key_L and Key_R. Both calculations use the Request MAC residue and Response MAC residue values as input data. In order to avoid key synchronisation problems, both key registers Key_L and Key_R are always be updated at the same time. This key update rule applies to both the terminal and the host.

D.6.5 Message Authentication Block (MAB) Generation

Calculation of the MAB is as defined in section D.4, using (MAC Processing Key)_L as the MAC Processing Key. The resulting MAB value is input to a subsequent calculation, which generates the Triple DES MAB for the message.

The input MAB is read into the data register of the DEA, the (MAC Processing Key)_R is read into the key register of the DEA and the DEA is set to decipher mode. The result of the DEA calculation is placed in the data register of the DEA.

The (MAC Processing Key)_L is then read into the key register of the DEA and the DEA is set to encipher mode. The result of the DEA calculation is the Triple-DES MAB (MAB_{3DES}).

If the resulting MAB_{3DES} block bits are labelled 1 to 64 from left to right, then bits 1 to 32 form the Triple-DES message authentication code (MAC_{3DES}) that replaces the Single-DES MAC, and bits 33 to 64 form the MAC residue.

D.6.6 PIN Cipher Block Generation

Encryption of the clear text PIN block is as defined in section D.5 using (PIN Processing Key)_L as the PIN Processing Key. The resulting PIN Cipher Block value is input to a subsequent calculation, which generates the Triple DES PIN cipher block for the transaction.

The input PIN Cipher Block is read into the data register of the DEA, the (PIN Processing Key)_R is read into the key register of the DEA and the DEA is set to decipher mode. The result of the DEA calculation is placed in the data register of the DEA.

The (PIN Processing Key)_L is then read into the key register of the DEA and the DEA is set to encipher mode. The result of the DEA calculation is the Triple-DES PIN Cipher Block (PIN Cipher Block_{3DES}) that replaces the Single-DES PIN Cipher Block.

Decryption of the Triple-DES PIN Cipher Block shall be achieved by reversing the process described above using the inverse DEA functions to produce the clear text PIN block.